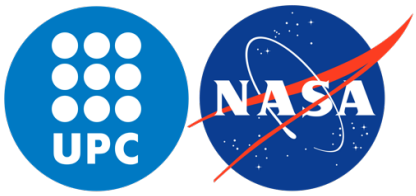
Unraveling the Core of Heliospheric Flux Ropes: Advanced Modeling of Magnetic Topologies and Thermodynamic Insights

NASA Goddard Space Flight Center

Heliophysics Science Division (HSD)

Universitat Politècnica de Catalunya

Degree in Aerospace Engineering - Degree in Engineering Physics



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Bachelor's Thesis

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May 2025

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Man must rise above the Earth, to the top of the atmosphere and beyond, for only thus will he fully understand the world in which he lives

---

Socrates (469-399 BC)

Abstract

English
Heliospheric Flux Ropes are magnetic confined field structures believed to act as the cores of coronal mass ejections (CMEs). Their study is essential for understanding space weather phenomena. In this work, a novel magnetic Flux Rope model based on a distorted geometry initially proposed by Nieves-Chinchilla et al. (2018) is presented, now extended through the inclusion of a radial current. This addition aims to explain magnetic field line discontinuities and interactions with external magnetized bodies. Furthermore, the full set of magnetohydrodynamic (MHD) equations is solved analytically in circular coordinates for the simplified case of a circular geometry. This solution could provide valuable constraints for future magnetic field fitting models and offers new insights into the internal structure of Flux Ropes.

Català:
Les cordes de flux o "Flux Ropes" heliosfèriques són estructures de camp magnètic confinat que es creu que actuen com a nuclis de les ejeccions de massa coronal (CMEs). El seu estudi és essencial per comprendre els fenòmens de meteorologia espacial. En aquest treball es presenta un nou model de Flux Rope magnètica basat en una geometria distorsionada proposada inicialment per Nieves-Chinchilla et al. (2018), i ampliat amb la inclusió d'una corrent radial. Aquesta incorporació pretén explicar discontinuïtats en les línies de camp magnètic i la interacció amb cossos magnetitzats externs. A més, es resol de forma analítica el conjunt complet d'equacions de magnetohidrodinàmica (MHD) en coordenades circulars per al cas simplificat d'una geometria circular. Aquesta solució podria proporcionar restriccions per a futurs models d'ajust de camp magnètic i ofereix nous coneixements sobre l'estructura interna de les Flux Ropes.

Castellano:
Las cuerdas de flujo o "Flux Ropes" heliosféricas son estructuras de campo magnético confinado que se cree que actúan como núcleos de las eyecciones de masa coronal (CMEs). Su estudio es fundamental para comprender los fenómenos de climatología espacial. En este trabajo se presenta un nuevo modelo de Flux Rope magnética basado en una geometría distorsionada propuesto inicialmente por Nieves-Chinchilla et al. (2018), y extendido mediante la adición de una corriente radial. Este nuevo enfoque busca explicar discontinuidades en las líneas de campo magnético y la interacción con cuerpos magnéticos externos con la propia "Flux Rope". Además, se resuelve de forma analítica el conjunto completo de ecuaciones de magnetohidrodinámica (MHD) en coordenadas circulares para el caso de una sección circular. Esta solución podría proporcionar restricciones para futuros modelos de ajuste de campo magnético y ofrece nuevas perspectivas sobre la estructura interna de las Flux Ropes.

Keywords— Flux Ropes, coronal mass ejections, elliptical-cylindrical model, radial current, magneto-hydrodynamics, thermodynamic modeling, magnetic field structure
Paraules Clau— cordes de flux magnètic, ejeccions de massa coronal, model el·líptic-cilíndric, corrent radial, magnetohidrodinàmica, modelització termodinàmica, estructura del camp magnètic
Palabras Clave— cuerdas de flujo magnético, eyecciones de masa coronal, modelo elíptico-cilíndrico, corriente radial, magnetohidrodinámica, modelado termodinámico, estructura del campo magnético

Acknowledgements

I would like to thank Dr.Teresa Nieves-Chinchilla and Dr.Laura Balmaceda for welcoming me and making this research stay possible, for their invaluable guidance, and for the many hours of discussion and mentorship in their offices. I am also grateful to the LASOSS team (Large-Scale Solar Structures coming from the Sun) for making me feel at home and for the feedback shared during our Thursday lunches.

I am especially thankful to NASA (National Aeronautics and Space Administration) for making a huge dream come true, and to George Mason University for hosting me as part of their academic exchange program and providing continuous support during my stay in the United States of America.

I am also grateful to CFIS program for its student mobility initiative, and to its current director Juan José Rue Perna and mobility deputy director Blas Echebarria for their efficient coordination and the personal care shown throughout the process. I would also like to thank former CFIS director Miguel Ángel Barja for his guidance, and Antoni Pascual for his patience during moments of uncertainty prior to the start of the program.

My deepest gratitude goes as well to *Fundació Privada Mir-Puig*, CFIS partners, and all donors of the crowdfunding program for making this experience financially possible and for believing in the vision of CFIS from the very beginning.

I would also like to thank UPC and its academic community for everything I have learned throughout these years, for their patience, motivation, and guidance and for empowering me to pursue every challenge I have set for myself.

Last but not least, I would like to thank my roommate Martí Massó for his support since the beginning of my degree and for surrounding some of the most important decision-making, ideas, and projects.

This thesis is dedicated to my family and friends, who have shown their unconditional support throughout this journey and to all those driven by the curiosity to better understand the world we live in.

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Introduction

Heliospheric Flux Ropes continue to be an active and constantly evolving area of heliophysics, representing together with CMEs some of the most complex magnetic structures in the solar wind. Despite decades of research, their internal configuration, origin and interaction with the solar environment continue to raise fundamental questions. Among the scientists who have contributed most to this field is Dr. Teresa Nieves-Chinchilla, whose extensive work on the observational and theoretical characterization of these structures has been key to advancing the understanding of interplanetary magnetic clouds and Flux Ropes.

Various models to study Flux Ropes have been developed over time, from the classical Lundquist solution Lundquist (1950), to circular cylindrical (CC) model formulations Nieves-Chinchilla et al. (2018, 2016) or more complex geometries such as Weiss et al. (2024) with the aim of capturing the fundamental physical mechanisms that govern these structures. However, there is always a trade-off between the degrees of freedom allowed in the model, the quality of in situ observations and the depth of physical interpretation that such models can provide.

This thesis aims to continue and extend the elliptic model proposed in Nieves-Chinchilla et al. (2018), generalizing the theoretical framework to explore whether relaxing certain constraints can provide a more complete description of magnetic phenomena that, to date, have not been explained by existing models. In particular, aspects such as discontinuities in the magnetic field and interactions between Flux Ropes and other external magnetic structures, whose existence has been observed but not fully described by conventional analytical approaches, are addressed.

This work is developed under a rigorous approach, imposing physical and mathematical consistency in every part of the development, and adding only one extra degree of freedom, which allows to keep the simplicity of the model without giving up its ability to be used. In addition, a natural question arises: is it possible to obtain new constraints by applying the complete set of magnetohydrodynamics equations to the CC model? For this purpose, the resistive and conductive cases of the Flux Rope are solved analytically, and synthetic trajectories are generated to obtain Flux Rope signatures.

The results obtained show a remarkable agreement with in situ observations, which reinforces the physical validity of the approach and provides a new perspective to interpret the dynamic behavior of heliospheric magnetic structures.

1 . Space Phenomena in the Solar Environment

1.1 The Sun: Our Closest Star

In addition to being the star closest to Earth, the Sun is the main body in the solar system because of its impact on interplanetary media and space weather. Studying the Sun is essential for understanding stellar physics and the behavior of cosmic plasmas more broadly. Numerous definitions of the Sun can be found in the literature Noyes (1982); Stix (2002); Lang (2009), but for the purposes of this work, it may be defined as a typical main sequence star of spectral type G2 V, with an absolute magnitude of approximately 4.8. From a physical point of view, solar activity is generally classified into two regimes: the **quiet Sun**, which can be approximated as a spherically symmetric ball of plasma, and the **active Sun**, characterized by asymmetries and the emergence of magnetically driven features such as sunspots, prominences, and coronal mass ejections (CMEs from now on). These dynamic structures play a crucial role in shaping the space environment and are of central importance in heliophysics research. Some of the most relevant physical parameters of the Sun are summarized in Table 1.1.

Property	**Value**
Age	4.6×10^9 yr
Mass ($M_\odot$)	1.99×10^{30} kg
Radius ($R_\odot$)	695.5 Mm $= 6.955 \times 10^8$ m
Mean density	1.4×10^3 kg m^{-3}
Mean distance from Earth (1 AU)	1.496×10^{11} m $= 215\, R_\odot$
Surface gravity ($g_\odot$)	274 m s^{-2}
Escape velocity (at surface)	618 km s^{-1}
Luminosity ($L_\odot$)	3.86×10^{26} W $= 3.86 \times 10^{33}$ erg s^{-1}
Effective temperature	5785 K
Rotation period (equatorial, synodic)	26.24 days
Mass-loss rate	10^9 kg s^{-1}

Table 1.1: Fundamental Physical Properties of the Sun extracted from Piel (2010)

1.2 Internal Structure of the Sun

The internal structure of the Sun is composed of a number of different concentric layers that each follow distinct physical processes. At the core up to a depth of approximately $0.25\,R_\odot$, there is nuclear fusion under extreme conditions of temperature ($T \sim 1.5\times10^7$ K) and density ($\rho \sim 1.6\times10^5$ kg m^{-3}), which produces the energy that powers the Sun. The proton-proton chain reaction is the one responsible for converting hydrogen into helium and giving the Sun the energy that needs. This reaction is characterized by:

$$4\,{}^1\mathrm{H} \rightarrow {}^4\mathrm{He} + 2e^+ + 2\nu_e + 26.7\,\mathrm{MeV}, \tag{1.1}$$

where ^{1}H represents a proton, ^{4}He a helium nucleus, e^+ a positron and ν_e an electron neutrino. This energy is transported outward through the radiative zone ($0.25\,R_\odot$ to $0.7\,R_\odot$) via radiative diffusion, where photons undergo countless scatterings over thousands of years. Beyond this lies the convection zone ($0.7\,R_\odot$ to the photosphere), where the opacity increases and energy is carried outward by turbulent convective. Above the photosphere are the outer atmospheric layers: the chromosphere, the transition region, and the corona; the latter being a hot diffuse plasma envelope extending millions of kilometers into space.

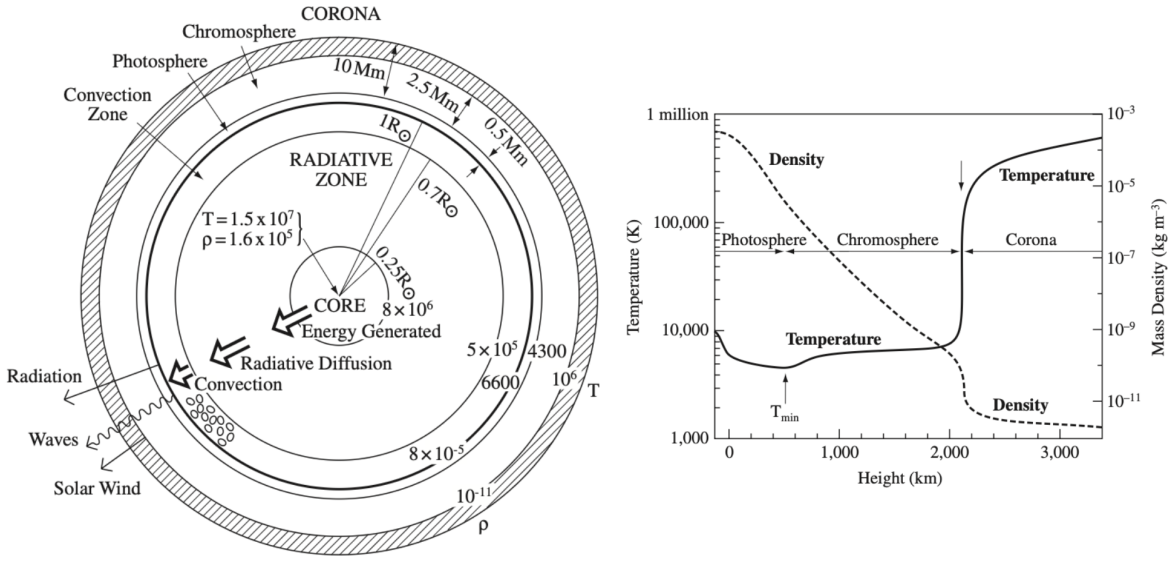


Figure 1.1: Internal structure and atmospheric properties of the Sun extracted from Piel (2010)

1.2.1 Coronal Mass Ejections (CMEs)

Coronal Mass Ejections (CMEs) are large-scale eruptions of solar plasma and confined magnetic fields expelled from the Sun's corona into interplanetary media. A single CME may release up to 10^{13} kg of material, consisting mainly of electrons, protons and alpha particles. These events hugely impact space weather and can have the potential to affect planetary magnetospheres, communication systems and satellites. CMEs originate in the solar corona above magnetically active regions. They are typically triggered by magnetic

reconnection, a physical process where stressed magnetic field lines break and reconnect; releasing a huge amount of energy. This results in the **sudden destabilization of the coronal structure and the ejection of plasma into the heliosphere**. The physical conditions of the corona, extreme low density and high temperature, makes the buildup and release of magnetic energy required to launch CMEs an easier task than other parts of the Sun.

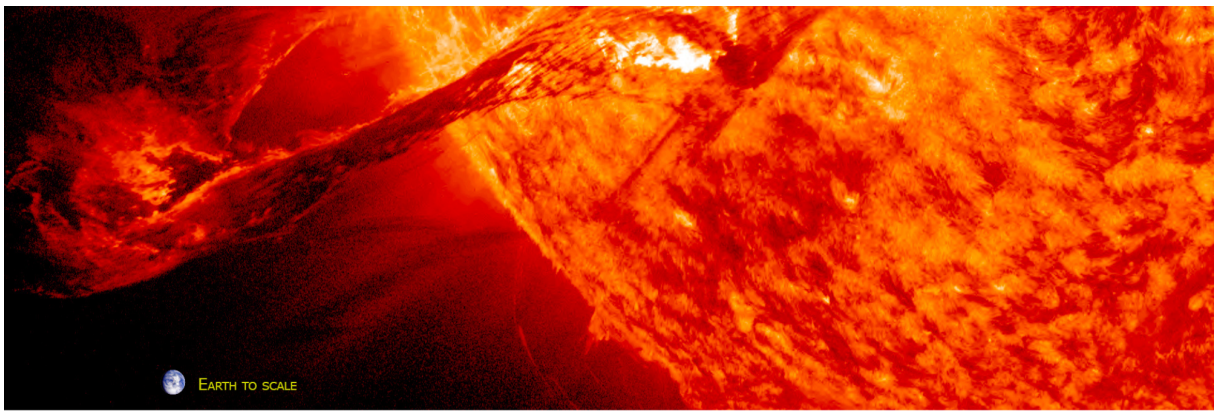


Figure 1.2: A Coronal Mass Ejection (CME) emerging from the Sun compared to the size of the Earth

1.3 Flux Ropes

Flux Ropes are magnetically enclosed structures characterized by magnetic field lines that are twisted around a central axis. They are of particular importance because it is believed that they frequently serve as the magnetic core of CMEs, playing a very important role in their propagation/dynamics and interaction with the heliospheric environment.

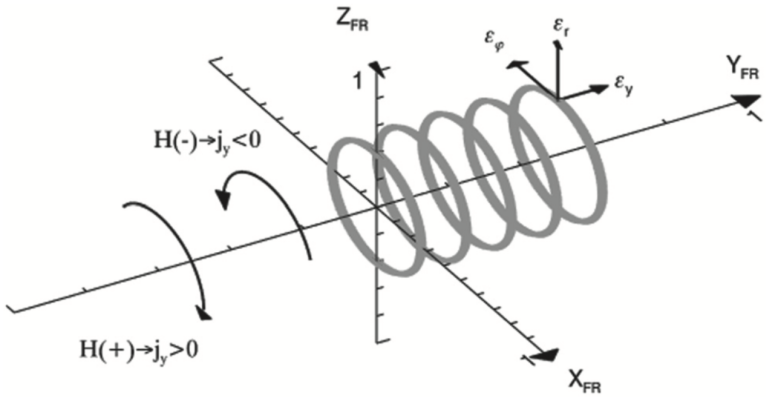


Figure 1.3: Magnetic field structure of a Flux Rope in a cylindrical circular geometry, adapted from Nieves-Chinchilla et al. (2016). The helicity sign convention and the associated basis for the mathematical description are also indicated

1.3.1 Characterization of Heliospheric CMEs and Flux Ropes

A huge amount of space missions such as **Solar Orbiter**, **WIND** and **Parker Solar Probe** provide in-situ and remote-sensing measurements that are essential for the iden-

tification and analysis of Flux Ropes. These instruments capture parameters such as the magnetic field magnitude, plasma density, temperature and thermal speed of the particles; allowing for a detailed reconstruction of the magnetic topology and dynamic properties of the Flux Rope. These observations enable the characterization of Flux Rope parameters such as axial and poloidal magnetic fluxes, helicity, and orientation with respect to the solar wind flow. One way to characterize ICME signatures at approximately one astronomical unit is provided by Zurbuchen and Richardson (2006), who compiled various criteria developed by multiple researchers; some of them are summarized below:

Signature	**Description**	**Reference**
B Rotation	$\gg 30°$, smooth	Klein and Burlaga (1982)
B Enhancement	> 10nT	Hirshberg and Colburn (1969)
B Variance decrease		Pudovkin et al. (1979) and Klein and Burlaga (1982)
Discontinuity of B at the boundaries		Pudovkin et al. (1979) and Klein and Burlaga (1982)
Field line draping around ICME		McComas et al. (1989) and Gosling and McComas (1987)
Plasma beta	$\beta = \frac{2\mu_0 \sum nK_BT}{\lvert\mathbf{B}\rvert^2} \ll 1$	Klein and Burlaga (1982) and Lepping et al. (1990)
Declining velocity profile/ expansion	Monotonic decrease	Klein and Burlaga (1982) and Russell and Shinde (2003)
Extreme density decrease	$\leq 1\text{cm}^3$	Richardson et al. (2000)
Proton temperature decrease	$\mathrm{T}_p \leq 0.5\mathrm{T}_{\mathrm{exp}}$	Richardson and Cane (1995) and Gosling et al. (1973)
Electron temperature upper limit	$\mathrm{T_e} < 6 \times 10^4\mathrm{K}$	Montgomery et al. (1974)
Electron and proton temperature inequality	$\mathrm{T_e} \gg \mathrm{T_p}$	Sittler and Burlaga (1998) and Richardson et al. (1997)

Table 1.2: In situ signatures of interplanetary CMEs applied at 1AU in the magnetic field and plasma dynamics extracted from Zurbuchen and Richardson (2006)

Using these criteria, experts are able to identify the characteristic features of coronal mass ejections (CMEs) and the possible presence of Flux Ropes by analyzing the data collected by various space missions. These, provide detailed measurements of magnetic field configurations, plasma composition, and other key in-situ parameters.

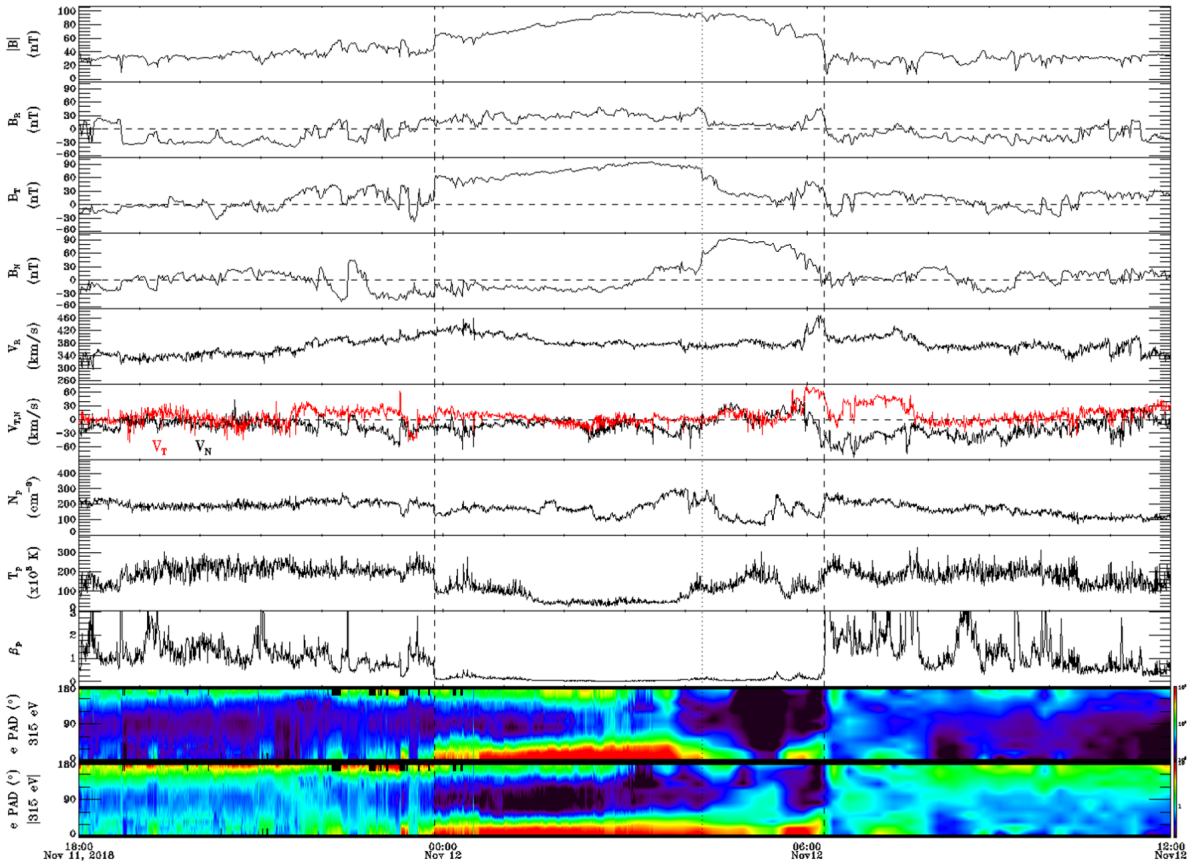


Figure 1.4: Example of in-situ solar wind data illustrating typical signatures of CMEs and embedded Flux Ropes, adapted from Nieves-Chinchilla et al. (2023b). The first dashed line shows the shock front, the dotted line shows the beginning of the Flux Rope and the second dashed line at 6:00 UT shows again the typical solar wind signature and that the magnetic object has been overcome.

1.3.2 Relevant Space Missions

Several space missions have provided essential data that give context and relevance to this work. These missions, launched in different years with varying scientific objectives, have been key in characterizing magnetic field measurements and validating physical models like the one shown in this document. In the following table, some of the missions are presented:

Mission	Launch Year	Purpose and Key Contributions
WIND	1994	Measures electromagnetic waves, particles and magnetic fields in the solar wind. Operational until approx. 2074
SOHO	1995	ESA/NASA mission focuses on solar flares, CMEs and solar wind from the solar atmosphere
MESSENGER	2004	Mercury mission; also contributed data on solar wind and magnetic fields during orbit

Continued on next page

Mission	Launch Year	Purpose and Key Contributions
STEREO	2006	Twin spacecraft provide 3D imaging of solar activity; STEREO A still active, B ceased in 2018
SDO	2010	Geosynchronous orbit; monitors solar flares, sunspots, and magnetic field evolution continuously in high resolution
Parker Solar Probe (PSP)	2018	Flies into solar corona; measures magnetic fields, plasma, and particles up to 6.26 million km from the Sun
Solar Orbiter	2020	ESA/NASA mission; combines in-situ and remote sensing, imaging solar poles and tracing solar wind sources

Table 1.3: Brief Summary of Space Missions Contributing to Flux Rope and CME Characterization

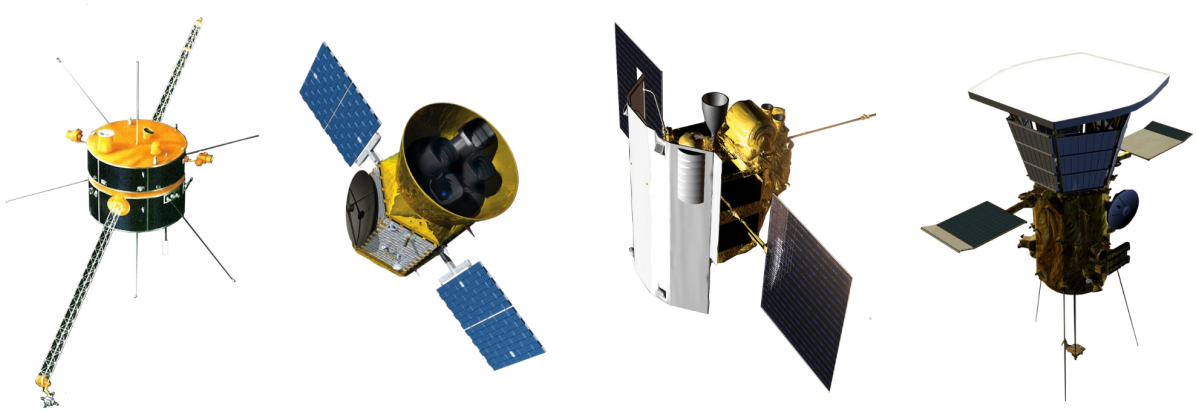


Figure 1.5: Illustration of key space missions contributing to heliospheric science. From left to right: WIND (1994), MESSENGER (2004), SOHO (1995), and Parker Solar Probe (2018)

2 . Description of Plasma

The way plasma is understood and modeled is presented in this section, where the fundamental concepts that will lead to the formal development of the theory throughout this document are introduced.

2.1 Definition of Plasma

Since the majority of ordinary matter in the universe exists as plasma, it is worthwhile to examine its nature. The simple definition of Goldston (2020) is adopted: a plasma is an ionized gas. When a solid is heated sufficiently that the thermal motion of its atoms breaks apart the crystal lattice and atoms vaporize faster than they can recombine, a charged gas forms. In other words, a plasma is essentially a "soup" of free charged particles. In this sense, a plasma must satisfy both the equations of electromagnetism, because it contains free charges and currents, and the equations of fluid mechanics, because it exhibits collective fluid-like behavior. This creates a need for a unified magnetohydrodynamic description of its dynamics

2.2 Electromagnetism Equations

The equations of electromagnetism govern the behavior of two fundamental fields in physics: the electric field $\mathbf{E}$ and the magnetic field $\mathbf{B}$. Each field is described by a pair of equations which together, with appropriate boundary conditions, uniquely determine the field configuration. No field can exist without simultaneously satisfying these equations, and among the infinitely many mathematical solutions that might satisfy a physical situation, the boundary conditions select the one that actually occurs. Notice that each pair involves both a divergence and a curl, because the divergence and curl of a vector field are independent pieces of information, both are needed to fully characterize the field. The presentation of the electromagnetic field equations are followed from a great classic text such as Pollack and Stump (2002).Maxwell's equations, in differential form are read as:

Gauss's Law (Electric):

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \tag{2.1}$$

Gauss's Law (Magnetism):

$$\nabla \cdot \mathbf{B} = 0 \tag{2.2}$$

Faraday's Law of Induction:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{2.3}$$

Ampère–Maxwell Law:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{2.4}$$

Applying the divergence theorem to the first two equations (2.1) and (2.2) gives their integral forms:

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{in}}}{\varepsilon_0} \tag{2.5}$$

$$\oint_S \mathbf{B} \cdot d\mathbf{A} = 0 \tag{2.6}$$

These expressions imply that the net flux through any closed surface is not arbitrary: for the electric field it is fixed by the total enclosed charge (up to the factor $1/\varepsilon_0$), whereas for the magnetic field it completely vanishes. In other words, electric monopoles exist as fundamental sources of $\mathbf{E}$, but there are no magnetic monopoles producing $\mathbf{B}$. Applying Stokes' theorem to the curl equations (2.3) and (2.4) yields their integral forms:

$$\oint_C \mathbf{E} \cdot d\boldsymbol{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{A} \tag{2.7}$$

$$\oint_C \mathbf{B} \cdot d\boldsymbol{l} = \mu_0 \int_S \mathbf{j} \cdot d\mathbf{A} + \mu_0 \varepsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{A} \tag{2.8}$$

These expressions relate the circulation of the vector fields around closed loops and are very useful for intuitively deducing information about the directions of currents or, more generally, the existence regimes of these fields in complex media; they are primarily used in the differential form.

2.3 Fluid Mechanics Equations

The fluid-mechanics equations that will be employed consist of mass conservation and linear momentum conservation. Other extensions, such as an equation of state or thermodynamic evolution, often rely on adiabatic, polytropic, or similar assumptions, which are chosen not to be imposed in order to keep the development as general as possible. Although angular momentum is also conserved, its conservation is implicit here since no

external torques act on the system. In the momentum equation, viscous stresses are neglected (justified by the high electrical conductivity and large magnetic Reynolds number in a flux rope). Gravitational forces are also non dominant in this context. The balance then reduces to the pressure gradient and the Lorentz force, fluid dyamic equations can be obtained from standard references such as Landau and Lifshitz (1959) or Bergadà Granyó (2012). These equations can be expanded, and the temporal, advective/convective, pressure gradient and Lorentz force terms can be identified:

Continuity Equation (Mass Conservation):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho\, \mathbf{v}) = 0$$
$$\implies \underbrace{\frac{\partial \rho}{\partial t}}_{\substack{\text{temporal}\\ \text{term}}} + \underbrace{\mathbf{v} \cdot \nabla \rho}_{\substack{\text{advective}\\ \text{term}}} + \underbrace{\rho \nabla \cdot \mathbf{v}}_{\substack{\text{dilatation}\\ \text{term}}} = 0 \tag{2.9}$$

Momentum Conservation Equation:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right) = -\nabla p + \mathbf{j} \times \mathbf{B}$$
$$\implies \underbrace{\rho \frac{\partial \mathbf{v}}{\partial t}}_{\substack{\text{temporal}\\ \text{term}}} + \underbrace{\rho\, (\mathbf{v} \cdot \nabla)\mathbf{v}}_{\substack{\text{convective}\\ \text{term}}} = \underbrace{-\nabla p}_{\substack{\text{pressure}\\ \text{gradient}}} + \underbrace{\mathbf{J} \times \mathbf{B}}_{\substack{\text{Lorentz}\\ \text{force}}} \tag{2.10}$$

As an equation of state, with thermal velocity distribution assumed to be Maxwellian (as explained below), the total thermal pressure for all particle species is given by

$$p \;=\; \sum_s n_s\, k_B\, T_s \tag{2.11}$$

where the index s runs over each species (electrons, protons, ions), n_s is the number density of species s and T_s its temperature. By combining the electromagnetic equations with the fluid dynamic equations, one obtains the **magnetohydrodynamic (MHD) framework**, one of the many equation systems used in MHD to describe plasma behavior.

2.4 Statistical Framework

A statistical description of the particle ensemble inside the flux rope is required to build the theoretical model. In particular, the classical regime, excluding the high energy tails, is chosen and the velocity distribution is governed by the Maxwell–Boltzmann law. It is given by

$$f_{\mathrm{MB}}(v) \;=\; \left(\frac{m}{2\pi k_B T}\right)^{3/2} \exp\!\left(-\frac{m\,v^2}{2\,k_B T}\right) \tag{2.12}$$

To capture these enhanced high energy tails, the kappa distribution can be used as discussed by Pierrard and Lazar (2010), defined by

$$f_\kappa(v) \;=\; \frac{n}{\theta^3}\,\frac{\Gamma(\kappa+1)}{\Gamma\!\left(\kappa-\frac{1}{2}\right)}\left(\frac{1}{\pi\kappa}\right)^{3/2}\left[1+\frac{v^2}{\kappa\,\theta^2}\right]^{-(\kappa+1)} \tag{2.13}$$

Here, the characteristic speed is defined as $\theta^2 = \frac{2\,k_B T}{m\,(2\kappa-3)}$, where n is the number density and $\kappa > \frac{3}{2}$ controls the slope of the power-law tail. As $\kappa \to \infty$, f_κ recovers the Maxwell–Boltzmann form; for finite (lower) κ, the distribution exhibits a heavier, power law tail, reflecting the enhanced presence of energetic particles. These distributions provide a theoretical justification for using the ideal gas law to describe the thermal motion of particles in the thermodynamic limit.

2.5 Debye Length

One of the plasma properties is that its particles rearrange themselves in the most efficient way to screen any imposed electric charge. The characteristic distance over which quasi-neutrality is restored is called the Debye length, denoted by λ_D. To derive this expression, a slightly modified approach proposed by Piel (2010) is followed. Shielding is not a static process but is governed by the thermal motion of plasma electrons and ions; higher temperature implies greater excitation of the charged particles and different shielding. Using the Boltzmann factor, one obtains a statistical distribution for the particle population subject to the electric potential $E = -\nabla\Phi$ as

$$\frac{n_i}{n_k} = \frac{g_i}{g_k}\exp\!\left(-\frac{W_i - W_k}{k_B T}\right) \xrightarrow[i=k=1]{\text{degeneracy}} n(\mathbf{r}) = n_0\exp\!\left(-\frac{q\Phi}{k_B T}\right) \tag{2.14}$$

For electrons and ions in a non-degenerate plasma

$$n_e(\mathbf{r}) = n_e^0\exp\!\left(\frac{e\Phi}{k_B T}\right) \tag{2.15}$$

$$n_i(\mathbf{r}) = n_i^0\exp\!\left(-\frac{e\Phi}{k_B T}\right) \tag{2.16}$$

Assuming that the **potential energy is small compared to the thermal energy**, $e\Phi \ll k_B T$, the Boltzmann relations are expanded to first order:

$$n_e(\mathbf{r}) \approx n_0\left(1+\frac{e\,\Phi(\mathbf{r})}{k_B T}\right) \tag{2.17}$$

$$n_i(\mathbf{r}) \approx n_0\left(1-\frac{e\,\Phi(\mathbf{r})}{k_B T}\right) \tag{2.18}$$

where $n_e^0 = n_i^0 = n_0$ is set under background quasi-neutrality. Now, a test charge Q is introduced at the origin. Poisson's equation becomes

$$\varepsilon_0 \, \nabla^2 \Phi(\mathbf{r}) = -Q \, \delta(\mathbf{r}) + e\big[n_e(\mathbf{r}) - n_i(\mathbf{r})\big] \tag{2.19}$$

Substituting the linearized densities gives

$$\varepsilon_0 \, \nabla^2 \Phi \approx -Q \, \delta(\mathbf{r}) + e \, n_0 \Big[\big(1 + \tfrac{e\Phi}{k_B T}\big) - \big(1 - \tfrac{e\Phi}{k_B T}\big)\Big] = -Q \, \delta(\mathbf{r}) + \frac{2 \, n_0 e^2}{k_B T} \, \Phi \tag{2.20}$$

Dividing through by ε_0 and defining

$$\frac{1}{\lambda_D^2} \equiv \frac{2 \, n_0 e^2}{\varepsilon_0 \, k_B T}$$

yields for $r > 0$

$$\nabla^2 \Phi(\mathbf{r}) - \frac{1}{\lambda_D^2} \, \Phi(\mathbf{r}) = -\frac{Q}{\varepsilon_0} \, \delta(\mathbf{r}) \tag{2.21}$$

from which the Debye length follows:

$$\boxed{\lambda_D = \sqrt{\frac{\varepsilon_0 \, k_B T}{2 \, n_0 e^2}}} \tag{2.22}$$

Thus, the plasma may exhibit slight deviations from quasi-neutrality on length scales of order $\sim \lambda_D$, but on **scales larger than the Debye length the plasma can be regarded as effectively neutral**.

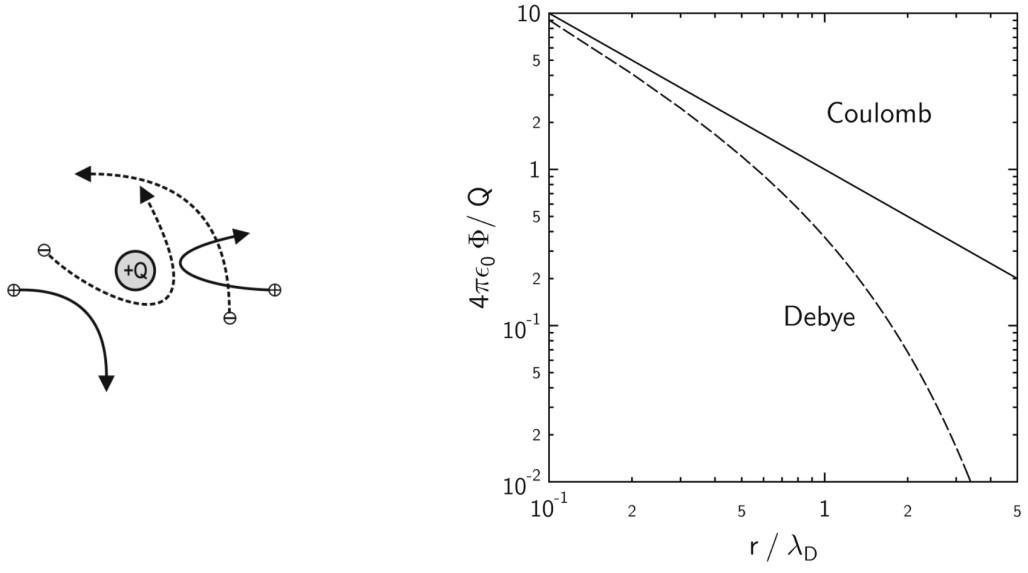


Figure 2.1: For $r > \lambda_D$ the potential decays much faster that a Couloumb potential, any deviation in a cloud of plasma several times longer that λ_D tends to become quasineutral. Images extracted from Piel (2010)

2.5.1 Previous Flux Rope Models

As mentioned in the introduction, various flux rope models have been proposed over the years in an effort to understand the internal configuration of these structures. These models aim to describe physical aspects such as magnetic field lines topology, current distributions, magnetic energy density and the Lorentz Forces. The following table presents a selection of relevant flux rope models along with their original authors, highlighting their main advantages and limitations. This includes the classical force-free Lundquist (1950) model, the Nieves-Chinchilla et al. (2016), Nieves-Chinchilla et al. (2018) and Nieves-Chinchilla et al. (2023a) as well as Weiss et al. (2024).

Model and Author(s)	Geometry	Advantages	Limitations
Lundquist Model Lundquist (1950)	Cylindrical, force-free	Analytical solution; widely used; matches many observations	Perfect **force-free** condition; **oversimplifies** real structures
Circular Cylindrical Model Nieves-Chinchilla et al. (2016)	Cylindrical, non-force-free	Includes **non-force-free** fields; better fits some in-situ data	Limited in describing **complex deformations**
Elliptical Model Nieves-Chinchilla et al. (2018)	Elliptical cross-section, non-force-free	Captures **aspect ratio deformation**; improves accuracy for distorted events	Computationally more intensive
Toroidal Model Nieves-Chinchilla et al. (2023a)	3D torus-like structure	Accounts for **curvature effects**; closer to CME geometry	Mathematically more complex; **harder to fit** observational data
Generalized Geometrical Model Weiss et al. (2024)	Arbitrary geometry	Flexible; adaptable to a variety of observations and orientations	**Difficult to constrain**; relies heavily on initial **assumptions** and data quality; **large number of free functions**

Table 2.1: Summary of flux rope models, their geometries, and main features.

3 . Fundamentals of Differential Geometry

This chapter provides the mathematical framework upon which the new Flux Rope model, featuring cylindrical geometry and non vanishing radial current is constructed. In addition to establishing the essential foundations and presenting the generalized expressions for linear differential operators in curvilinear spaces, it also serves as a valuable reminder of how lucky we are by living in a world of orthogonal coordinates.

3.1 Non-Orthogonal Coordinates: Contravariant and Covariant Components

Understanding the concept of covariance and contravariance can be rather subtle. At its core, it arises from the realization that a reference system does not need to be orthogonal, that is, the basis vectors generating the coordinate system are not necessarily orthogonal to each other. Let us consider a vector subspace generated in $\mathbb{R}^3$ by the basis $\{e_1, e_2, e_3\}$. In this setting, the "classical" way to navigate through this space is by using the generating functions associated with these basis vectors. Within such a system, the interpretation of motion or displacement is particularly intuitive, as the coefficients accompanying each basis vector correspond directly to displacements along their respective directions. These components have so far been naturally interpreted in this framework and will henceforth be referred to as **covariant vector components**.

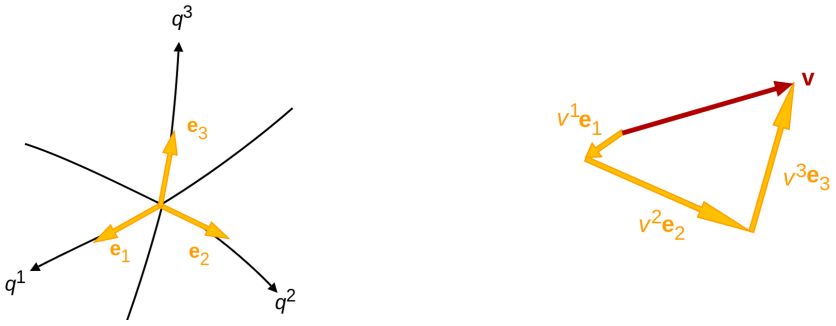


Figure 3.1: Geometric interpretation of covariant vector components in a non-orthogonal coordinate system. The components are obtained as projections of the vector onto the basis directions, corresponding to displacements along each generating vector of the coordinate system.

Let us now imagine that these generating vectors are in fact the tangent vectors to a surface defined by all local possible combinations of the coordinates. In this setting, the space is no longer a single flat vector subspace but rather a collection of infinitely many tangent planes, one at each point of the surface, through which one can reach any point in space using the associated tangent vector; this will be called **contravariant vector component**.

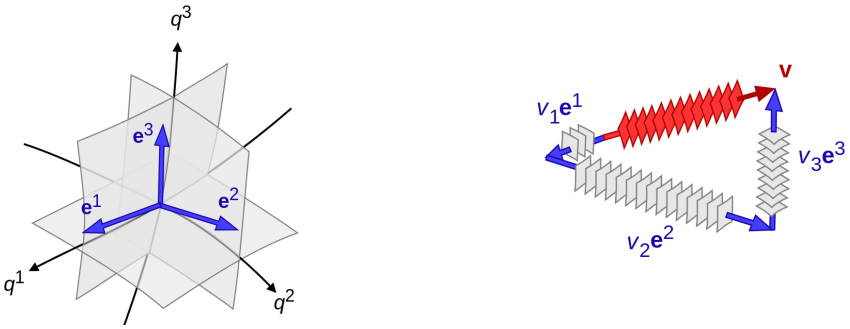


Figure 3.2: Geometric representation of contravariant vector components in a non-orthogonal coordinate system. The components are expressed as scalar weights along the basis vectors, indicating how the vector is constructed through linear combinations of the tangent directions of the coordinate grid.

This shift in perspective is crucial: when the coordinate system is curved or non-orthogonal, expressing a vector in terms of the original basis no longer captures all of its geometric characteristics. Instead, a dual basis is introduced, which allows to express vectors in a way that naturally transform under coordinate changes.

3.1.1 Invariance of Physical Quantities

Let $\mathbf{V}$ be a vectorial function representing a physical quantity, regardless of whether it is expressed in terms of a contravariant or covariant basis the physical magnitude itself must remain invariant. That is, while the components of $\mathbf{V}$ may change depending on the chosen coordinate system or representation, the underlying physical entity does not.

$$\mathbf{V} = \sum_i V^i \mathbf{e}_i = \sum_i V_i \mathbf{e}^i \tag{3.1}$$

From this point onward, the Einstein summation convention will be adopted. According to this convention, any repeated index that appears once as a subscript and once as a superscript in a single term implies summation over that index, without the need to explicitly write the summation symbol. This notation significantly simplifies expressions involving tensors and is widely used in differential geometry and theoretical physics. So, last expression can be writen as:

$$\mathbf{V} = V^i \mathbf{e}_i = V_i \mathbf{e}^i \tag{3.2}$$

3.1.2 Construction of the Covariant and Contravariant Basis

Let us consider a curvilinear coordinate system in $\mathbb{R}^n$ defined by a transformation $\mathbf{R}(u^1, ..., u^i, ..., u^n)$. The tangent vector associated with the coordinate u^i is then given by the partial derivative $\partial\mathbf{R}/\partial u^i$. Consequently, the tangent basis, also referred to as the covariant basis vectors, is defined as:

$$\mathbf{e}_i = \frac{\partial \mathbf{R}}{\partial u^i} \tag{3.3}$$

This representation is highly intuitive, as it corresponds to the classical reference frame: one in which movement occurs tangentially along coordinate curves and the coefficients in a vector expression indicate how many times the vector advances in each of those directions. To obtain the vectors that navigate normally to the surfaces, referred to as contravariant basis vectors, the concept of the gradient is used. The gradient represents the direction of maximal increase of a differentiable scalar field; in the context of curvilinear coordinates, it provides a natural way to define the dual basis:

$$\mathbf{e}^i = \nabla u^i \tag{3.4}$$

This means that the contravariant basis vectors are perpendicular to the surfaces of constant u^i, and they form the dual basis to the covariant vectors $\mathbf{e}_i$ defined previously.

3.1.3 Completeness Relation

Applying the chain rule, the differential of a curvilinear coordinate u^i can be written as:

$$du^i = \frac{\partial u^i}{\partial \mathbf{R}} \cdot d\mathbf{R} = \nabla u^i \cdot d\mathbf{R} = \mathbf{e}^i \cdot d\mathbf{R} \tag{3.5}$$

where $\mathbf{e}^i = \nabla u^i$ denotes the contravariant basis vector associated with the coordinate u^i. The differential displacement in the manifold, $d\mathbf{R}$, can be expressed in terms of the covariant basis $\{\mathbf{e}_j\}$ as:

$$d\mathbf{R} = \frac{\partial \mathbf{R}}{\partial u^j}\, du^j = \mathbf{e}_j\, du^j \tag{3.6}$$

Substituting (3.6) into (3.5), we obtain:

$$du^i = \mathbf{e}^i \cdot d\mathbf{R} = \mathbf{e}^i \cdot (\mathbf{e}_j\, du^j) = (\mathbf{e}^i \cdot \mathbf{e}_j)\, du^j \tag{3.7}$$

Since this equation must hold for arbitrary differentials du^j, we conclude:

$$\delta^i_j = \mathbf{e}^i \cdot \mathbf{e}_j \tag{3.8}$$

which expresses the duality (orthonormality) condition between the contravariant and covariant bases.

This leads to the completeness relation, which in tensor notation reads:

$$\boxed{\sum_i \mathbf{e}_i \otimes \mathbf{e}^i = \mathbb{I}} \tag{3.9}$$

stating that the composition of covariant and contravariant bases reconstructs the identity operator in the tangent space.

3.2 The Metric Tensor

Since non-orthogonal space will be used, it is essential to introduce a mathematical object capable of capturing the distortion of the space. This object, known as the metric tensor, encodes the scalar products between the covariant or contravariant basis vectors in an organized way. The metric plays a central role in performing scalar operations such as inner products and in defining distances within the space. The (i, j)-th component of the covariant metric tensor is given by the scalar product between the corresponding covariant basis vectors:

$$g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j = \frac{\partial \mathbf{R}}{\partial u^i} \cdot \frac{\partial \mathbf{R}}{\partial u^j} \tag{3.10}$$

This expression quantifies the angle and relative length between the directions associated with the curvilinear coordinates u^i and u^j, and it fully characterizes the local geometry of the manifold. In $\mathbb{R}^3$, the covariant metric tensor takes the form of a symmetric 3×3 matrix, whose components are given by the scalar products of the covariant basis vectors:

$$g_{ij} = \begin{pmatrix} \mathbf{e}_1 \cdot \mathbf{e}_1 & \mathbf{e}_1 \cdot \mathbf{e}_2 & \mathbf{e}_1 \cdot \mathbf{e}_3 \\ \mathbf{e}_2 \cdot \mathbf{e}_1 & \mathbf{e}_2 \cdot \mathbf{e}_2 & \mathbf{e}_2 \cdot \mathbf{e}_3 \\ \mathbf{e}_3 \cdot \mathbf{e}_1 & \mathbf{e}_3 \cdot \mathbf{e}_2 & \mathbf{e}_3 \cdot \mathbf{e}_3 \end{pmatrix} \tag{3.11}$$

Since the dot product is symmetric, the metric satisfies $g_{ij} = g_{ji}$. In orthonormal coordinate systems this matrix reduces to the identity, however, in curvilinear or non-orthogonal coordinates the off-diagonal terms generally differ from zero and encode the geometric distortion of the basis. Similarly, the contravariant metric tensor is defined as described in

$$g^{ij} = \mathbf{e}^i \cdot \mathbf{e}^j = \nabla u^i \cdot \nabla u^j \tag{3.12}$$

3.2.1 Transformation of Components

The relation between the covariant and contravariant basis vectors can be naturally obtained by inserting the dual basis and using the identity $\mathbf{e}^i \cdot \mathbf{e}_j = \delta^i_j$. For instance:

$$\mathbf{e}^i = \mathbf{e}^i(\mathbf{e}^j \cdot \mathbf{e}_j) = g^{ij}\, \mathbf{e}_j \tag{3.13}$$

where g^{ij} are the components of the contravariant metric tensor. In a completely analogous way, the covariant basis vectors can be written as:

$$\mathbf{e}_i = \mathbf{e}_i(\mathbf{e}_j \cdot \mathbf{e}^j) = g_{ij}\,\mathbf{e}^j \tag{3.14}$$

Analogously to the transformation of the basis vectors, the components of a vector can be expressed in either covariant or contravariant form using the metric tensor. For the covariant components:

$$V_i = \mathbf{V} \cdot \mathbf{e}_i = V^j\,\mathbf{e}_j \cdot \mathbf{e}_i = V^j g_{ji} \tag{3.15}$$

Similarly, the contravariant components are given by:

$$V^i = \mathbf{V} \cdot \mathbf{e}^i = V_j\,\mathbf{e}^j \cdot \mathbf{e}^i = V_j g^{ji} \tag{3.16}$$

The **metric tensor serves as the bridge for transforming vector components** between covariant and contravariant forms.

3.2.2 Inverse of the Metric Tensor

To find the inverse of the metric tensor, the identity given in equation (3.14) is used, which can be rewritten as:

$$\mathbf{e}_i \cdot \mathbf{e}^k = g_{ij}\mathbf{e}^j \cdot \mathbf{e}^k \tag{3.17}$$

By the definition of the Kronecker delta in this context, the following identity is obtained:

$$\delta_i^k = g_{ij}g^{jk} \tag{3.18}$$

This relation is particularly useful, as it provides a direct matrix formulation for computing the inverse metric tensor. Specifically, it confirms that the matrices g_{ij} and g^{ij} are mutual inverses, that is:

$$g^{ij} = (g_{ij})^{-1} \tag{3.19}$$

In practical terms, once the components of the covariant metric tensor g_{ij} are known, its inverse g^{ij} can be obtained by matrix inversion techniques. Furthermore, the determinant of the metric, denoted $g = \det g_{ij}$, and its inverse $\det g^{ij} = \dfrac{1}{\det g_{ij}}$, will play a crucial role in the formulation of volume elements and in the definition of linear differential operators and the measurement of distances and volumes in curved spaces.

3.3 The Jacobian and the Metric Tensor

Before introducing the concept of measurements in a curvilinear space, it is essential to define a fundamental mathematical object in coordinate transformations: the Jacobian. Let the following coordinate transformation be considered:

$$\mathbf{R}(u_1, u_2, u_3) : \begin{cases} x = x(u_1, u_2, u_3) \\ y = y(u_1, u_2, u_3) \\ z = z(u_1, u_2, u_3) \end{cases} \tag{3.20}$$

The Jacobian, by definition, is the determinant of the matrix formed by the partial derivatives of the Cartesian coordinates with respect to the curvilinear coordinates:

$$J(u^1, u^2, u^3) = \begin{vmatrix} \frac{\partial x}{\partial u^1} & \frac{\partial x}{\partial u^2} & \frac{\partial x}{\partial u^3} \\ \frac{\partial y}{\partial u^1} & \frac{\partial y}{\partial u^2} & \frac{\partial y}{\partial u^3} \\ \frac{\partial z}{\partial u^1} & \frac{\partial z}{\partial u^2} & \frac{\partial z}{\partial u^3} \end{vmatrix} \tag{3.21}$$

This scalar quantity essentially measures the distortion between vector spaces. In other words, it captures the possible deformation that a differential element may undergo. In the case of $\mathbb{R}^3$, this refers to the deformation of a differential volume element. With this interpretation the Jacobian can be expressed as:

$$J = \frac{\partial \mathbf{R}}{\partial u^1} \cdot \left(\frac{\partial \mathbf{R}}{\partial u^2} \times \frac{\partial \mathbf{R}}{\partial u^3} \right) \tag{3.22}$$

This expression represents the volume variation of a differential element. If the determinant of the metric tensor is taken, it is obtained that g is given by:

$$g = \det[g_{ij}] = \left(\det \left[\frac{\partial \mathbf{R}}{\partial u^i} \right] \right) \left(\det \left[\frac{\partial \mathbf{R}}{\partial u^j} \right] \right) \tag{3.23}$$

This product can be explicitly written as:

$$g = \begin{vmatrix} \frac{\partial x}{\partial u^1} & \frac{\partial x}{\partial u^2} & \frac{\partial x}{\partial u^3} \\ \frac{\partial y}{\partial u^1} & \frac{\partial y}{\partial u^2} & \frac{\partial y}{\partial u^3} \\ \frac{\partial z}{\partial u^1} & \frac{\partial z}{\partial u^2} & \frac{\partial z}{\partial u^3} \end{vmatrix} \cdot \begin{vmatrix} \frac{\partial x}{\partial u^1} & \frac{\partial y}{\partial u^1} & \frac{\partial z}{\partial u^1} \\ \frac{\partial x}{\partial u^2} & \frac{\partial y}{\partial u^2} & \frac{\partial z}{\partial u^2} \\ \frac{\partial x}{\partial u^3} & \frac{\partial y}{\partial u^3} & \frac{\partial z}{\partial u^3} \end{vmatrix} \tag{3.24}$$

This expression is equivalent to the squared magnitude of the triple scalar product between the basis vectors of the curvilinear system:

$$\boxed{g = \left(\frac{\partial \mathbf{R}}{\partial u^1} \cdot \left(\frac{\partial \mathbf{R}}{\partial u^2} \times \frac{\partial \mathbf{R}}{\partial u^3} \right) \right)^2 = J^2} \tag{3.25}$$

Thus, the determinant of the metric tensor, g, is the square of the Jacobian, J and will be a key result moving forward, as it provides a direct connection between the metric determinant, the Jacobian, and the foundation for defining measures in curvilinear coordinates as was elegantly discussed in D'haeseleer (1991).

3.4 Dot-products and cross-products in curvilinear coordinates

When the metric tensor is not diagonal, cross terms will be encountered in both scalar and vector products, terms that were previously absent due to the orthogonality of the basis vectors. In this chapter, the generalization of these concepts is introduced, which play a fundamental role when measuring geometric quantities in space.

3.4.1 Levi-Civita Symbols

The Levi-Civita symbols is defined in $\mathbb{R}^3$ as:

$$\begin{cases} \varepsilon_{ijk} = \dfrac{\mathbf{e}_i \cdot (\mathbf{e}_j \times \mathbf{e}_k)}{\sqrt{g}} \\ \varepsilon^{ijk} = \mathbf{e}^i \cdot (\mathbf{e}^j \times \mathbf{e}^k)\sqrt{g} \end{cases} \tag{3.26}$$

These objects take the following values:

$$\varepsilon_{ijk},\, \varepsilon^{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3) \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3) \\ 0 & \text{if any two indices are equal} \end{cases} \tag{3.27}$$

3.4.2 Dot product

In Euclidean space, the dot product measures the projection of one vector onto another and is widely used to quantify concepts such as orthogonality. While this definition is straightforward in Cartesian coordinates, a more general formulation is required when working within curvilinear coordinate systems. By expanding the dot product in terms of the contravariant components of the vectors and the metric tensor, one obtains:

$$\mathbf{A} \cdot \mathbf{B} = A^i \mathbf{e}_i \cdot B^j \mathbf{e}_j = A^i B^j (\mathbf{e}_i \cdot \mathbf{e}_j) = A^i B^j g_{ij} \tag{3.28}$$

Alternatively, one could expand the covariant components of the vectors to obtain:

$$\mathbf{A} \cdot \mathbf{B} = A_i B_j g^{ij} \tag{3.29}$$

These will be the two most common expressions used throughout this document to compute scalar products. However, in cases where one wishes to avoid the explicit use of the metric tensor in the scalar product, the following identity can be employed:

$$\boxed{\mathbf{A} \cdot \mathbf{B} = \left(A_i g^{ij}\right) B_j = A^j B_j} \tag{3.30}$$

3.4.3 Cross product

Given two vectors, the cross product produces a third vector that is orthogonal to the plane defined by the original vectors, with a magnitude proportional to the area of the parallelogram they span and a direction determined by the right-hand rule. When working in curvilinear coordinate systems, especially in the context of differential geometry, a more general formulation is required. The derivation of the cross product follows almost directly from the definition of the Levi-Civita symbol as presented in equation (3.26). The cross product is defined as:

$$\mathbf{A} \times \mathbf{B} = A_j \mathbf{e}^j \times B_k \mathbf{e}^k = A_j B_k (\mathbf{e}^j \times \mathbf{e}^k) \tag{3.31}$$

According to the definition of the Levi-Civita symbol in (3.26), this can be expressed as:

$$\frac{\varepsilon^{ijk}}{\sqrt{g}} = \mathbf{e}^i \cdot (\mathbf{e}^j \times \mathbf{e}^k) \tag{3.32}$$

Using the index raising and lowering properties from (3.8):

$$\frac{\varepsilon^{ijk}}{\sqrt{g}} \mathbf{e}_i = \mathbf{e}^j \times \mathbf{e}^k \tag{3.33}$$

Thus, the following result is obtained:

$$\mathbf{A} \times \mathbf{B} = \frac{\varepsilon^{ijk}}{\sqrt{g}} A_j B_k \mathbf{e}_i \tag{3.34}$$

That is, the contravariant component of the cross product is given by:

$$(\mathbf{A} \times \mathbf{B})^i = \frac{\varepsilon^{ijk}}{\sqrt{g}} A_j B_k \tag{3.35}$$

Contravariant components of each physical quantity naturally appear in linear differential operators, in this context:

$$(\mathbf{A} \times \mathbf{B})^i = \frac{\varepsilon^{ijk}}{\sqrt{g}} g_{jm} A^m g_{kn} B^n \tag{3.36}$$

To convert from contravariant to covariant components, the metric tensor is applied, leading to the following relation:

$$(\mathbf{A} \times \mathbf{B})^i = g^{ij} (\mathbf{A} \times \mathbf{B})_j \tag{3.37}$$

Therefore, the final expression widely used throughout this document will be:

$$\boxed{\mathbf{A} \times \mathbf{B} = \frac{\varepsilon^{ijk}}{\sqrt{g}} g_{jm} g_{kn} A^m B^n \mathbf{e}_i} \tag{3.38}$$

This formulation will be employed to compute Lorentz forces, differential area elements, volume elements and to characterize measurements within the curvilinear coordinate system.

3.5 Measurements in Curvilinear Coordinates

Obtaining the necessary tools for measuring space is fundamental in the context of physics: whether it involves calculating volumes, areas, or lengths within a given system. Physical quantities derived from these geometric measures, such as flux, mass, and others, often serve as constraints in proposed models. Therefore, having precise control over spatial measurement is essential for accurate physical modeling.

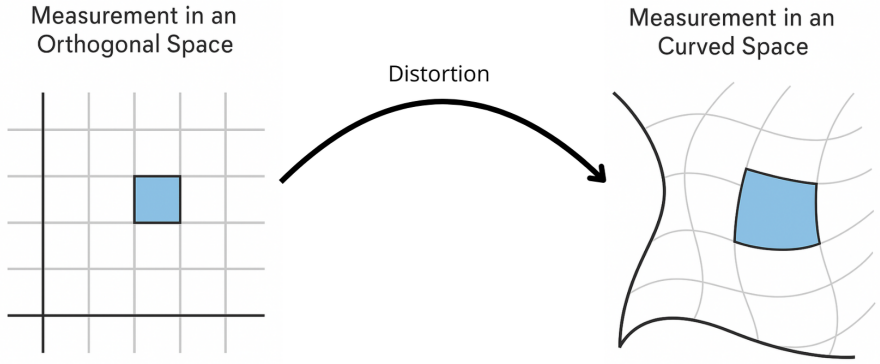


Figure 3.3: Comparison between spatial measurement in an Eucledian space and a curved space. On the left, measurements such as areas and volumes are straightforward due to the flat and orthogonal nature of the coordinate system. On the right, in a curved coordinate system, the same region is deformed.

In Euclidean geometry, these calculations are straightforward due to the flat and orthogonal nature of the space, where distances, areas, and volumes can be determined directly without additional considerations. However, this simplicity disappears when dealing with a curved space. This raises a fundamental question: How can geometric quantities be measured when operating within a curved coordinate system?

3.5.1 Infinitesimal Distance

Given $\mathbf{R}(u^1, \ldots, u^i, \ldots, u^n)$, if one aims to measure the differential displacement $d\mathbf{R}$ along the i-th direction, denoted as $dl(i)$:

$$dl(i) = |d\mathbf{R}(i)| = \sqrt{d\mathbf{R}(i) \cdot d\mathbf{R}(i)} \tag{3.39}$$

Using the previous results, $d\mathbf{R}(i) = \mathbf{e}_i \, du^i$, and therefore:

$$dl(i) = \sqrt{g_{ii}} \, du^i \tag{3.40}$$

This expression provides the general form for calculating the differential arc length along the i-th coordinate direction in a curvilinear coordinate system, as can be seen, the metric

tensor accounts for the geometric deformations of the space. In vector form, following the arc coordinate direction, this relation becomes:

$$\boxed{d\mathbf{l}(i) = \sqrt{g_{ii}}\, du^i\, \hat{\mathbf{e}}_i} \tag{3.41}$$

3.5.2 Surface Area Element

To compute the differential surface element, the cross product of the two basis vectors spans a surface of constant u^i. The area element corresponding to the i-constant surface is given by:

$$dS(i) = |\mathbf{e}_j \times \mathbf{e}_k|\, du^j\, du^k = \sqrt{g}\, |\nabla u^i|\, du^j\, du^k \tag{3.42}$$

In vector form, the differential surface element can be expressed as:

$$\boxed{d\mathbf{S}(i) = \sqrt{g}\, du^j\, du^k \mathbf{e}^i} \tag{3.43}$$

This formulation provides both the magnitude and orientation of the surface element, where $\mathbf{e}^i$ denotes the unit vector normal to the surface u^i = constant, it will be fundamental in evaluating surface integrals such as flux computations

3.5.3 Volume Element

Proceeding further, starting from the definition of the differential volume element in its most general form:

$$dV = d^3\mathbf{R} = (\mathbf{e}_i \times \mathbf{e}_j) \cdot \mathbf{e}_k\, du^i\, du^j\, du^k \tag{3.44}$$

By applying the definition of the Levi-Civita symbol as presented in equation (3.26), and considering that the basis vectors form a right-handed system, it is obtained:

$$\boxed{dV = \sqrt{g}\, du^i\, du^j\, du^k} \tag{3.45}$$

Here, $\sqrt{g}$ represents the square root of the determinant of the metric tensor, which encapsulates the geometric properties of the curvilinear coordinate system. This expression provides the general form for calculating volume elements in non-Euclidean spaces, and it is fundamental for performing volume integrals in curvilinear frameworks.

3.6 Concept of the Covariant Derivative

Let us consider a differentiable vector field denoted by $\mathbf{B}$, which depends on position such that $\mathbf{B} = \mathbf{B}(\mathbf{R}(u^1, u^2, u^3))$. This vector field can be expressed in terms of its components and basis vectors as:

$$\mathbf{B} = B^i \mathbf{e}_i \tag{3.46}$$

Taking the partial derivative of this expression with respect to a given coordinate direction u^k yields:

$$\frac{\partial \mathbf{B}}{\partial u^k} = \frac{\partial}{\partial u^k} \left(B^i \mathbf{e}_i \right) \tag{3.47}$$

Applying the product rule, it is obtained:

$$\frac{\partial \mathbf{B}}{\partial u^k} = \frac{\partial B^i}{\partial u^k} \mathbf{e}_i + B^i \frac{\partial \mathbf{e}_i}{\partial u^k} \tag{3.48}$$

The j-th contravariant component of this derivative can then be written as:

$$\boxed{\left(\frac{\partial \mathbf{B}}{\partial u^k} \right)^j = \frac{\partial B^j}{\partial u^k} + B^i \left\{ \frac{\partial \mathbf{e}_i}{\partial u^k} \cdot \mathbf{e}^j \right\}} \tag{3.49}$$

For simplicity, the covariant derivative $\left(\frac{\partial \mathbf{B}}{\partial u^k} \right)^j$ is often denoted as $\mathbf{B}^j_{,k}$. It is important to note that the covariant derivative does not only measure how a physical quantity changes along a given direction, but also accounts for how the basis vectors themselves vary in that direction; key aspect in curvilinear coordinate systems.

3.7 Derivation of Gradient, Curl, and Divergence in Non-Orthogonal Coordinates

In this section, the fundamental operations of the nabla operator on scalar and vector fields are derived. These operations are essential for formulating the governing equations of Maxwell's equations and fluid mechanics, which will be addressed in subsequent chapters.

3.7.1 Gradient

The gradient can be interpreted as the direction of the maximum rate of increase of a scalar function. In generalized coordinates, it takes the form:

$$\boxed{\nabla \Phi = \frac{\partial \Phi}{\partial u^i} \nabla u^i} \tag{3.50}$$

3.7.2 Divergence

The divergence of a vector field in general coordinates can be elegantly derived using fundamental vector calculus identities, as shown in D'haeseleer (1991). Specifically, the following properties are used:

$$\nabla \times \nabla\Phi \equiv 0 \quad \text{and} \quad \nabla \cdot (\nabla \times \mathbf{A}) \equiv 0 \tag{3.51}$$

These expressions state that the curl of the gradient of a scalar field is zero, and that the divergence of the curl of a vector field is also zero. Using the Levi-Civita symbols (3.26), it can be shown that:

$$\frac{\mathbf{e}_i}{\sqrt{g}} = \mathbf{e}^j \times \mathbf{e}^k = \nabla u^j \times \nabla u^k = \nabla \times (u^j \nabla u^k) \tag{3.52}$$

In other words, the covariant and contravariant basis vectors, $\mathbf{e}_i \leftrightarrow \mathbf{e}^j$, can be expressed in terms of a curl, where i, j, and k follow a cyclic order. Therefore, by applying (3.51):

$$\nabla \cdot \left(\frac{\mathbf{e}_i}{\sqrt{g}} \right) = \nabla \cdot \left[\nabla \times (u^j \nabla u^k) \right] = 0 \tag{3.53}$$

With this result, the divergence of a vector field $\mathbf{A}$ can be derived as follows:

$$\nabla \cdot \mathbf{A} = \nabla \cdot (A^i \mathbf{e}_i) = \nabla \cdot \left(\sqrt{g} A^i \frac{\mathbf{e}_i}{\sqrt{g}} \right) = \sqrt{g} A^i \nabla \cdot \left(\frac{\mathbf{e}_i}{\sqrt{g}} \right) + \frac{\mathbf{e}_i}{\sqrt{g}} \cdot \nabla(\sqrt{g} A^i) \tag{3.54}$$

Since the first term vanishes:

$$\nabla \cdot \mathbf{A} = \frac{\mathbf{e}_i}{\sqrt{g}} \cdot \nabla(\sqrt{g} A^i) \tag{3.55}$$

Finally, by applying the definition of the nabla operator in curvilinear coordinates, one can get:

$$\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^i} (\sqrt{g} A^i)\, \mathbf{e}^i \cdot \mathbf{e}_i \tag{3.56}$$

Contracting indices and using the properties of the Kronecker delta,the final expression is obtained:

$$\boxed{\nabla \cdot \mathbf{A} = \frac{1}{\sqrt{g}} \frac{\partial}{\partial u^i} \left(\sqrt{g} A^i \right)} \tag{3.57}$$

3.7.3 Curl

To demonstrate this relation, it is essential to apply the definition of the operator itself, together with the identities presented in (3.51). For a vector field $\mathbf{A}$, one obtains:

$$\nabla \times \mathbf{A} = \nabla \times \left(A_j \mathbf{e}^j \right) = A_j (\nabla \times \mathbf{e}^j) + \nabla A_j \times \mathbf{e}^j = A_j (\nabla \times \nabla u^j) + \nabla A_j \times \mathbf{e}^j \tag{3.58}$$

By the property seen in (3.51) this leads to

$$\nabla \times \mathbf{A} = \nabla A_j \times \mathbf{e}^j = \frac{\partial A_j}{\partial u^i} \left(\mathbf{e^i} \times \mathbf{e}^j \right) \tag{3.59}$$

By applying the definition of the Levi-Civita symbol, as presented in (3.26), the expression becomes:

$$\boxed{\nabla \times \mathbf{A} = \frac{\varepsilon^{ijk}}{\sqrt{g}} \frac{\partial A_j}{\partial u^i} \mathbf{e}_k} \tag{3.60}$$

3.8 Scale Factors and the Physical Notion of Normalization

This chapter aims to provide an explanation of the coefficients accompanying the covariant and contravariant basis vectors. As introduced in (3.2), a vector in arbitrary coordinates can be expressed as:

$$\mathbf{V} = V^i \mathbf{e}_i = V_i \mathbf{e}^i \tag{3.61}$$

It is important to clarify that the coefficients V_i and V^i do not necessarily have a direct physical meaning. These coefficients are mathematical constructs employed to derive the final physical quantities of interest. They correspond to measurable physical magnitudes, only when the basis is orthonormal. Otherwise, it is the vector $\mathbf{V}$, treated as a whole entity, that retains the true physical significance. This distinction arises because, in general, the basis vectors are not required to be normalized. As a result, the coefficients lose their direct association with physical measurement. In some contexts, particularly in works such as those by Nieves-Chinchilla et al. (2018), the concept of a *scaled basis* is introduced. This involves normalizing the basis vectors, which can be rewritten as:

$$\mathbf{e} = h \frac{\hat{\mathbf{e}}}{h} \tag{3.62}$$

where h is known as the **scale factor**. Accordingly, the vector can be expressed as:

$$\mathbf{V} = V_i \mathbf{e}^i = (V_i \, h^i) \hat{\mathbf{e}}^i \tag{3.63}$$

In this document, a non-normalized basis will be adopted, consistent with standard treatments in differential geometry. However, specific remarks regarding scale factors will be made when relevant, particularly in contexts where physical interpretation requires careful attention to normalization. Ultimately, the scale factor can be understood as a measure of how space is no longer divided into segments of equal length. Due to the intrinsic deformations of the chosen coordinate system, some segments may effectively occupy more physical extent than others. It is the scale factor that governs this behavior. One can imagine a discretized space, a kind of fabric being deformed. The elements, now discrete, no longer retain uniform size. The scale factor determines how each of these segments

stretches or compresses, providing the necessary correction to interpret measurements within the deformed geometry.

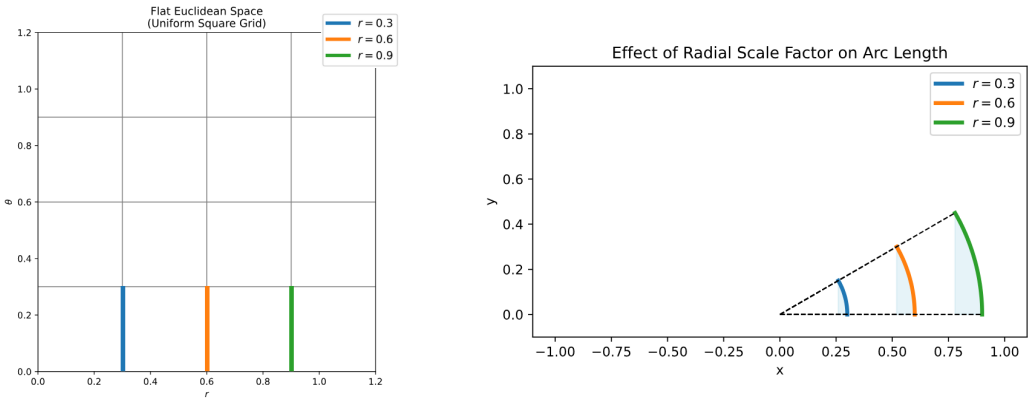


Figure 3.4: Comparison between a Euclidean space and a deformed space.

The previous figure illustrates how the fabric of space changes depending on the reference system employed. In the first case, an Euclidean reference system is observed, where each segment preserves its length uniformly. When considering the boundary of a differential area element, moving along r, one notices that the length of the colored sections remains constant. This behavior is a consequence of the fact that the underlying metric is diagonal and, specifically, corresponds to the identity matrix:

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{3.64}$$

Thus, the differential line element, as previously introduced in (3.40), simplifies to:

$$dl = 1 \cdot d\theta = d\theta \tag{3.65}$$

Taking a closer look at the differential line element in circular cylindrical coordinates, it is evident by inspection that the length element changes as one moves away from the origin. The metric in cylindrical coordinates takes the form:

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & r^2 \end{pmatrix} \tag{3.66}$$

In this case, when computing the differential arc length along the angular direction, it is obtained:

$$dl = r\, d\theta \tag{3.67}$$

This expression reveals a direct proportionality to the radius, indicating that the length of the angular differential element is not constant, but rather grows linearly with distance from the center. This is a geometric consequence of the nontrivial scale factor introduced by the curvilinear coordinate system.

4 . Modeling of a Flux Rope with Constant Elliptical Cross-Section and Radial Current

The model proposed by Nieves-Chinchilla et al. (2018), which describes the structure of a magnetic Flux Rope with elliptical-cylindrical geometry, serves as the foundation for the present work. The present study represents a significant advancement by **introducing, for the first time, a nonzero radial current** into such modern frameworks. This new approach has unique challenges: particularly in the formulation and solution of the associated partial differential equations that will be solved in the next pages.

Within this context a rigorous mathematical formulation that incorporates a new generating function is developed, specifically tailored to accommodate the presence of a radial current. Two foundational assumptions are established to ensure the internal coherence of the extended model: the adoption of a distorted elliptical geometry and the imposition of invariance in the y-direction, $\partial_y \square = 0$.

This work aspires to deepen the theoretical understanding of the physical mechanisms governing Flux Ropes. The primary motivation behind this study lies in exploring whether this new approach can explain magnetic field discontinuities, interactions between Flux Ropes, and other phenomena that cannot be explained by current models.

4.1 Equations to be Solved for the Physical Model

The primary objective is to model the magnetic field configurations that may arise within the Flux Rope (FR); this structure is assumed to behave as a magnetically coherent body. The governing equations for this physical model are those that constrain the magnetic field. Specifically, any admissible magnetic field configuration within the FR must satisfy the following two conditions:

- ◇ The divergence-free condition:

$$\nabla \cdot \mathbf{B} = 0 \tag{4.1}$$

- ◇ The curl condition relating the magnetic field to the current density:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \tag{4.2}$$

Any magnetic field configuration must be compatible with both conditions but will be

the initial conditions what will determine it. The current density is introduced as the generating function for the magnetic field through its appearance in the curl equation. Following the relations derived in (3.57) and (3.60), and working in generalized coordinates using contravariant components, these laws can be written as:

$$\nabla \cdot \mathbf{B} = \frac{1}{\sqrt{g}} \partial_i \left(\sqrt{g} B^i \right) \tag{4.3}$$

$$(\nabla \times \mathbf{B})^i = \frac{\varepsilon^{ijk}}{\sqrt{g}} \partial_j \left(g_{kl} B^l \right) \tag{4.4}$$

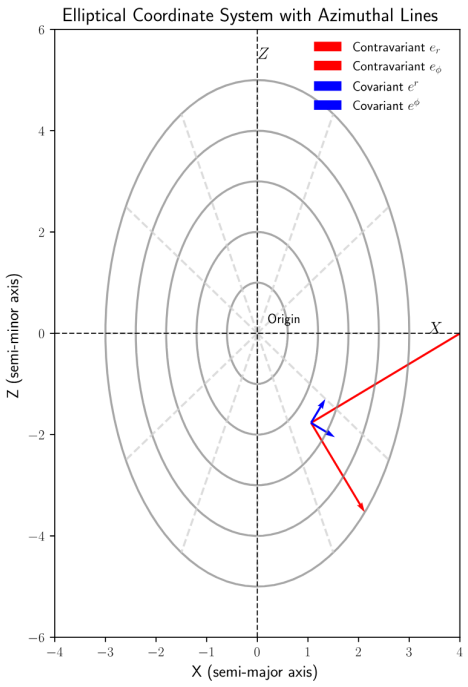
where $\sqrt{g}$ is the Jacobian, and ε^{ijk} is the Levi-Civita symbol in three dimensions.

4.2 Generalized Elliptical-Cylindrical System Framework

Building on the foundation of previous papers and adopting a cylindrical shaped geometry coordinate system, the coordinates are defined as follows:

$$\mathbf{R}(r, y, \varphi) : \begin{cases} x = \delta\, r\, \cos\varphi \\ y = y \\ z = r\, \sin\varphi \end{cases}$$

Here, r represents the radial coordinate, y the axial coordinate, and φ the poloidal coordinate. The parameter δ characterizes the ellipticity of the cross section, with the maximum elongation occurring along the z-axis.



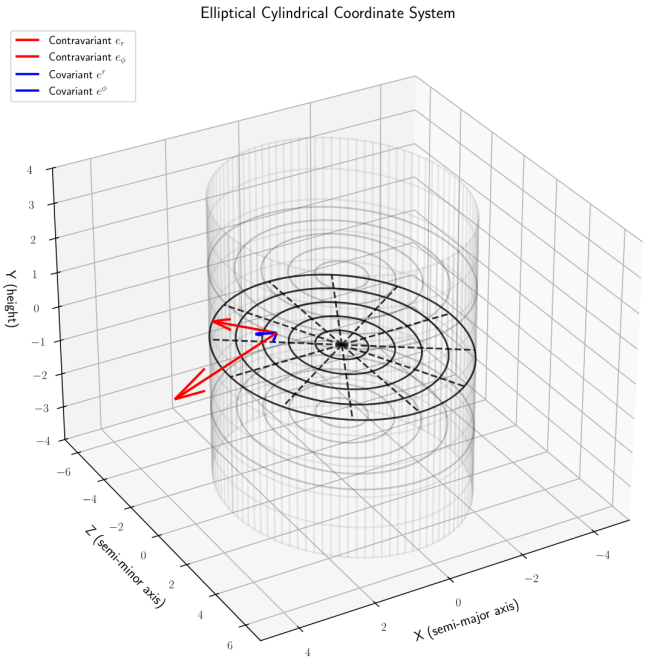


Figure 4.1: Elliptical cylindrical reference system used for deriving the equations of the model. On the left, a cross section of the Flux Rope at constant y-planes is shown, while on the right, a 3D representation is provided.

Furthermore, the distortion parameter δ is defined by the ratio:

$$\delta = \frac{R}{d} \tag{4.5}$$

where d and R represent the semi-major and semi-minor axes of the ellipse, respectively, with d aligned along the z-axis and L denotes the length of the system along the y-direction. At this point, the domains of the variables involved are established as:

$$r \in [0, d] \tag{4.6}$$
$$\varphi \in [0, 2\pi) \tag{4.7}$$
$$y \in [0, L] \tag{4.8}$$

It is worth noting that the entire space can be accessed using the reference system described. Once fixed, the parameter δ determines how radial displacements correspond to the level curves of an underlying elliptical shape, as illustrated in Figure 4.2. In this sense, the coordinate system can be thought of as a modified circular framework, where the deviation from circular symmetry is governed by the parameter δ. This distortion defines the shape of the transverse cross section, for instance:

Value of δ	**Cross-Section Geometry**
$\delta = 1$	Circle
$\delta \neq 1$	Ellipse
$\delta = \delta(r, \theta)$	Arbitrary (e.g., torus, warped surfaces)

Table 4.1: Summary of the effect of the distortion parameter δ on the cross-sectional geometry of the Flux Rope. A value of $\delta = 1$ corresponds to a perfect circle, $\delta \neq 1$ leads to an elliptical deformation, and a non-constant $\delta = \delta(r, \theta)$ allows for more complex, arbitrary shapes such as toroidal or warped structures.

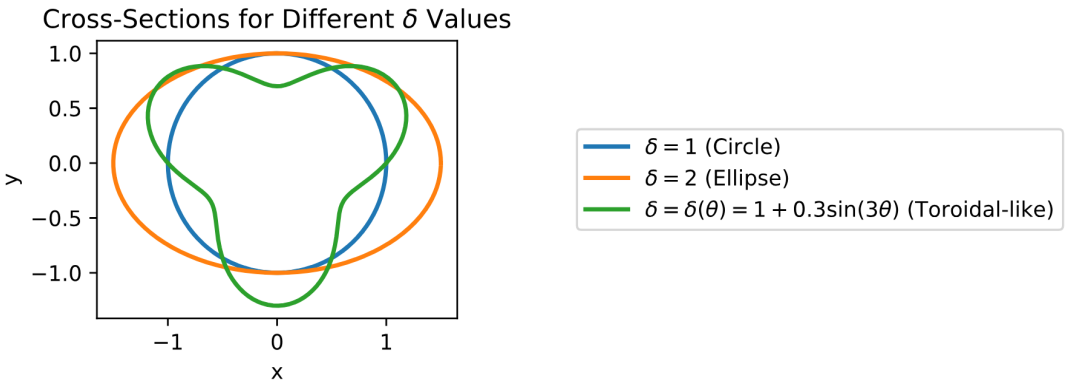


Figure 4.2: Illustration of the effect of the distortion parameter δ on the transverse cross-section. Three cases are shown: a circular section ($\delta = 1$), an elliptical section ($\delta = 2$), and a toroidal-like distorted section ($\delta = \delta(\varphi) = 1 + 0.3\sin(3\varphi)$).

4.3 Derivation of Differential Geometry Elements

In this reference system, the non-unitary covariant basis vectors can be constructed as:

$$\mathbf{e}_i = \partial_i \mathbf{R}$$

Consequently, the following non-normalized covariant basis vectors are obtained for the non-orthogonal curvilinear system:

$$\begin{aligned} \mathbf{e}_r &= (\delta \cos\varphi, 0, \sin\varphi) \\ \mathbf{e}_y &= (0, 1, 0) \\ \mathbf{e}_\varphi &= r(-\delta \sin\varphi, 0, \cos\varphi) \end{aligned} \tag{4.9}$$

The covariant metric tensor can then be written as:

$$g_{ij} = \begin{pmatrix} g_{rr} & g_{ry} & g_{r\varphi} \\ g_{yr} & g_{yy} & g_{y\varphi} \\ g_{\varphi r} & g_{\varphi y} & g_{\varphi\varphi} \end{pmatrix} = \begin{pmatrix} g_{rr} & 0 & g_{r\varphi} \\ 0 & g_{yy} & 0 \\ g_{\varphi r} & 0 & g_{\varphi\varphi} \end{pmatrix}$$

In the principal directions, the diagonal components of the metric tensor are:

$$\begin{aligned} g_{rr} &= \mathbf{e}_r \cdot \mathbf{e}_r = \left(\delta^2 \cos^2\varphi + \sin^2\varphi\right) \\ g_{yy} &= \mathbf{e}_y \cdot \mathbf{e}_y = 1 \\ g_{\varphi\varphi} &= \mathbf{e}_\varphi \cdot \mathbf{e}_\varphi = r^2\left(\delta^2 \sin^2\varphi + \cos^2\varphi\right) \end{aligned} \tag{4.10}$$

The off diagonal components of the metric tensor take the form:

$$g_{r\varphi} = g_{\varphi r} = \mathbf{e}_r \cdot \mathbf{e}_\varphi = r(1-\delta^2)\sin\varphi\cos\varphi \tag{4.11}$$

Finally, the determinant of the metric tensor is given by:

$$\sqrt{g} = J = \delta r$$

The presence of non-zero off diagonal components in the metric tensor implies that the coordinate system is not orthogonal. In this case, the coupling term $g_{r\varphi} \neq 0$ reveals that the directions r and φ are not perpendicular.

4.4 Solution to the Gauss's Law for Magnetism

The divergence-free condition leads to the following equation:

$$\partial_r(\delta r B^r) + \partial_\varphi(\delta r B^\varphi) = 0 \tag{4.12}$$

In general, the radial component B^r may depend on both r and φ.

4.4.1 Magnetic Monopole Annihilation

Due to the divergence-free condition of the magnetic field, there can be no sources or sinks of magnetic flux and the Flux Rope is interpreted as a magnetically closed volume. Therefore, at the boundary $r = R(\varphi)$, the radial component must vanish. Although a non-trivial magnetic field configuration that satisfies this condition could, in principle, be constructed, symmetry arguments and consistency with previously established models like Lundquist (1950) justify fixing $B^r = 0$ identically throughout the domain. Consequently, we conclude that:

$$\partial_\varphi(rB^\varphi) = 0 \quad \Rightarrow \quad B^\varphi = B^\varphi(r) \tag{4.13}$$

This provides a first constraint on the admissible forms of the magnetic field. In particular, the azimuthal component must depend solely on the radial coordinate.

$$\boxed{\mathbf{B}(r,\varphi) = B^y(r,\varphi)\,\mathbf{e}_y + B^\varphi(r)\,\mathbf{e}_\varphi} \tag{4.14}$$

4.5 Solution of Ampère's Law

As discussed in Nieves-Chinchilla et al. (2018), the assumption $j^r = 0$ was an arbitrary simplification intended to reduce the complexity of the system and the number of free parameters. However, in the present work, this constraint is relaxed by allowing the radial current to be non-zero. As will be shown later in equation (4.21), there exists an intrinsic relationship between the components of the current density and the main objective of this work is to explore the physical implications of lifting this constraint. The system of equations derived from Ampère's Law in the chosen coordinate system becomes:

$$\partial_\varphi(B^y) = -\mu_0\sqrt{g}\,j^r \tag{4.15}$$
$$\partial_\varphi(g_{r\varphi}B^\varphi) - \partial_r(g_{\varphi\varphi}B^\varphi) = \mu_0\sqrt{g}\,j^y \tag{4.16}$$
$$\partial_r(g_{yy}B^y) = \mu_0\sqrt{g}\,j^\varphi \tag{4.17}$$

Before solving the system, it is worth noting that by manipulating equations (4.15) and (4.17), one can obtain an implicit relation between the radial and poloidal components of the current density. This relation corresponds to the current continuity condition, derived from applying the divergence operator to Ampère's Law:

$$\nabla\cdot(\nabla\times\mathbf{B}) = \nabla\cdot(\mu_0\mathbf{j}) \tag{4.18}$$

Using the vector identity:

$$\nabla\cdot(\nabla\times\mathbf{B}) = 0 \tag{4.19}$$

we find that:

$$\mu_0 (\nabla \cdot \mathbf{j}) = 0 \quad \Rightarrow \quad \boxed{\nabla \cdot \mathbf{j} = 0} \tag{4.20}$$

Therefore, by solving Ampère's Law, the condition that the current density must be divergence-free is enforced, there can be no current monopoles. This implies that, out of the three equations derived from Ampère's Law, two will determine the poloidal and axial components of the magnetic field while the third will serve as a constraint equation on the admissible current distributions:

$$\partial_r(\delta r j^r) + \partial_\varphi(\delta r j^\varphi) = 0 \tag{4.21}$$

This leads to the important conclusion that **not every arbitrary current distribution can give rise to a magnetic field compatible with a Flux Rope**. There exists, in fact, a deeper condition that must be satisfied to ensure physical and geometrical consistency within the model.

Solution of the Axial Magnetic Field

By applying the integral operator to both sides of equation (4.17), the axial magnetic field is obtained:

$$\boxed{B^y(r,\varphi) = B_0^y + \mu_0 \delta \int_0^r j^\varphi(r',\varphi)\, r'\, dr'} \tag{4.22}$$

Here, B_0^y represents the value of the axial magnetic field at the origin and is treated as a constant of integration.

Solution of the Poloidal Magnetic Field

To solve equation (4.16), the expression is expanded, arriving at the following simplified form:

$$\partial_r B^\varphi + \frac{\chi}{r} B^\varphi = -\frac{\mu_0 \delta}{h^2 r} j^y \tag{4.23}$$

Here, $h^2 = \dfrac{g_{\varphi\varphi}}{r^2}$ and $\chi = \dfrac{\delta^2 + 1}{h^2}$ are defined. The general solution is constructed as the product of a homogeneous solution and a particular solution:

$$B^\varphi(r) = B_h^\varphi \cdot u_p \tag{4.24}$$

For the homogeneous part:

$$\partial_r(B^\varphi) + \frac{\chi}{r} B^\varphi = 0 \tag{4.25}$$

The solution to this differential equation is given by:

$$B_h^{\varphi} = B_0^{\varphi}\, r^{-\chi} \tag{4.26}$$

For the particular solution,the method of variation of parameters is applied:

$$\partial_r u_p = -\frac{\mu_0 \delta}{h^2}\, r^{\chi-1} j^y \tag{4.27}$$

Integrating, one can obtain:

$$u_p = -\frac{\mu_0 \delta}{g_{\varphi\varphi}} r^2 \int_0^r r'^{\chi-1} j^y(r', \varphi)\, dr' \tag{4.28}$$

Therefore, the full solution for the poloidal magnetic field, as described in equation (4.24), takes the form:

$$\boxed{B^{\varphi}(r) = -B_0^{\varphi}\, \frac{\mu_0 \delta}{g_{\varphi\varphi}\, r^{\chi-2}} \int_0^r r'^{\chi-1} j^y(r', \varphi)\, dr'} \tag{4.29}$$

It is important to note that the poloidal magnetic field depends exclusively on the radial coordinate. The presence of an axial current density $j^y(\varphi)$ introduces a mechanism through which the poloidal component compensates for the metric induced variation in B^{φ}. As a result, $j^y(\varphi)$ should not be interpreted as an additional degree of freedom in the system, but rather as a mathematical consequence required to ensure the internal consistency of the model and the satisfaction of the divergence-free condition given in equation (4.14). As will be shown later, this function is entirely determined by the constraints imposed by flux conservation.

Solution of the Divergence-Free Condition for the Current

Solving the divergence-free condition for the current is straightforward by applying the integral operator. The radial current is expressed as a functional of the poloidal current as follows:

$$\boxed{j^r(r, \varphi) = -\frac{1}{r}\partial_{\varphi} \int_0^r r'\, j^{\varphi}(r', \varphi)\, dr'} \tag{4.30}$$

4.6 Solutions Obtained for the Magnetic Fields and Currents

The resolution of the previous equations ensures the physical compatibility required for the existence of a magnetic field given a specified current distribution. In particular, all divergence and curl conditions are satisfied and the magnetic field is finally obtained in the form:

$$\boxed{\mathbf{B}(r,\varphi) = \left(B_0^y + \mu_0\delta \int_0^r j^\varphi(r',\varphi)\, r'\, dr'\right)\mathbf{e}_y - \left(B_0^\varphi \frac{\mu_0\delta}{g_{\varphi\varphi}\, r^{\chi-2}} \int_0^r r'^{\chi-1} j^y(r',\varphi)\, dr'\right)\mathbf{e}_\varphi} \tag{4.31}$$

while the current vector takes the form:

$$\boxed{\mathbf{j}(r,\varphi) = -\left(\frac{1}{r}\partial_\varphi \int_0^r r'\, j^\varphi(r',\varphi)\, dr'\right)\mathbf{e}_r + j^y(r,\varphi)\mathbf{e}_y + j^\varphi(r,\varphi)\mathbf{e}_\varphi} \tag{4.32}$$

From the formulation, it is evident that a prior knowledge of two independent components of current is enough to obtain both the distribution of current and the magnetic field across the Flux Rope. It is imposed that the current functions may be articulated as separable functions, referred to them as *generating functions*. Such an assumption is properly substantiated, as it suggests that the radial and poloidal dependencies are either independent or only weakly coupled. In fact, previously developed models, such as the one introduced in Nieves-Chinchilla et al. (2016); Lundquist (1950); Nieves-Chinchilla et al. (2018), are recovered precisely under this condition of variable separability. Thus:

$$j^r(r,\varphi) = -\frac{\partial_\varphi\left(j_2^\varphi(\varphi)\right)}{r}\int_0^r r j_1^\varphi(r), dr \tag{4.33}$$

$$j^y(r,\varphi) = j_1^y(r) j_2^y(\varphi) \tag{4.34}$$

$$j^\varphi(r,\varphi) = j_1^\varphi(r) j_2^\varphi(\varphi) \tag{4.35}$$

The only unknown functions are $j_1^y(r)$, $j_1^\varphi(r)$, and $j_2^\varphi(\varphi)$, since $j_2^y(\varphi)$ is constrained to cancel the poloidal dependence of B^φ, as will be derived in subsequent sections.

4.7 Development of the Generating Functions

To select the generating functions of the model, they must be sufficiently versatile to represent any physically admissible current configuration while also being "smooth enough" to be considered an observable physical quantity. Accordingly, the following requirements are imposed:

$$\begin{cases} j_1(r) \in \mathcal{C}^\infty([0,R]) \cap L^1([0,R]) \\ j_2(\varphi) \in \mathcal{C}^\infty([0,2\pi]) \cap L^1([0,2\pi]) \\ j_2(\varphi + 2\pi) = j_2(\varphi) \end{cases} \tag{4.36}$$

Choice of the Radial Part

A Taylor series expansion centered at r = 0 is adopted, as it satisfies the above conditions and provides a straightforward representation of the radial component. The computation of the coefficients is also direct. The expansion is given by:

$$j_1(r) = \sum_{k=0}^{\infty} a_k r^k = a_0 + a_1 r + a_2 r^2 + \dots \tag{4.37}$$

The coefficients are explicitly given by:

$$a_k = \frac{1}{k!} \frac{d^k j_1(r)}{dr^k}\bigg|_{r=0} \tag{4.38}$$

Choice of the Azimuthal Part

For the azimuthal component, a Fourier series expansion is employed, which naturally satisfies the periodicity condition and ensures smoothness, differentiability, and integrability. Specifically, the azimuthal part of the current, $j_2(\varphi)$, is expressed as a complex Fourier series:

$$j_2(\varphi) = \sum_{l=-\infty}^{\infty} b_l e^{il\varphi} \tag{4.39}$$

Here, b_l are the Fourier coefficients that represent the amplitude of each harmonic mode and the index $l \in \mathbb{Z}$ ensures that the frequencies are integer multiples, which leads to the2π-periodic nature of the angular coordinate φ. The coefficients are computed using the standard Fourier projection formula:

$$b_l = \frac{1}{2\pi} \int_0^{2\pi} j_2(\varphi) e^{-il\varphi} d\varphi \tag{4.40}$$

The final expressions for the current components, obtained from the choice of generating functions, are given by:

$$\begin{cases} j^r(r,\varphi) = \left(\sum_{l=-\infty}^{+\infty} l\, b_l^{\varphi} e^{i(l\varphi + 3\pi/2)} \right) \left(\sum_{k=0}^{+\infty} a_k^{\varphi} \frac{r^{k+1}}{k+2} \right) \\ j^y(r,\varphi) = \left(\sum_{k=0}^{+\infty} a_k^y r^k \right) j_{\varphi}^y(\varphi) \\ j^{\varphi}(r,\varphi) = \left(\sum_{k=0}^{+\infty} a_k^{\varphi} r^k \right) \left(\sum_{l=-\infty}^{+\infty} b_l^{\varphi} e^{il\varphi} \right) \end{cases} \tag{4.41}$$

4.7.1 Physical Consistency and constraints to the generating functions

Existence of Directional Limits of the Currents

In the cross-section of our Flux Rope, the directional limits of the current components defined within the slice must exist. In other words, the currents $j^r(r,\varphi)$ and $j^{\varphi}(r,\varphi)$ must

converge to a unique value as one approaches the axis of the domain from any direction within the plane. Furthermore, in order to avoid an infinite twisting of the current at the center, the azimuthal component j^{φ} is required to vanish at the origin. This condition is expressed as:

$$\lim_{r\to 0} j^{\varphi}(r,\varphi) = 0 \tag{4.42}$$

This leads to the constraint $k \geq 1$ in the Taylor expansion of the radial coefficient a_k^{φ}, ensuring that no constant or singular term survives at the origin.

Current components:

$$j^{r}(r,\varphi) = \left(\sum_{l=-\infty}^{+\infty} l\, b_l^{\varphi}\, e^{i(l\varphi+3\pi/2)}\right)\left(\sum_{k=1}^{+\infty} a_k^{\varphi}\,\frac{r^{k+1}}{k+2}\right) \tag{4.43}$$

$$j^{y}(r,\varphi) = \left(\sum_{k=0}^{+\infty} a_k^{y}\, r^{k}\right) j_2^{y}(\varphi) \tag{4.44}$$

$$j^{\varphi}(r,\varphi) = \left(\sum_{k=1}^{+\infty} a_k^{\varphi}\, r^{k}\right)\left(\sum_{l=-\infty}^{+\infty} b_l^{\varphi}\, e^{il\varphi}\right) \tag{4.45}$$

Magnetic field components:

$$B^{r}(r,\varphi) = 0 \tag{4.46}$$

$$B^{y}(r,\varphi) = B_0^{y} - \mu_0\delta\left(\sum_{l=-\infty}^{+\infty} b_l^{\varphi}\, e^{il\varphi}\right)\left(\sum_{k=1}^{+\infty} a_k^{\varphi}\,\frac{r^{k+2}}{k+2}\right) \tag{4.47}$$

$$B^{\varphi}(r,\varphi) = -B_0^{\varphi}\mu_0\delta\,\frac{j_2^{y}(\varphi)}{g_{\varphi\varphi}}\left(\sum_{k=0}^{+\infty} a_k^{y}\,\frac{r^{k+2}}{\chi+k}\right) \tag{4.48}$$

4.8 Implications of the Existence of a Radial Current

Let us take a closer look at equation (4.34), where the radial current is related to the azimuthal current through:

$$j^{r}(r,\varphi) = -\frac{\partial_{\varphi} j_2^{\varphi}(\varphi)}{r}\int_0^{r} r'\, j_1^{\varphi}(r')\, dr' \tag{4.49}$$

Compared to the previous model in Nieves-Chinchilla et al. (2018), a new current component emerges. Specifically, the term $j_2^{\varphi}(\varphi)$ introduces an explicit azimuthal dependence in the azimuthal current, which was previously assumed to be purely radial. It is important to emphasize that the only physically consistent condition for eliminating the radial current j^{r} is:

$$\boxed{\partial_\varphi j_2^\varphi(\varphi) = 0} \tag{4.50}$$

This implies that the azimuthal current becomes purely radial in its dependence, aligning fully with earlier models and recovering their expected behavior. In other words, **the presence of a non-constant azimuthal current component $j_2^\varphi(\varphi)$ necessarily leads to the appearance of a radial current component**. Compared to the model introduced in Nieves-Chinchilla et al. (2018), which assumes $j^r = 0$, the present formulation introduces an additional degree of freedom via the poloidal dependence of j^φ:

EC with $j^r = 0$	**EC with $j^r \neq 0$**
$j_1^\varphi(r)$, $j_1^y(r)$	$j_1^\varphi(r)$, $j_2^\varphi(\varphi)$, $j_1^y(r)$

Table 4.2: Comparison of generating current components in the EC model with and without radial current

Does the construction of the radial current make physical sense? To preserve the divergence-free condition of the current, meaning the absence of current monopoles, how can one allow for a radial current component that appears to point strictly outward? At first glance, this may seem contradictory. However, intuitively, if some current lines flow outward while others flow inward, the net current flux can still cancel out. As a result, the total divergence remains zero and no monopoles of current are introduced. The condition that must therefore be satisfied is:

$$\int_0^{2\pi} j^r(R, \varphi)\, R\, d\varphi = 0 \tag{4.51}$$

For a poloidal current profile expressed as a Fourier series, this condition becomes:

$$\Rightarrow \quad b_0^\varphi = 0 \tag{4.52}$$

This result has a clear physical interpretation. By eliminating the zeroth Fourier mode, it is guaranteed that the poloidal function has zero mean, that is, it contains no constant offset. The resulting current distribution is composed entirely of sine and cosine modes that are 2π-periodic and symmetric. Such a construction naturally leads to alternating behavior: positive current flow over one half of the domain and negative flow over the other, producing a net zero radial current flux through the boundary. Most elegantly, it is observed that this cancellation is automatically embedded in the actual model due to the derivative acting on the Fourier series:

$$\left(\sum_{l=-\infty}^{+\infty} l\, b_l^\varphi\, e^{i(l\varphi + 3\pi/2)} \right) \quad \Rightarrow \quad \text{zeroth mode automatically canceled} \tag{4.53}$$

The presence of the factor l eliminates the $l = 0$ term entirely, thereby ensuring that the radial current satisfies the physical condition of zero net current without requiring additional constraints.

4.9 Physical Quantities

A physical quantity is defined in this context as a measurable property derived from the magnetic fields and currents previously obtained. These quantities help characterize the Flux Rope and, in many cases, impose physically meaningful constraints. Some particularly relevant quantities that will be examined include magnetic fluxes, Lorentz forces, the twist number, magnetic energy density, and specific properties that determine whether the Flux Rope is force-free.

4.9.1 Magnetic Field Fluxes in the Flux Rope

Magnetic flux is a fundamental quantity that characterizes the amount of vector field passing through a given surface. It is defined as the sum of the magnetic field contributions over all points on a surface. In the continuum limit, this is expressed as:

$$\Phi(i) = \iint_S \mathbf{B} \cdot d\mathbf{S}(i) \tag{4.54}$$

When working in an arbitrary metric, careful attention must be given to the definition of the surface over which the vector field is evaluated. The vector associated with a surface of constant u^i, $\mathbf{e}^i$, accounts for the contravariant component of the field contributing to the total flux. Therefore, the flux through the surface of constant u^i is given by:

$$\boxed{\Phi(i) = \iint_S \mathbf{B} \cdot \mathbf{e}^i \, dS(i) = \iint_S B^i \, dS(i) = \iint_S B^i \, \sqrt{g} \, du^j \, du^k} \tag{4.55}$$

The axial magnetic flux is then computed by integrating the axial component of the magnetic field B^y over the cross section perpendicular to the axial direction. This is expressed as:

$$\Phi(y) = \int_0^{2\pi} \int_0^R B^y(r, \varphi) \, \sqrt{g} \, dr \, d\varphi \tag{4.56}$$

Evaluating this expression yields

$$\boxed{\Phi(y) = \pi R^2 \delta B_0^y + 2\pi \mu_0 \delta^2 b_0^\varphi \left(\sum_{k=0}^{+\infty} a_k^\varphi \frac{R^{k+4}}{(k+4)(k+2)} \right)} \tag{4.57}$$

This result shows that only the first term in the Foruier expansion contributes to the total flux, a consequence of the Kronecker delta arising from the angular integration in the orthogonal basis. Similarly, the poloidal magnetic flux is computed by integrating the poloidal component B^φ over the radial and axial directions:

$$\Phi(\varphi) = \int_0^L \int_0^R B^\varphi(r) \, \sqrt{g} \, dy \, dr \tag{4.58}$$

which leads to the expression

$$\boxed{\Phi(\varphi) = -\mu_0 \delta^2 L B_0^{\varphi} \frac{R^2}{g_{\varphi\varphi}(R,\varphi)} j_2^y(\varphi) \left(\sum_{k=0}^{+\infty} a_k^y \frac{R^{k+2}}{(k+2)(k+\chi)} \right)} \tag{4.59}$$

These fluxes are invariants and are conserved along the entire length of the Flux Rope. That is, even if the cross section is distorted, the poloidal flux remains constant across all cross sections. Likewise, the axial flux being invariant along y indicates that the amount of magnetic field entering one end of the Flux Rope along the axial direction equals the amount exiting through the other end.

4.9.2 Determination of the $j_2^y(\varphi)$ Function

The fundamental principle in this context is that the introduction of a radial current, delineated by component $j_2^{\varphi}(\varphi)$, exclusively influences the axial magnetic field, which is derived from the poloidal current shown by 4.31. As a result, the poloidal flux remains constant to this additional current, maintaining its role as an invariant within both the EC model characterized by zero radial current ($j^r = 0$) Nieves-Chinchilla et al. (2018) and the EC model with $j^r \neq 0$(this model). Since the flux must be preserved from the circular (Nieves-Chinchilla et al. (2016)) to elliptical model, one observes that the circular flux is obtained when $\chi = 2$. Under these circumstances, the total poloidal flux of the CC model is obtained as:

$$\Phi(\varphi) \;=\; -\,\mu_0\, L\, B_0^{\varphi} \left(\sum_{k=0}^{+\infty} a_k^y \frac{R^{k+2}}{(k+2)^2} \right) \tag{4.60}$$

This total poloidal flux must remain invariant. Therefore, within the newly proposed framework, preserving the poloidal flux translates into:

$$\sum_{k=0}^{+\infty} a_k^y \frac{R^{k+2}}{(k+2)^2} \;=\; \delta^2 \frac{R^2}{g_{\varphi\varphi}\big(R,\varphi\big)}\, j_2^y(\varphi) \sum_{k=0}^{+\infty} a_k^y \frac{R^{k+2}}{(k+2)\,\big(k+\chi\big)} \tag{4.61}$$

From this relation, one may isolate $j_{\varphi}^y(\varphi)$, which, in accordance with the theory, is contingent solely upon φ. Consequently, we derive:

$$\boxed{j_2^y(\varphi) \;=\; \frac{\left(\sum_{k=0}^{+\infty} a_k^y \frac{R^{k+2}}{(k+2)^2} \right) g_{\varphi\varphi}\big(R,\varphi\big)}{\delta^2\, R^2 \left(\sum_{k=0}^{+\infty} a_k^y \frac{R^{k+2}}{(k+2)\,\big(k+\chi\big)} \right)}} \tag{4.62}$$

As can be seen, $j_2^y(\varphi)$ depends exclusively on φ, as required, and it fully cancels the φ-dependence of the magnetic field component B^{φ}.

4.9.3 Lorentz Forces

The Lorentz force is defined as

$$\mathbf{f}(\mathbf{r}) = \mathbf{j} \times \mathbf{B} \tag{4.63}$$

and represents the magnetic force exerted on a charged particle located at an arbitrary position $\mathbf{r}$. In the context of an arbitrary metric, the contravariant components of the Lorentz force density are given by:

$$\mathbf{f}^i = \frac{\varepsilon^{ijk}}{\sqrt{|g|}} g_{jm} g_{kn} j^m B^n \tag{4.64}$$

Using this formulation, the explicit expressions for the components of the Lorentz force density are derived as follows:

$$\begin{aligned} f^r &= \frac{g_{yy} g_{\varphi\varphi}}{\delta r} \left(j^y B^\varphi - j^\varphi B^y \right) \\ f^y &= \frac{1}{\delta r} \left((g_{\varphi\varphi} j^\varphi + g_{\varphi r} j^r) g_{r\varphi} B^\varphi - (g_{r\varphi} j^\varphi + g_{rr} j^r) g_{\varphi\varphi} B^\varphi \right) \\ f^\varphi &= \frac{g_{rr} g_{yy}}{\delta r} j^r B^y \end{aligned} \tag{4.65}$$

These forces are responsible for confining the Flux Rope within a finite volume, counteracting the pressure gradient that tends to expand it.

4.9.4 Twist Number

The twist number $\mathcal{N}$ quantifies how many turns magnetic field lines perform around the axial direction as one moves along the length of the Flux Rope. It is defined as the ratio of the azimuthal displacement to the axial displacement of a magnetic field line:

$$\mathcal{N} = \frac{1}{2\pi} \int_0^L \frac{B^\varphi}{r B^y} \, dy \tag{4.66}$$

The quantity inside the integral represents the local twist rate. In our case, where the system is invariant along the axial direction y, the expression simplifies to:

$$\boxed{\mathcal{N}(r, \varphi) = \frac{L}{2\pi r} \frac{B^\varphi(r)}{B^y(r, \varphi)}} \tag{4.67}$$

It should be noted that in this case the twist number depends not only on the radial position but also exhibits a poloidal dependence. Although it may seem abstract or counterintuitive that the number of turns can vary at a fixed radius, **the physical implication is that the pitch angle of the Flux Rope will not remain constant along a given magnetic field line, it will vary**. This result is highly significant and **arises from the introduction of a radial current in the model**. In contrast to previous models where the pitch angle was constant, the **presence of a radial current** leads to a more realistic and dynamic configuration.

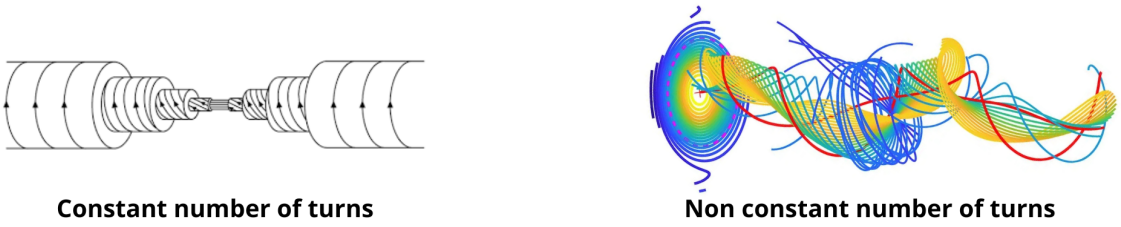


Figure 4.3: Comparison between previous analytical Flux Rope models, where the twist number depended only on the radial position, and the present model, in which the twist number exhibits both radial and angular dependence. This leads to a pitch angle that varies with position along the magnetic field lines.

4.9.5 Magnetic Field Line Parametrization

To obtain the parametric curve of a magnetic field line, an infinitesimal magnetic field vector can be decomposed into its r and φ components. This is illustrated in the following figure:

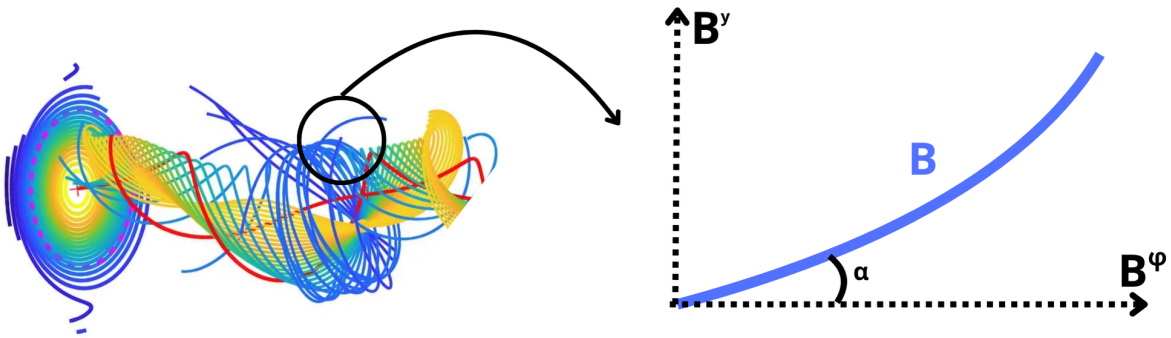


Figure 4.4: Differential magnetic field element in the Flux Rope projected onto the principal components of the covariant basis.

This general case, for an arbitrary pitch angle, can be described by the following relation:

$$\tan\alpha = \frac{B^y}{B^\varphi} = \frac{dy}{r\,d\varphi} \implies dy = \frac{B^y}{B^\varphi}\, r\, d\varphi \tag{4.68}$$

Integrating over the axial coordinate y yields:

$$y(r,\varphi) = \int_0^\varphi \frac{B^y}{B^\varphi}\, r\, d\varphi \tag{4.69}$$

This expression provides a direct relationship between the vertical growth of the helical Flux Rope and the ratio of the magnetic field components. Consequently, the parametrized magnetic field line is given by:

$$\boxed{\mathbf{s}(r,\varphi) = \left(\delta\, r\cos\varphi,\ \int_0^{\varphi} \frac{B^y}{B^\varphi}\, r\, d\varphi,\ r\sin\varphi\right)} \tag{4.70}$$

where the parameter domains for a Flux Rope of length L are:

$$r \in [0, d] \tag{4.71}$$
$$\varphi \in [0, 2\pi) \tag{4.72}$$

Pitch Angle

From expression (4.68), the pitch angle is directly defined as:

$$\alpha(r,\varphi) = \arctan\left(\frac{B^y(r,\varphi)}{B^\varphi(r)}\right) \tag{4.73}$$

4.9.6 Force-freeness

Recall that the Lorentz force describes the magnetic force experienced by a charged particle in the presence of a non-zero magnetic field and electric current. The magnitude of this force can vary depending on the region in which the particle is located, as the force depends on spatial position. To study this effect quantitatively, it is useful to define a measurable quantity that indicates in which regions the Lorentz force is dominant, and where its influence is negligible. The absolute value of the Lorentz force is defined as:

$$|\mathbf{j}\times\mathbf{B}| = |\mathbf{j}|\,|\mathbf{B}|\,\sin\omega \tag{4.74}$$

where ω represents the angle between the current density vector $\mathbf{j}$ and the magnetic field vector $\mathbf{B}$. When this angle is zero, the vectors are aligned, and the Lorentz force vanishes. In contrast, when there is a misalignment between $\mathbf{j}$ and $\mathbf{B}$, a non-zero force arises, reaching its maximum when the angle is $\omega = \pi/2$. Since the sine function is bounded between -1 and 1,$\sin\omega$ is defined as a quality factor that measures the degree of alignment between the magnetic field and the current. Values of $\sin\omega$ close to zero indicate that the field is locally more force-free, whereas higher values (closer to 1) mean stronger local misalignment and deviation from the force-free condition.

$$\sin(\omega) = \frac{|\mathbf{j}\times\mathbf{B}|}{|\mathbf{j}|\cdot|\mathbf{B}|} = \frac{\sqrt{g_{uv}f^u f^v}}{\sqrt{g_{ij}j^i j^j}\sqrt{g_{nm}B^n B^m}} \tag{4.75}$$

4.9.7 Magnetic Energy Density

Another important parameter to characterize within the Flux Rope is the spatial distribution of magnetic energy. The energy stored per unit volume in a differential volume element is given by:

$$u_{\text{mag}} = \frac{|\mathbf{B}|^2}{2\mu_0} \tag{4.76}$$

Mapping the magnetic energy density across the Flux Rope is crucial for understanding energetic processes, particularly in regions where the magnetic field strength is high. These regions are typically associated with greater energy storage and, potentially, with the onset of magnetic reconnection or other dynamic events. In this model, where the metric is non-trivial, this is accounted by computing the magnetic energy density as:

$$\boxed{u_{\text{mag}}(r,\varphi) = \frac{1}{2\mu_0}\left(B^y(r,\varphi)^2 + g_{\varphi\varphi}(r,\varphi)\, B^\varphi(r,\varphi)^2\right)} \tag{4.77}$$

Visualizing u_{mag} as a contour map reveals the regions of maximum energy concentration, and provides insights into where magnetic tension and pressure are strongest. This, in turn, can help identify zones more likely to host instabilities or energy release events.

4.10 Magnetic Helicity

This section is a particularly **challenging yet rewarding part of this work: the analytical determination of magnetic helicity** for the elliptical cylindrical model with non null radial current($j^r \neq 0$). Achieving a closed-form solution for the helicity in such a geometry is a significant result, as it involves a non-trivial combination of vector calculus and advanced partial differential equations tools that had to be mastered during the course of this research. But before diving into the derivation: what is helicity?

4.10.1 What is Magnetic Helicity?

Magnetic helicity is a topological invariant that quantifies the degree of linkage or twisting of magnetic field lines within a given volume. It is formally defined as:

$$\mathcal{H} = \iiint_V \mathbf{A} \cdot \mathbf{B}\, dV \tag{4.78}$$

where $\mathbf{A}$ is the magnetic vector potential such that $\mathbf{B} = \nabla \times \mathbf{A}$. Helicity has dimensions of Weber^2 and is conserved under ideal MHD.

4.10.2 Why is Helicity Useful?

In solar physics, for instance, helicity is used to trace the increase of magnetic energy in coronal mass ejections (CMEs) and flares to determine reconnection regimes. In this work, obtaining the magnetic helicity analytically not only validates the internal consistency of the model but also enables a deeper understanding of how geometry and magnetic topology are interconnected.

4.10.3 Solution to the helicity

Gauge Invariance

If the volume V is magnetically closed, the helicity H is said to be gauge invariant. In other words, the value of H remains unchanged under the transformation of the vector potential $\mathbf{A}$ as:

$$\mathbf{A}' = \mathbf{A} + \nabla\xi \tag{4.79}$$

where ξ is an arbitrary scalar function. To confine the magnetic field within the Flux Rope, a single assumption is made: at the endpoints of the Flux Rope, the axial magnetic field vanishes, since the poloidal field wraps around and the radial component is null. In this sense,

$$\lim_{y\to\pm\infty} B^y \to 0 \tag{4.80}$$

The long range behavior of the axial magnetic field is not necessary, only in its local behavior, since slice of length L of the Flux Rope is what will be studied. Where L is much smaller than the total length of the CME or Flux Rope.

Calculation of the Vector Potential A from B

The magnetic field $\mathbf{B}$ is related to the vector potential $\mathbf{A}$ by:

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{4.81}$$

This relationship, however, does not uniquely define $\mathbf{A}$, as any gradient $\nabla\xi$ of a scalar function can be added to $\mathbf{A}$ without changing the physical content of $\mathbf{B}$. To eliminate this ambiguity, a gauge condition is introduced.

Gauge Choice

To simplify the governing equations without modifying the physical content of the problem, the following condition on the divergence of the vector potential is imposed:

$$\boxed{\nabla \cdot \mathbf{A} = \frac{\partial_r\left(\delta r A^r\right) - \delta r \partial_r\left(g_{rr}A^r + g_{r\varphi}A^\varphi\right) + \delta r \partial_\varphi\left(A^\varphi(1+g_{\varphi\varphi}) + g_{\varphi r}A^r\right)}{\delta r}} \tag{4.82}$$

This condition suggests the existence of a scalar function $\mathfrak{D}(r,\varphi)$ satisfying:

$$\begin{aligned} \partial_\varphi \mathfrak{D}(r,\varphi) &= g_{rr}A^r(r,\varphi) + g_{r\varphi}A^\varphi(r,\varphi) \\ \partial_r \mathfrak{D}(r,\varphi) &= g_{\varphi r}A^r(r,\varphi) + g_{\varphi\varphi}A^\varphi(r,\varphi) \end{aligned} \tag{4.83}$$

This means that the function $\mathfrak{D}(r,\varphi)$ encodes directional derivatives that reconstruct the original components of the vector potential.

Resolution of the Differential Equation

In cylindrical coordinates (r, y, φ), the curl of $\mathbf{A}$ can be expressed as:

$$\begin{aligned} B^r \delta r &= -\partial_\varphi (g_{yy} A^y) \\ B^y \delta r &= \partial_\varphi (g_{rr} A^r + g_{r\varphi} A^\varphi) - \partial_r (g_{\varphi r} A^r + g_{\varphi\varphi} A^\varphi) \\ B^\varphi \delta r &= \partial_r (g_{yy} A^y) \end{aligned} \tag{4.84}$$

From the first equation in (4.84), $\partial_\varphi (g_{yy} A^y) = 0$, which implies that A^y is independent of φ, and hence can be written as:

$$A^y = A^y(r) \tag{4.85}$$

Using the third equation, an explicit expression for A^y is found:

$$\boxed{A^y(r) = \delta \int_0^r r B^\varphi(r)\, dr} \tag{4.86}$$

The remaining components A^r and A^φ are derived from the second equation in (4.84), making use of the variable substitution introduced above. This leads to:

$$B^y(r, \varphi)\delta r = \partial_\varphi^2 \mathfrak{D}(r, \varphi) - \partial_r^2 \mathfrak{D}(r, \varphi) \tag{4.87}$$

Solution of the PDE

This partial differential equation can be solved by decomposing the solution into a homogeneous and a particular part:

$$\mathfrak{D}(r, \varphi) = \mathfrak{D}_h(r, \varphi) + \mathfrak{D}_p(r, \varphi) \tag{4.88}$$

Homogeneous Solution $\mathfrak{D}_h(r, \varphi)$

The homogeneous part corresponds to the general solution of the wave equation and is given by:

$$\mathfrak{D}_h(r, \varphi) = g(r - \varphi) + h(r + \varphi) \tag{4.89}$$

In the absence of boundary conditions or an initial value problem, the functions g and h remain arbitrary.

Particular Solution $\mathfrak{D}_p(r, \varphi)$

To determine the particular solution, a Green's function approach is employed. The linear operator is defined as:

$$\mathfrak{L} = \partial_\varphi^2 - \partial_r^2 \tag{4.90}$$

The Green's function $G(r, \varphi; r', \varphi')$ satisfies:

$$\mathfrak{L}G(r, \varphi; r', \varphi') = \delta(r - r')\delta(\varphi - \varphi') \tag{4.91}$$

The solution $\mathfrak{D}_p(r, \varphi)$ is then given by:

$$\mathfrak{D}_p(r, \varphi) = \delta \int_0^r r'\, dr' \int_{-\infty}^{+\infty} G(r, \varphi; r', \varphi') B^y(r', \varphi')\, d\varphi' \tag{4.92}$$

Applying a Fourier transform yields:

$$\tilde{G}(s, \omega; r', \varphi') = \frac{e^{-isr'} e^{-i\omega\varphi'}}{s^2 - \omega^2} \tag{4.93}$$

Inverting the transform, we obtain:

$$G(r, \varphi; r', \varphi') = \frac{1}{4\pi^2} \int_{-\infty}^{+\infty} e^{i\omega(\varphi - \varphi')}\, d\omega \int_{-\infty}^{+\infty} \frac{e^{is(r-r')}}{s^2 - \omega^2}\, ds \tag{4.94}$$

Evaluating the inner integral via the residue theorem, the pole is placed in the upper half plane in order to enforce causality in the solution.

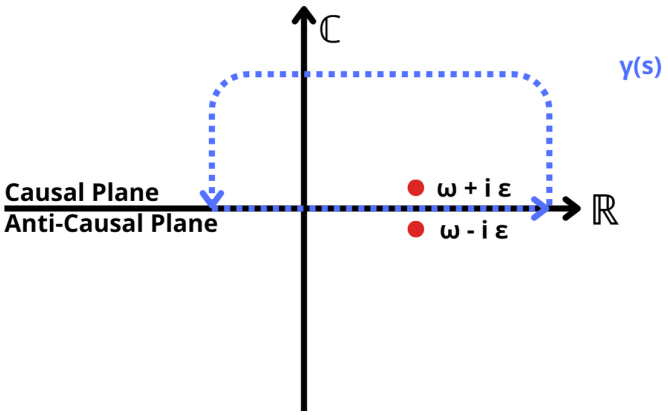


Figure 4.5: Causal contour $\gamma(s)$ enclosing the upper half-plane pole at $s = \omega + i\varepsilon$. The real axis is traversed from $-R$ to R, and the path is closed by a semicircle of radius R in the upper half-plane.

The parametrized contour $\gamma(s)$ is defined as a piecewise function composed of:

$$\gamma(s) = \lim_{R \to +\infty} \begin{cases} 4Rs - R, & 0 \le s \le \frac{1}{2} \quad \text{(real axis: from } -R \text{ to } R\text{)} \\ Re^{i\pi(2s-1)}, & \frac{1}{2} < s \le 1 \quad \text{(semicircular arc: from } R \text{ to } -R\text{)} \end{cases} \tag{4.95}$$

This leads to the following result:

$$\lim_{\varepsilon \to 0} \int_{-\infty}^{+\infty} \frac{e^{is(r-r')}}{s^2 - (\omega + i\varepsilon)^2}\, ds = \frac{\pi\, i}{\omega} e^{i\omega(r-r')} \tag{4.96}$$

This result holds under the assumption that $r - r' > 0$, which ensures that the integrand vanishes on the arc at infinity in the upper half plane and justifies the use of a causal contour. Taking the limit $R \to \infty$ and $\varepsilon \to 0^+$ guarantees that the pole at $s = \omega + i\varepsilon$ is enclosed within the contour, and the arc integral vanishes due to Jordan's lemma. This structure guarantees that the system's response is zero for $r < r'$, thus preserving the principle of causality. The remaining integal can be computed in a very similar way:

$$G(r, \varphi; r', \varphi') = \frac{i}{4\pi} \int_{-\infty}^{+\infty} \frac{e^{i\omega(r-r'+\varphi-\varphi')}}{\omega}\, d\omega \tag{4.97}$$

Solving the integral, one can find:

$$\boxed{G(r, \varphi; r', \varphi') = \frac{1}{2}\left[u(r' - r + \varphi' - \varphi) - u(r - r' + \varphi - \varphi')\right]} \tag{4.98}$$

This Green's function can be interpreted as a characteristic function that is non-zero only within a well-defined domain in the (r', φ') space. In other words, the function is supported on the region bounded by the two lines:

$$\varphi' = \varphi \pm (r - r') \tag{4.99}$$

This delimits a *causal cone* in the (r', φ') plane, where the influence of a source point (r', φ') can reach the observation point (r, φ). Outside this region, the Green's function vanishes, reflecting causality in the propagation of information or fields.

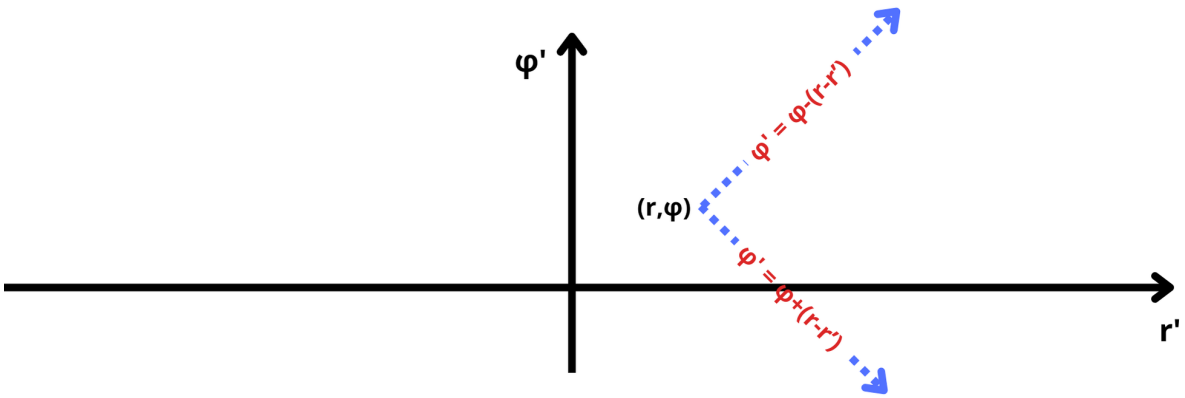


Figure 4.6: Support of the Green's function $G(r, \varphi; r', \varphi')$ in the (r', φ') plane for fixed (r, φ).

This results in the particular solution:

$$\mathfrak{D}_p(r, \varphi) = \delta \int_0^r \frac{r'}{2}\, dr' \int_{-\infty}^{+\infty} \left[u(r' - r + \varphi' - \varphi) - u(r - r' + \varphi - \varphi')\right] B^y(r', \varphi')\, d\varphi' \tag{4.100}$$

The Heaviside functions effectively restrict the domain of integration in φ' to the interval where the Green's function is non-zero. This allows us to rewrite the expression in compact form:

$$\mathfrak{D}_p(r,\varphi) = \frac{\delta}{2}\int_0^r r'\,dr' \int_{\varphi-(r-r')}^{\varphi+(r-r')} B^y(r',\varphi')\,d\varphi' \tag{4.101}$$

Complete Solution and Vector Potential Components

Combining both contributions, the total scalar potential is:

$$\boxed{\mathfrak{D}(r,\varphi) = g(r-\varphi) + h(r+\varphi) + \frac{\delta}{2}\int_0^r r'\,dr' \int_{\varphi-(r-r')}^{\varphi+(r-r')} B^y(r',\varphi')\,d\varphi'} \tag{4.102}$$

From this, the components of $\mathbf{A}$ can be written as:

$$\boxed{\mathbf{A} = \left\{ \begin{array}{l} \dfrac{g_{\varphi\varphi}}{\delta r}\,\partial_\varphi \mathfrak{D}(r,\varphi) - \dfrac{g_{r\varphi}}{\delta r}\,\partial_r \mathfrak{D}(r,\varphi) \\ \delta \displaystyle\int_0^r r B^\varphi(r)\,dr \\ \dfrac{g_{rr}}{\delta r}\,\partial_r \mathfrak{D}(r,\varphi) - \dfrac{g_{r\varphi}}{\delta r}\,\partial_\varphi \mathfrak{D}(r,\varphi) \end{array} \right\}_{(\mathbf{e}_r,\mathbf{e}_y,\mathbf{e}_\varphi)}} \tag{4.103}$$

Final Expression for the Helicity

Finally, the helicity is given by:

$$\boxed{\mathcal{H} = \int_0^{2\pi}\int_0^L\int_0^R \left[A^r g_{r\varphi} B^\varphi + A^y B^y + A^\varphi g_{\varphi\varphi} B^\varphi\right] \delta r\,dr\,dy\,d\varphi} \tag{4.104}$$

4.11 Versatility of the New Approach: Recovering Classical Models

One of the main advantages of having generalized previous models of Flux Rope geometry, originally constrained to circular cross sections or first order distortions such as the elliptic, is that they can be recovered by applying specific constraints that characterize them. These classical configurations are uniquely defined by their expressions for the magnetic field, current densities, and magnetic fluxes, from which all other relevant quantities are derived.

4.11.1 Recovering the Circular-Cylindrical Model of Nieves-Chinchilla et al. (2016) without Radial Current

To retrieve the circular cylindrical model originally proposed in Nieves-Chinchilla et al. (2016), two specific conditions must be imposed that restrict the generalized framework accordingly:

1. **Circular cross-section:** $\delta = 1$, which enforces a circular geometry by making the semi-major and semi-minor axes equal. In this case, the elliptic distortion is eliminated and the cross-section reduces to a perfect circle.
2. **Radial current-free condition:** As this model assumes no radial current, the condition of vanishing radial current density is imposed, explicitly given by:

$$\partial_\varphi j_2^\varphi(\varphi) = 0 \tag{4.105}$$

By applying these constraints to our generalized expressions, the circular-cylindrical (CC) model is recovered from the literature. For the fields:

Analytical Expressions for Current and Magnetic Field Components for Nieves-Chinchilla et al. (2016)

Current components:

$$j^r(r,\varphi) = 0 \tag{4.106}$$

$$j^y(r,\varphi) = \left(\sum_{k=0}^{+\infty} a_k^y\, r^k\right) \tag{4.107}$$

$$j^\varphi(r,\varphi) = \left(\sum_{k=1}^{+\infty} a_k^\varphi\, r^k\right) \tag{4.108}$$

Magnetic field components:

$$B^r(r,\varphi) = 0 \tag{4.109}$$

$$B^y(r,\varphi) = B_0^y - \mu_0 \left(\sum_{k=1}^{+\infty} a_k^\varphi \frac{r^{k+2}}{k+2}\right) \tag{4.110}$$

$$B^\varphi(r,\varphi) = -\mu_0 \frac{B_0^\varphi}{r^2} \left(\sum_{k=0}^{+\infty} a_k^y \frac{r^{k+2}}{k+2}\right) \tag{4.111}$$

4.11.2 Recovering the Elliptic-Cylindrical Model from Nieves-Chinchilla et al. (2018) Without Radial Current

To recover this model,the radial current constraint has to be imposed. Again, this means constraining the condition given in (4.50). The field components then become:

Analytical Expressions for Current and Magnetic Field Components for Nieves-Chinchilla et al. (2018)

Current components:

$$j^r(r,\varphi) = 0 \tag{4.112}$$

$$j^y(r,\varphi) = \left(\sum_{k=0}^{+\infty} a_k^y\, r^k\right) j_2^y(\varphi) \tag{4.113}$$

$$j^\varphi(r,\varphi) = \left(\sum_{k=1}^{+\infty} a_k^\varphi\, r^k\right) \tag{4.114}$$

Magnetic field components:

$$B^r(r,\varphi) = 0 \tag{4.115}$$

$$B^y(r,\varphi) = B_0^y - \mu_0\delta\left(\sum_{k=1}^{+\infty} a_k^\varphi \frac{r^{k+2}}{k+2}\right) \tag{4.116}$$

$$B^\varphi(r,\varphi) = -B_0^\varphi\mu_0\delta\,\frac{j_2^y(\varphi)}{g_{\varphi\varphi}}\left(\sum_{k=0}^{+\infty} a_k^y \frac{r^{k+2}}{\chi+k}\right) \tag{4.117}$$

4.12 Minimum Energy Configuration

It is now interesting to ask whether the model proposed represents a minimum energy state of the system. A minimum energy state is the configuration to which any physical system tends to evolve given enough time. Although it is modeling a Flux Rope at an arbitrary stage of its evolution, it is not expected to correspond to the minimum energy configuration.

4.12.1 Constrained Minimization of Magnetic Energy

The problem to be solved consists in minimizing the magnetic energy of the system under the constrain of a fixed magnetic configuration, characterized by a constant helicity. This assumption is physically justified by the fact that working in a superconducting regime, where magnetic reconnection is absent, and under time independent conditions, helicity is conserved. The constrained minimization problem using a Lagrange multiplier λ is formulated:

$$\delta\left[E - \lambda\mathcal{H}\right] = 0 \tag{4.118}$$

where E is the total magnetic energy and $\mathcal{H}$ is the magnetic helicity. Computing the variation of the functional as:

$$\delta\left[\frac{1}{2\mu_0}\int_V |\mathbf{B}|^2\, dV - \lambda\int_V \mathbf{A}\cdot\mathbf{B}\, dV\right] = 0 \tag{4.119}$$

The vector potential is perturbed in a way thet $\mathbf{A} \to \mathbf{A} + \delta\mathbf{A}$, and recall that $\mathbf{B} = \nabla \times \mathbf{A}$, so:

$$\delta E = \frac{1}{\mu_0} \int_V (\nabla \times \mathbf{A}) \cdot (\nabla \times \delta\mathbf{A})\, dV \tag{4.120}$$

$$\delta \mathcal{H} = \int_V (\delta\mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \nabla \times \delta\mathbf{A})\, dV \tag{4.121}$$

Using the identity:

$$\int_V \mathbf{A} \cdot \nabla \times \delta\mathbf{A}\, dV = \int_V \delta\mathbf{A} \cdot \nabla \times \mathbf{A}\, dV + \underbrace{\int_S (\mathbf{A} \times \delta\mathbf{A}) \cdot d\mathbf{S}}_{\text{boundary terms}} \tag{4.122}$$

and assuming appropriate boundary conditions, specifically, that the system is magnetically confined such that the variation vanishes at the boundary ($\delta\mathbf{A} \times \mathbf{n} = 0$ on S), the surface term disappears, and it is obtained:

$$\delta \mathcal{H} = 2 \int_V \delta\mathbf{A} \cdot \mathbf{B}\, dV \tag{4.123}$$

Similarly, for the energy:

$$\delta E = \frac{1}{\mu_0} \int_V (\nabla \times \mathbf{A}) \cdot (\nabla \times \delta\mathbf{A})\, dV = \frac{1}{\mu_0} \int_V \delta\mathbf{A} \cdot \nabla \times \nabla \times \mathbf{A}\, dV \tag{4.124}$$

Adding everything together:

$$\delta \left[E - \lambda \mathcal{H} \right] = \int_V \delta\mathbf{A} \cdot \left(\frac{1}{\mu_0} \nabla \times \nabla \times \mathbf{A} - 2\lambda \mathbf{B} \right) dV = 0 \tag{4.125}$$

Since $\delta\mathbf{A}$ is arbitrary, in the admissible space, and the integrand must vanish, the only possible solution is:

$$\nabla \times \nabla \times \mathbf{A} = 2\mu_0 \lambda \nabla \times \mathbf{A} \tag{4.126}$$

This leads to:

$$\boxed{\nabla \times \mathbf{B} = 2\mu_0 \lambda \mathbf{B}} \tag{4.127}$$

This equation describes a linear force-free field, meaning that the current density is aligned with the magnetic field:

$$\mathbf{j} = \frac{1}{\mu_0} \nabla \times \mathbf{B} = 2\lambda \mathbf{B} \tag{4.128}$$

Therefore, any magnetic structure that relaxes in a closed system without changing its helicity will tend towards a configuration where $\mathbf{j} \parallel \mathbf{B}$.

4.12.2 Interpretation for the Proposed Model

In the context of our Flux Rope model, if the system is allowed to relax, through processes such as magnetic reconnection, the magnetic configuration will naturally evolve toward the minimum energy state. The model presented here, however, does not assume a force-free condition a priori, as the Flux Rope may be observed in an arbitrary, non-relaxed configuration while propagating through interplanetary space. Therefore, the general solution proposed does not necessarily represent a minimum energy state, but it recovers such a configuration as a limiting case by enforcing the condition $\nabla \times \mathbf{B} = \alpha \mathbf{B}$. This is particularly relevant in the scenario where the Flux Rope gradually merges with the ambient solar wind, loses its distinct identity, and the Lorentz forces weaken significantly. In that asymptotic regime, the system may be well approximated by a force-free, relaxed configuration; consistent with the minimum energy state derived above.

Conclusion on the Applicability of Minimum Energy States

This analysis clarifies that this new model does not, and is not intended to, represent a minimum energy state. In fact, such an assumption would not hold as the Flux Rope propagates through the interplanetary medium, where the internal magnetic energy remains significantly higher than that of a relaxed configuration. In this regime, force-free fields fail to explain the observed magnetic confinement and the geometric structures such as those proposed in the model by Weiss et al. (2024). Therefore, **it is not possible to impose a constraint based on minimum magnetic energy in this context**. The magnetic field corresponding to a constant-helicity minimum energy configuration is a force-free solution, which is physically incompatible with the observed states of the system under consideration.

5 . Reconstruction of a Flux Rope from Synthetic Data

In this chapter, the reconstruction of the magnetic field structure and associated physical quantities of various Flux Rope models is presented, derived from the generalized formulation developed in this thesis. The reconstruction is carried out using averaged synthetic data. At the beginning of each section, the set of physical constraints applied to recover particular model configurations is specified, highlighting how different assumptions influence the resulting magnetic topology.

5.1 Circular Cross-Section Flux Rope Without Radial Current

This model, originally proposed by Nieves-Chinchilla et al. (2016), can be recovered, as discussed in previous sections, by setting $\delta = 1$, which ensures a circular cross-section, and by imposing that the poloidal current is independent of the poloidal coordinate. Under these assumptions, the generating functions of the model reduce to:

$$j^r(r) = 0 \tag{5.1}$$
$$j^y(r) = j_1^y(r) \tag{5.2}$$
$$j^\varphi(r) = j_1^\varphi(r) \tag{5.3}$$

That is, only two generating functions (degrees of freedom) remain in the model. These functions are typically constrained via in situ data fitting from space missions such as WIND, Solar Orbiter, Parker Solar Probe, and others. In this case, the generating functions proposed by Nieves-Chinchilla et al. (2016) with $m = n = \tau = 1$ are adopted, described by the following expressions:

$$j^r(r) = 0 \tag{5.4}$$
$$j^y(r) = \beta_1\, r \tag{5.5}$$
$$j^\varphi(r) = -\alpha_1\, r \tag{5.6}$$

By selecting representative average values for the coefficients, the following physical and

geometrical input parameters for the model are obtained:

Quantity	Expression / Value
Physical Parameters	
Magnetic Helicity	-6.42×10^{33} Wb^2
β_1	1.00×10^{-22} A/m^3
α_1	1.00×10^{-32} A/m^3
Distortion parameter δ	1.0
Axial Field Amplitude B_0^y	$1.33 \times 10^{-9}\pi$ T
Geometrical Parameters	
Minor Semi-axis R	1.00×10^{10} m
Major Semi-axis a	1.00×10^{10} m
Length L	1.00×10^{11} m
Analytical Expressions	
Magnetic Field $B^{\varphi}(r, \varphi)$	$-1.33 \times 10^{-29}\pi \cdot r$ T
Magnetic Field $B^{y}(r, \varphi)$	$\pi\,(1.33 \times 10^{-9} - 1.33 \times 10^{-39} \cdot r^3)$ T
Current Density $j^{r}(r, \varphi)$	0 A/m^2
Current Density $j^{\varphi}(r, \varphi)$	$-1.00 \times 10^{-32} \cdot r$ A/m^2
Current Density $j^{y}(r, \varphi)$	$1.00 \times 10^{-22} \cdot r$ A/m^2

Table 5.1: Complete list of input parameters and analytical expressions of the model for the Circular Cylindrical case developed by Nieves-Chinchilla et al. (2016).

With the synthetic data obtained, it is possible to fully characterize the Flux Rope, including its vector fields and the physical quantities previously described.

5.1.1 Magnetic Fields in the Cross-Sectional Plane

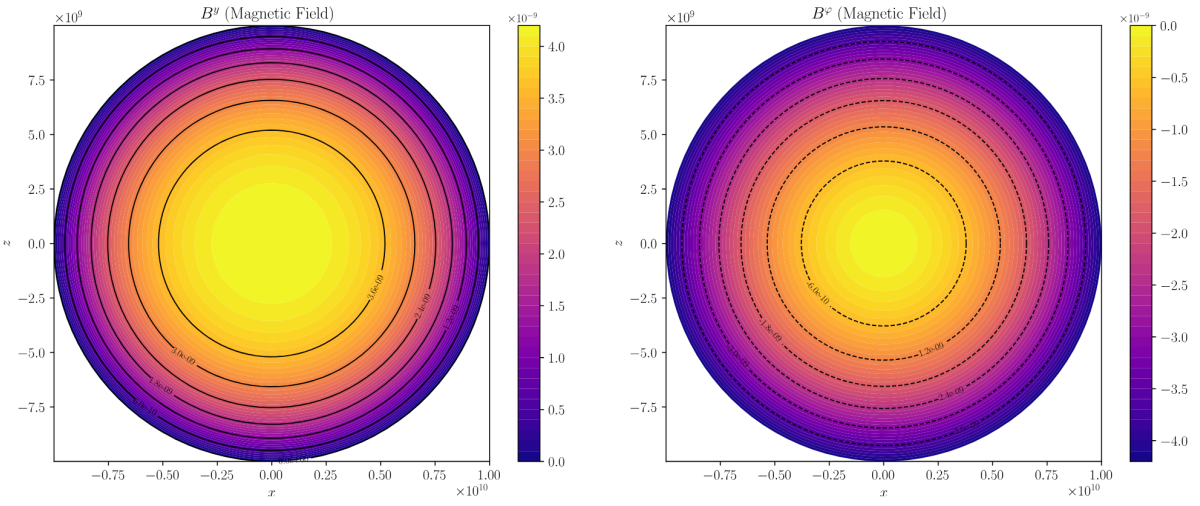


Figure 5.1: Magnetic field components in the slice of the synthetic circular cylindrical Flux Rope. Contour lines reflect the radial symmetry.

The magnetic fields exhibit a radial structure, directly resulting from the geometry of the system. The contour lines of both field components form concentric circles across the entire domain, more specifically:

- ◇ **Axial field**: maximal at the core and progressively decreases toward the boundaries, also displaying a radial distribution.
- ◇ **Poloidal field**: consistently negative throughout the domain, indicating a clockwise orientation. It reaches its maximum intensity at the center and decreases toward the edges, forming a purely radial field.

5.1.2 Current Density Fields in the Cross-Section

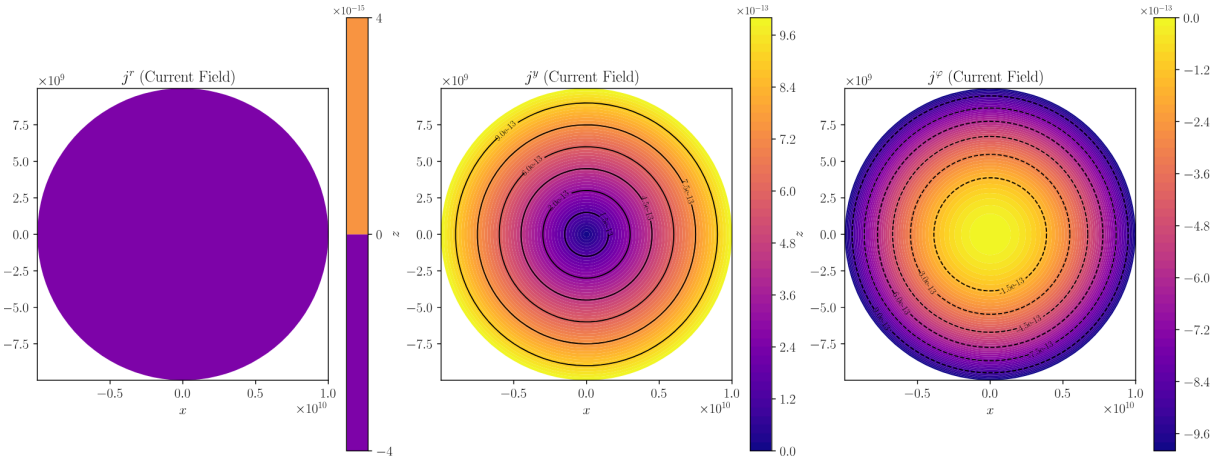


Figure 5.2: Current density components in the slice of the synthetic circular cylindrical Flux Rope.

Since the poloidal current depends solely on the radial coordinate, this condition is both necessary and sufficient to ensure the absence of radial current in the cross section, confirmed by its null value in the figure. The remaining components depend exclusively on the radial coordinate as well.

- ◇ **Axial current**: maximum at the edges and minimum at the center, with circular contour lines.
- ◇ **Poloidal current**: minimum at the boundaries, with a radial distribution.

5.1.3 Lorentz Forces in the Cross-Section

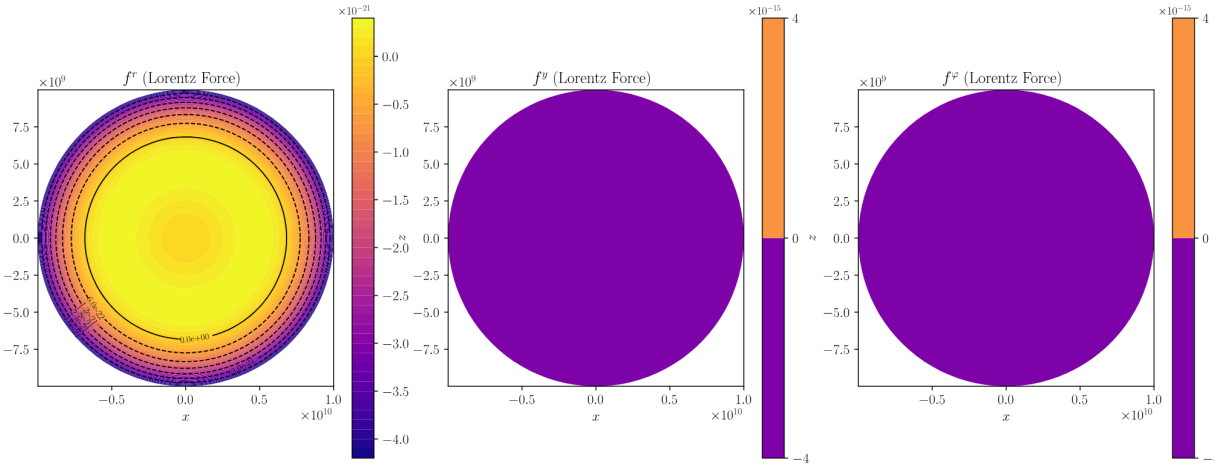


Figure 5.3: Radial component of the Lorentz force in the slice of the synthetic circular cylindrical Flux Rope. The force is negative and centrally directed, opposing the pressure gradient and reinforcing magnetic confinement. It is minimal near the core and reaches its peak at the edge of the structure.

As expected, the only significant Lorentz force present in the cross-sectional plane is a centrally directed radial force that encloses the Flux Rope. This negative radial component reflects the balance between the outward pressure gradient, which tends to expand the structure, and the magnetic forces, which act to confine it. The force is nearly zero at the center and becomes concentrated near the boundary, where confinement is strongest.

5.1.4 Perpendicular Current, Force-Freeness, and Magnetic Energy Density

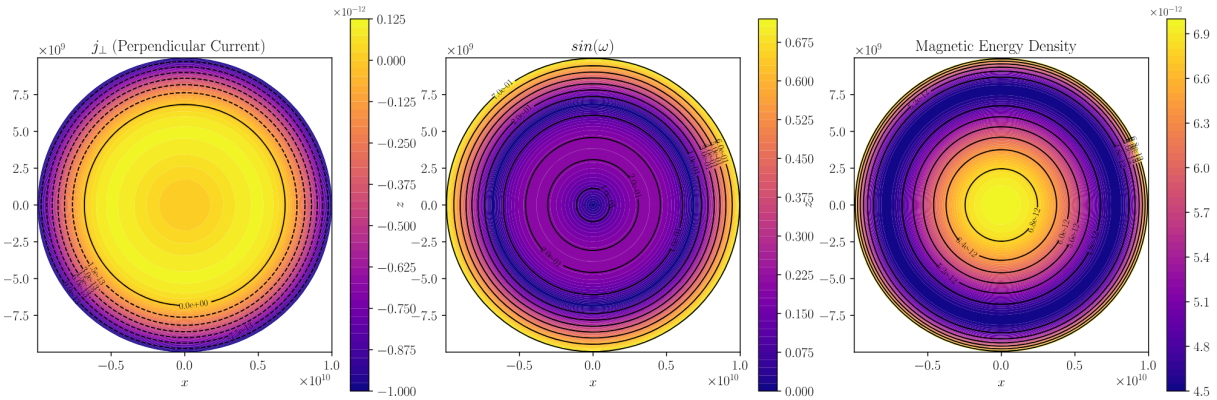


Figure 5.4: Distribution of perpendicular current $j_\perp$, misalignment factor $\sin(\omega)$, and magnetic energy density in the synthetic circular cylindrical Flux Rope. The core exhibits strong force-freeness (low misalignment), while the edges show significant deviation from the force-free condition. The energy density peaks at the center and also forms high-energy annular structures near the boundary.

One of the most notable features in these plots is the strong force-freeness observed at the core of the Flux Rope, with increasing misalignment toward the boundaries. This structure is consistent with the behavior of the radial Lorentz force discussed earlier and reflects the magnetic confinement of the structure. Regarding magnetic energy density, it is observed that it is primarily concentrated in the central region, with additional high energy annular rings forming near the outer boundary.

5.1.5 3D Simulation of Magnetic Field Lines Within the Flux Rope

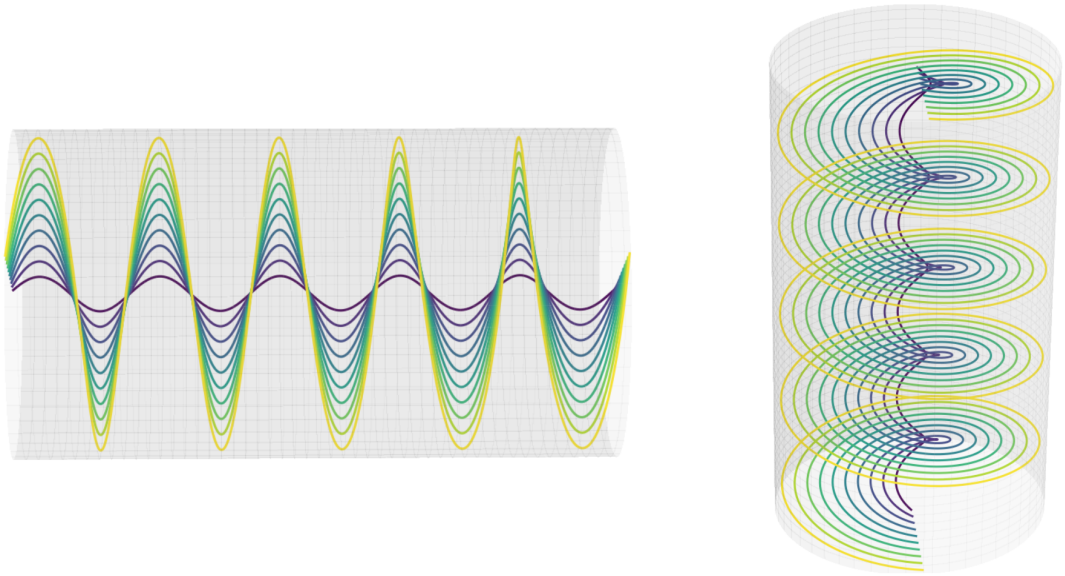


Figure 5.5: Top and side views of the confined synthetic circular cylindrical Flux Rope, illustrating the axial propagation of magnetic field lines.

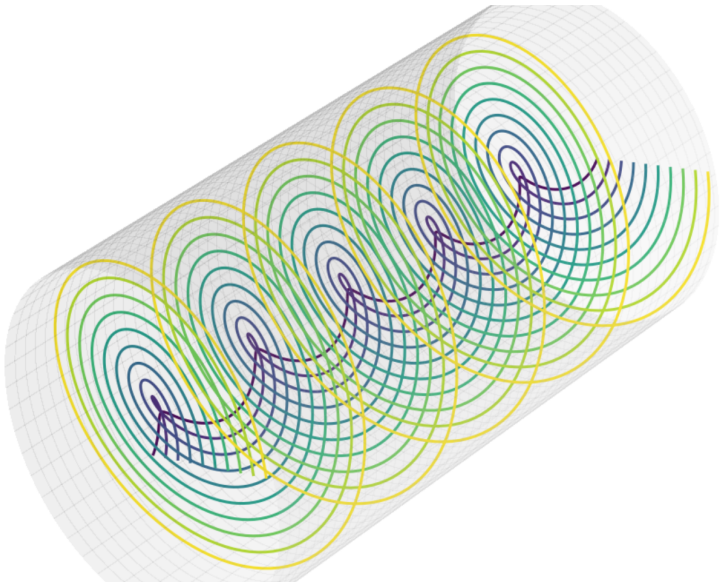


Figure 5.6: Oblique perspective of the Flux Rope showing the helical structure of the internal magnetic field lines.

The images illustrate the internal evolution of the magnetic field within the Flux Rope, specifically its propagation along the axial y-direction. One can note that the magnetic structures form helices whose geometry depends on the radial distance from the central axis. The pitch angle remains constant over the magnetic spline.

5.1.6 Pitch Angle Variation of the Field Lines

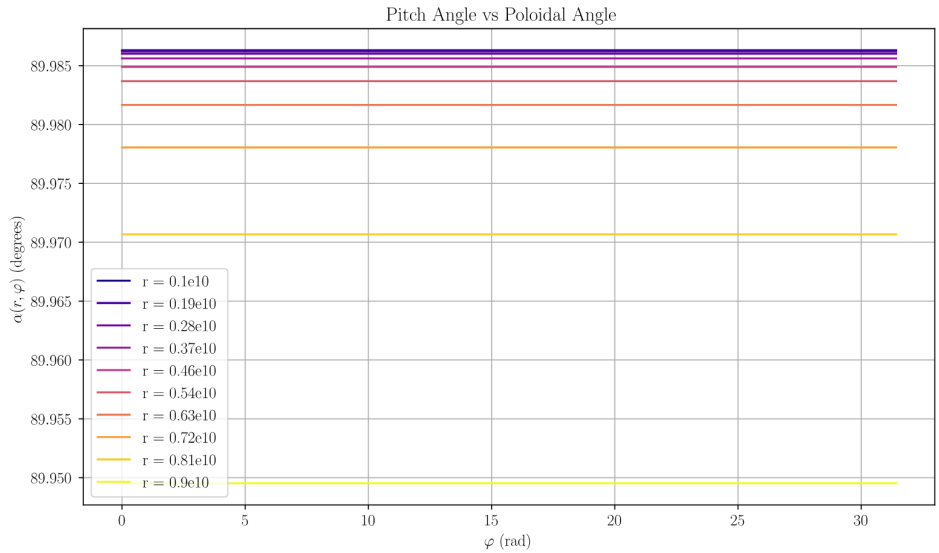


Figure 5.7: Pitch angle $\alpha(r, \varphi)$ as a function of poloidal angle φ for various radial positions within the Flux Rope. The angle remains effectively constant with respect to φ but decreases with radial distance.

The plot shows the pitch angle $\alpha(r, \varphi)$ in degrees as a function of the poloidal angle φ for a range of radial positions within the Flux Rope. Each colored curve corresponds

to a different fixed value of r, as indicated in the legend. A key observation is that for each radius, the pitch angle remains constant along φ. However, when comparing across different radii, the pitch angle decreases monotonically with the radial position. Specifically:

- ◇ Near the core ($r = 0.1 \times 10^{10}$ m), the pitch angle is approximately 89.985°.
- ◇ At the outer boundary ($r = 0.9 \times 10^{10}$ m), the angle decreases to roughly 89.950°, indicating more tightly wound helices with a strong poloidal component.

5.1.7 Synthetic trajectories of the satelite and field magnitude along them

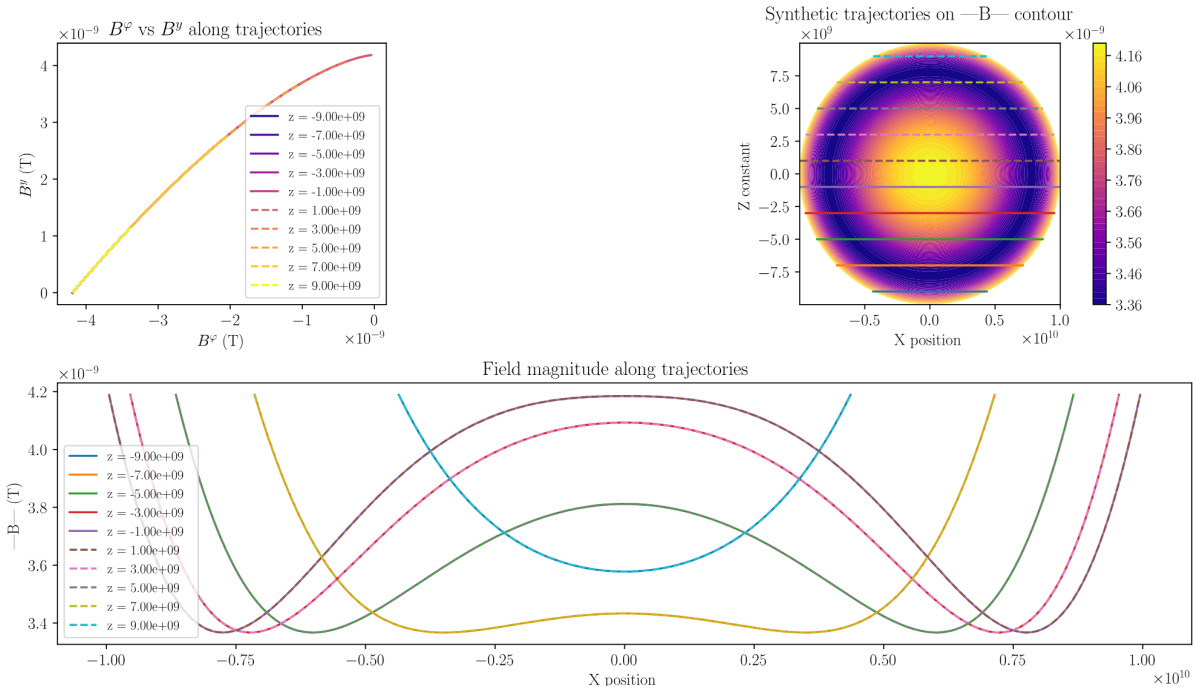


Figure 5.8: Synthetic spacecraft trajectories and magnetic field magnitude measurements in the circular cylindrical model. The lower plot shows the magnetic field magnitude along the x-axis for constant z-slices. The upper-left panel displays B^φ vs. B^y, highlighting characteristic features useful for in-situ identification of Flux Ropes.

In this plot, synthetic trajectories generated by a satellite crossing the y = const plane are observed. For a fixed value of z, the satellite moves along the x-axis and measures the magnetic field magnitude, which is displayed in the lower panel. Since the magnetic field distribution is purely radial, the resulting profiles are symmetric with respect to the origin. Consequently, for constant values of $|z|$, the magnetic field magnitude keeps the shape. In the upper-left corner, a plot of B^φ versus B^y is shown, forming segmented aligned curves. These curves are particularly useful when analyzing in-situ data, as they allow for the identification of typical Flux Rope signatures and internal magnetic structures, which can then be used by observers to characterize the event.

5.2 Elliptical-Section Flux Rope without Radial Current

In this case, the same generating functions used in the previous circular cylindrical model are employed. However, it is now observed that both the currents and the fields are modified due to the elliptical geometry. This is a direct consequence of the metric and the flux conservation discussed in 4.9.2.

Quantity	**Expression / Value**
	Physical Parameters
Magnetic Helicity	-2.26×10^{32} Wb^2
β_1	1.00×10^{-22} A/m^3
α_1	1.00×10^{-32} A/m^3
Distortion parameter δ	0.70
Axial Field Amplitude B_0^y	$9.33 \times 10^{-10}\,\pi$ T
	Geometrical Parameters
Minor semi-axis R	7.00×10^{9} m
Major semi-axis a	1.00×10^{10} m
Length L	1.00×10^{11} m
	Analytical Expressions
Magnetic Field $B^{\varphi}(r,\varphi)$	$-4.19 \times 10^{-29}\, r$ T
Magnetic Field $B^{y}(r,\varphi)$	$\pi\left(9.33 \times 10^{-10} - 9.33 \times 10^{-40}\, r^3\right)$ T
Current Density $j^{r}(r,\varphi)$	0.00 A/m^2
Current Density $j^{\varphi}(r,\varphi)$	$-1.00 \times 10^{-32}\, r$ A/m^2
Current Density $j^{y}(r,\varphi)$	$\dfrac{1.71 \times 10^{-15}\, r\left(2.18 \times 10^{-07} - 4.46 \times 10^{-08} \sin^2\varphi\right)}{\pi}$ A/m^2

Table 5.2: Complete list of physical quantities, geometric values and analytical expressions for the Elliptical Cylindrical case with null radial current.

5.2.1 Magnetic Fields in the Cross-Sectional Plane

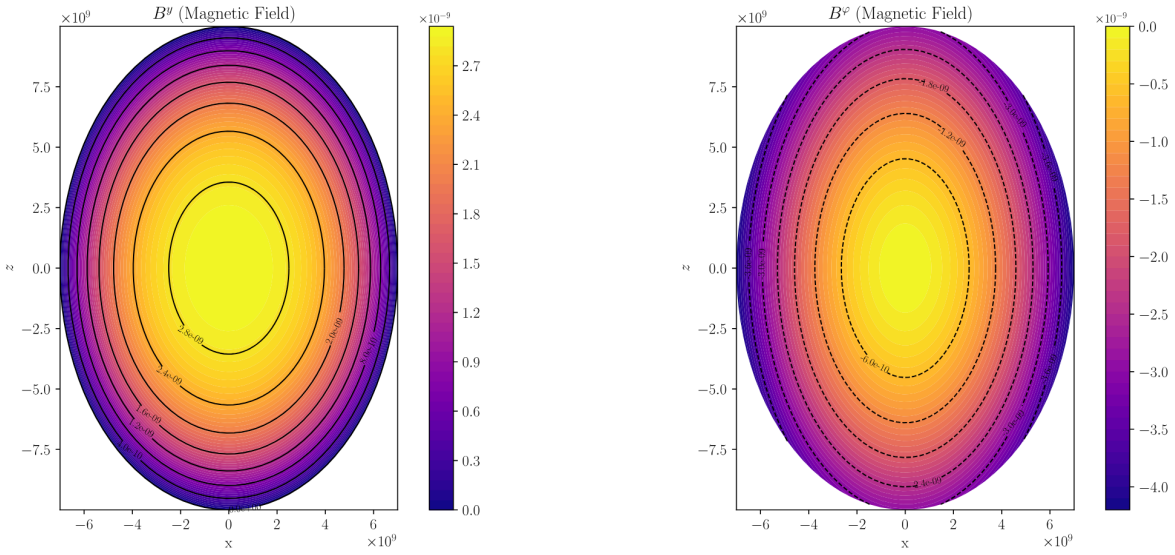


Figure 5.9: Magnetic field components in the transverse slice of the synthetic elliptical cylindrical Flux Rope.

The magnetic field lines show a similar pattern: both components decrease from a maximum value at the center to a minimum value at the boundaries, resembling those observed in the previous circular cylindrical model. However, some differences are found in each field component:

- ◇ **Axial field:** The axial component displays a purely radial distribution with concentric elliptical contour lines. The plot shown corresponds to the quantity $B^y_{\text{plot}} = B^y(r) \cdot \sqrt{g_{yy}}$, which remains unaffected due to the unitary scale factor in this coordinate.
- ◇ **Poloidal field:** This component also shows a purely radial profile in its contravariant form, however, the plotted contour lines appear distorted. This is because, despite the poloidal field being defined as $B^\varphi(r)$, the plotted quantity includes the coordinate scale factor, yielding $B^\varphi_{\text{plot}} = B^\varphi(r) \cdot \sqrt{g_{\varphi\varphi}}$, which introduces an angular and radial dependence in the representation.

5.2.2 Current Density Fields in the Cross-Section

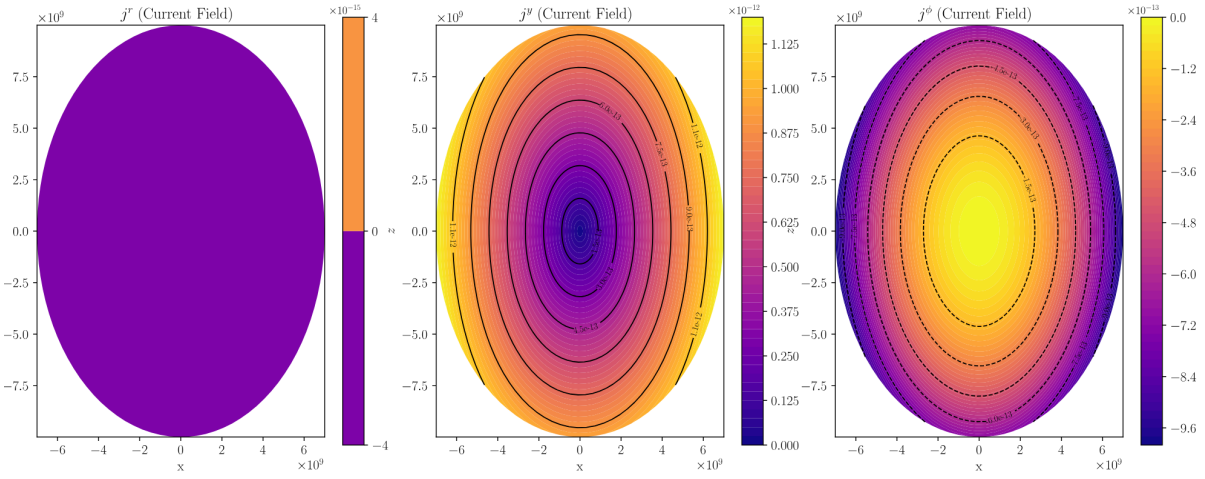


Figure 5.10: Current density components in the transverse slice of the synthetic elliptical cylindrical Flux Rope. The plot confirms the absence of radial current due to the radial-only dependence of the poloidal current.

In this case, although the generating functions are the same as those used in the circular-cylindrical model, the distortion introduced by the elliptical geometry causes the axial current to acquire a poloidal character in order to preserve the condition $B^\varphi = B^\varphi(r)$. Similarly as in the magnetic fields, although the poloidal current is defined with purely radial dependence, the presence of the coordinate scale factor introduces a combined radial–poloidal dependence in its plotted representation. Additionally, the behavior of the current components shows the following characteristics:

- ◇ **Axial current:** It reaches a minimum at the center and increases toward the edges.
- ◇ **Poloidal current:** It reaches a maximum at the center and decreases toward the edges.

5.2.3 Lorentz Forces in the Cross-Section

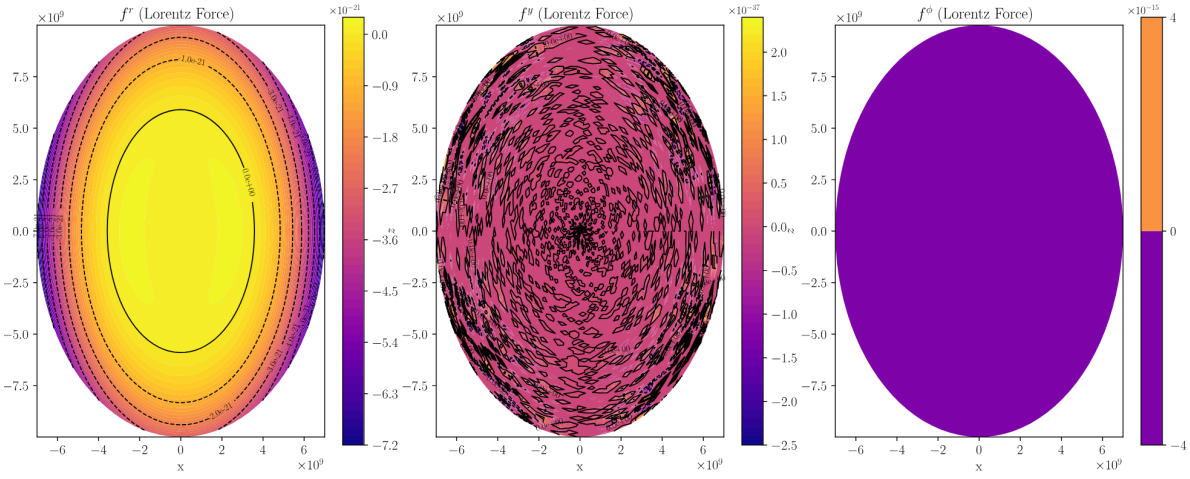


Figure 5.11: Radial component of the Lorentz force in the transverse section of the synthetic elliptical cylindrical Flux Rope.

The plot shows a non-force-free configuration with only a radial force component present. The apparent marks in the axial component are due to resolution limitations of the visualization library and do not indicate any actual force component. The radial force behavior closely recovers the behavior displayed by the circular-cylindrical model, showing a central force that confines the Flux Rope. However, it is now evident that the radial force increases significantly at the more elongated edges of the ellipse. This arises because, deforming a circular cross section into an elliptical, inherently requires applying greater force along the elongated sides and the figure clearly illustrates this behavior; just as trying to deform a plastic tube.

5.2.4 Perpendicular Current, Force-Freeness, and Magnetic Energy Density

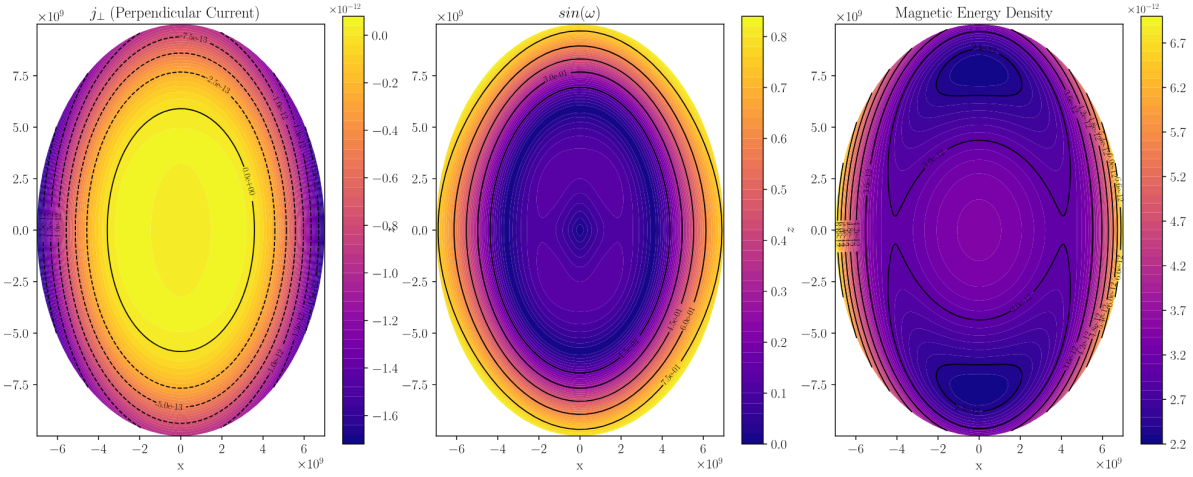


Figure 5.12: Distribution of perpendicular current $j_\perp$, misalignment factor $\sin(\omega)$, and magnetic energy density in the synthetic elliptical cylindrical Flux Rope. The core exhibits strong force-freeness, while the edges show significant deviation from the force-free condition. The energy density is concentrated at the boundaries.

Despite the distortion of the cross section, which no longer exhibits a circular shape, certain features preserve their essence. One such characteristic is that if the core of the circular cylindrical configuration is force-free, then the core of the elliptical configuration will also exhibit force-freeness. As can be seen from the middle plot, a strong force-free behavior in the core while confinement forces appear near the boundary. The magnetic energy distribution also supports this as can be seen stronger at the elongated edges, where the Flux Rope must exert greater confinement, whereas the central region shows a lower magnetic energy, consistent with a less intensive force configuration.

5.2.5 Pitch Angle Variation of the Field Lines

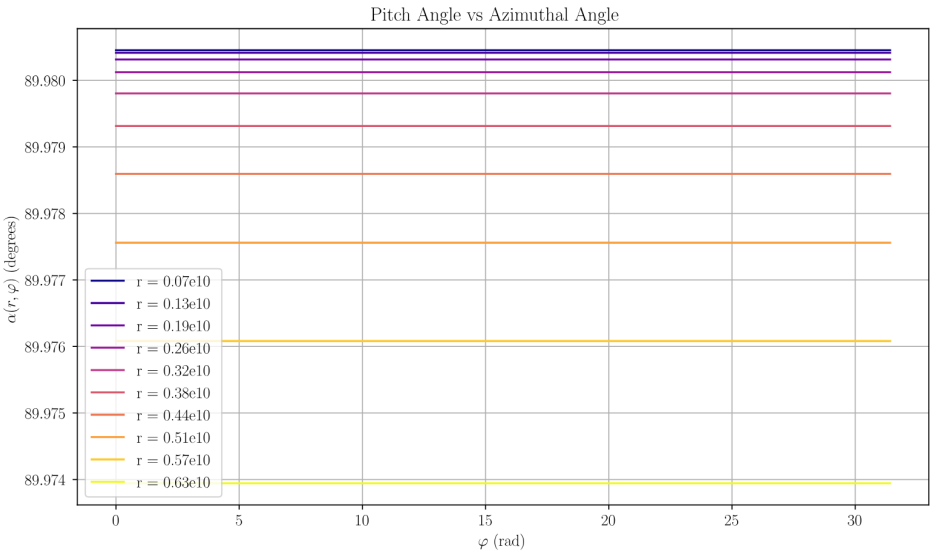


Figure 5.13: Pitch angle $\alpha(r, \varphi)$ as a function of poloidal angle φ for various radial positions within the Flux Rope.

The figure displays the pitch angle $\alpha(r, \varphi)$ as a function of the azimuthal angle φ for various radial positions within the Flux Rope. As observed, the pitch angle remains constant across all azimuthal values and exhibits only a slight radial variation. As seen in the circular-cylindrical, this shows that the field lines from perfect spirals with constant pitch angle along the trajectory.

5.2.6 Synthetic trajectories of the satelite and field magnitude along them

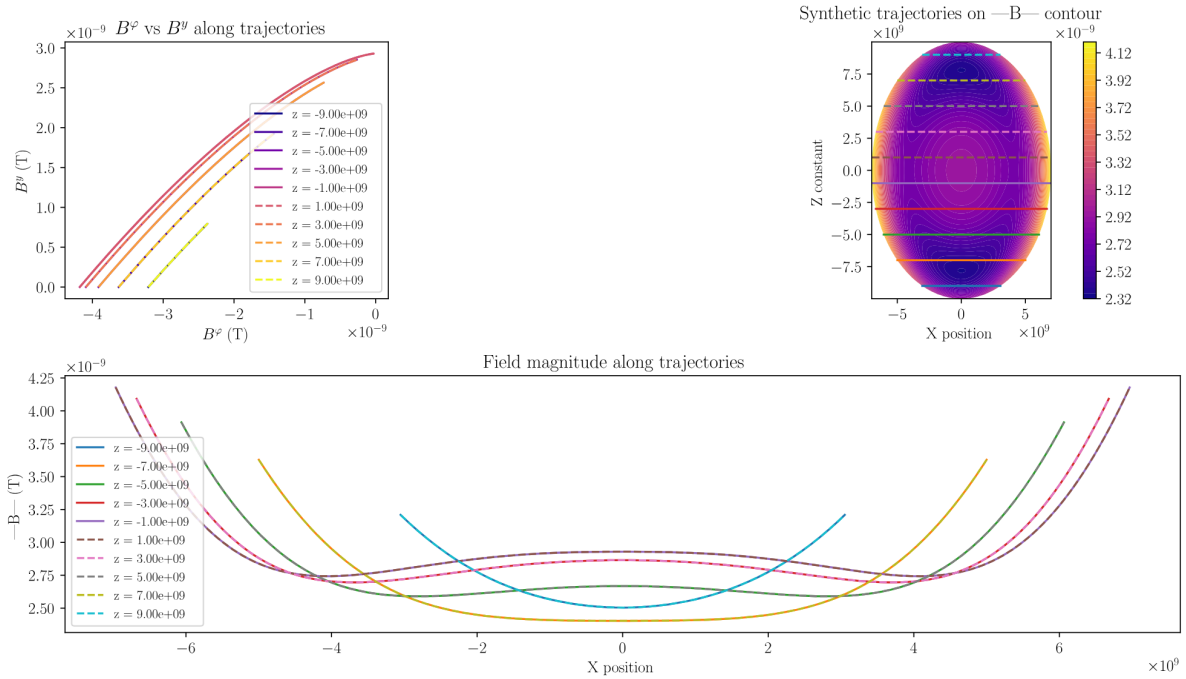


Figure 5.14: Synthetic spacecraft trajectories and corresponding magnetic field magnitude measurements in the elliptical cylindrical model. The lower plot shows the magnetic field magnitude along the x-axis for constant z-slices. The upper-left panel displays B^{φ} vs. B^{y}.

The plot illustrates the synthetic trajectory of a virtual spacecraft traveling through the same path used in the circular-cylindrical case. The motion is constrained to planes of constant z and y, while the x coordinate varies. The field components extracted along these paths show complete symmetry with respect to the origin, as expected from the only radial nature of the fields. However, a key difference arises in the B^{y} vs B^{φ} representation (top-left panel), where the curves no longer collapse into a single aligned trajectory. Instead, they form distinct paths depending on the spacecraft's z-position. This is a consequence of the geometric deformation introduced by the elliptical cross-section, which breaks the perfect symmetry found in the circular case. Despite this, the general trend of the magnetic field magnitude remains qualitatively similar to the circular-cylindrical configuration: it increases with radial distance and maintains a consistent concave profile across the synthetic trajectories.

5.3 Elliptical-Section Flux Rope with Radial Current: Interaction Between Flux Ropes

In this section, a synthetic simulation is carried out for a Flux Rope with elliptical cross-section and a **radial current component**. This approach represents not only a **novel analytical step beyond existing models, but also an attempt to explain certain phenomena that current theoretical frameworks fail to capture**. This direction was initially suggested by Dr. Teresa Nieves-Chinchilla at the beginning of my research stay. The goal of this section is to **simulate the interaction between two magnetic Flux Ropes**. To do so, a new approach is introduced by incorporating a non-zero radial current. If one Flux Rope is interacting with another in the interplanetary medium, it is expected to observe a localized alteration in the poloidal magnetic field, the remaining degree of freedom, which leads to the generation of a radial current. This radial current is interpreted as the physical manifestation of the compression exerted by one Flux Rope onto another in a confined region, producing a net radial flow of particles toward the interacting Flux Rope. In this context, the previously defined generating functions for $j_1^y(r)$ and $j_1^{\varphi}(r)$ are used while a perturbation is introduced in the poloidal component of the poloidal current, centered around $\varphi = \pi$. To model this perturbation, a trial Gaussian function of the following form is defined:

$$j_2^{\varphi}(\varphi) = e^{-\sigma(\varphi-\pi)^2} \tag{5.7}$$

where σ is a parameter that controls the localization of the perturbation, larger values of σ yield narrower, more sharply concentrated profiles.

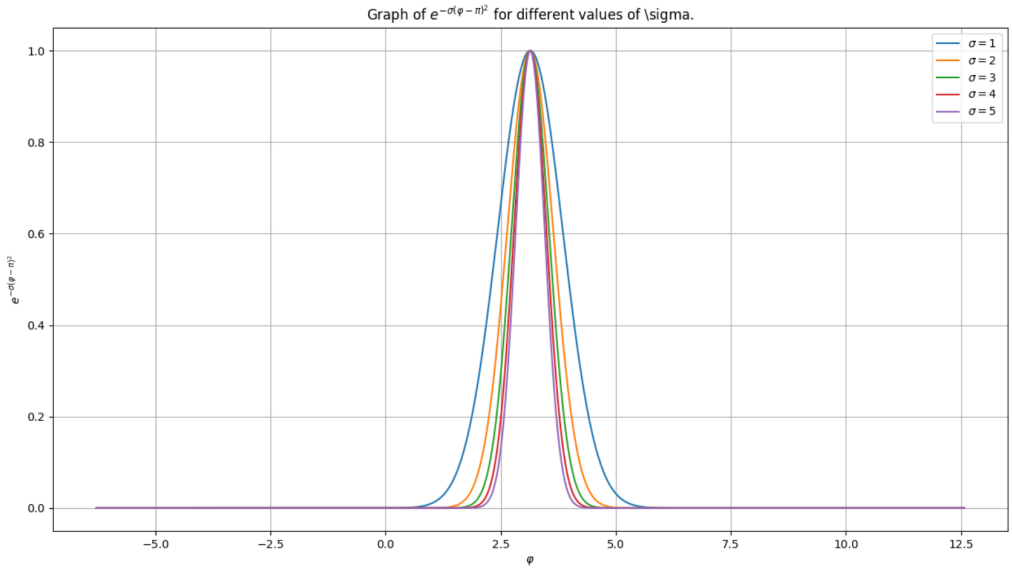


Figure 5.15: Gaussian test function used to simulate a localized poloidal current perturbation centered at $\varphi = \pi$. The width of the perturbation is controlled by the parameter σ and its maximum aplitude is 1.

To incorporate this function into the framework described in Section 4.7, it must be expressed in terms of its Fourier series. Therefore, the Gaussian function is first periodized with $\sigma = 5$, resulting in the following profile:

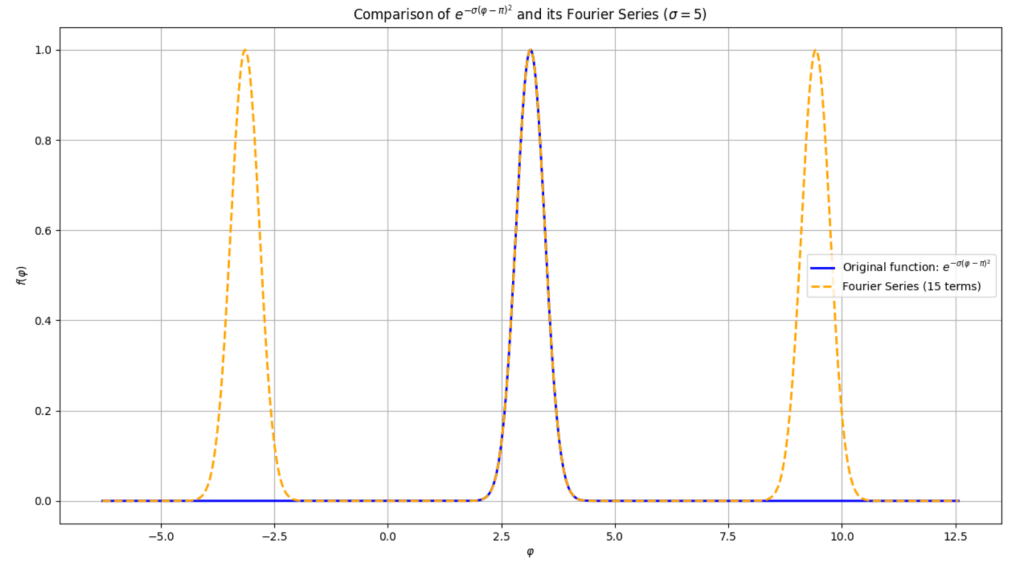


Figure 5.16: Periodized version of the Gaussian perturbation with $\sigma = 5$, used for its expansion in Fourier series.

With this expression, the analytical magnetic fields and currents, together, with the rest of the generating functions seen previously and collected in Nieves-Chinchilla et al. (2018) with $\tau = m = n = 1$ are obtained.

Quantity	Expression / Value
Physical Parameters	
Magnetic Helicity	$\sim -1 \times 10^{32}$ Wb^2
β_1	1.00×10^{-22} A/m^3
α_1	1.00×10^{-32} A/m^3
Distortion parameter δ	0.70
Axial Field Amplitude B_0^y	$9.33 \times 10^{-10}\,\pi$ T
Geometrical Parameters	
Minor semi-axis R	7.00×10^9 m
Major semi-axis a	1.00×10^{10} m
Length L	1.00×10^{11} m

Table 5.3: Complete list of physical quantities and geometric values for the Elliptical Cylindrical case with radial current (note that the generating functions as well as fields are too long to fit in a table since include 15 terms of a Fourier series).

5.3.1 Magnetic Fields in the Cross-Sectional Plane

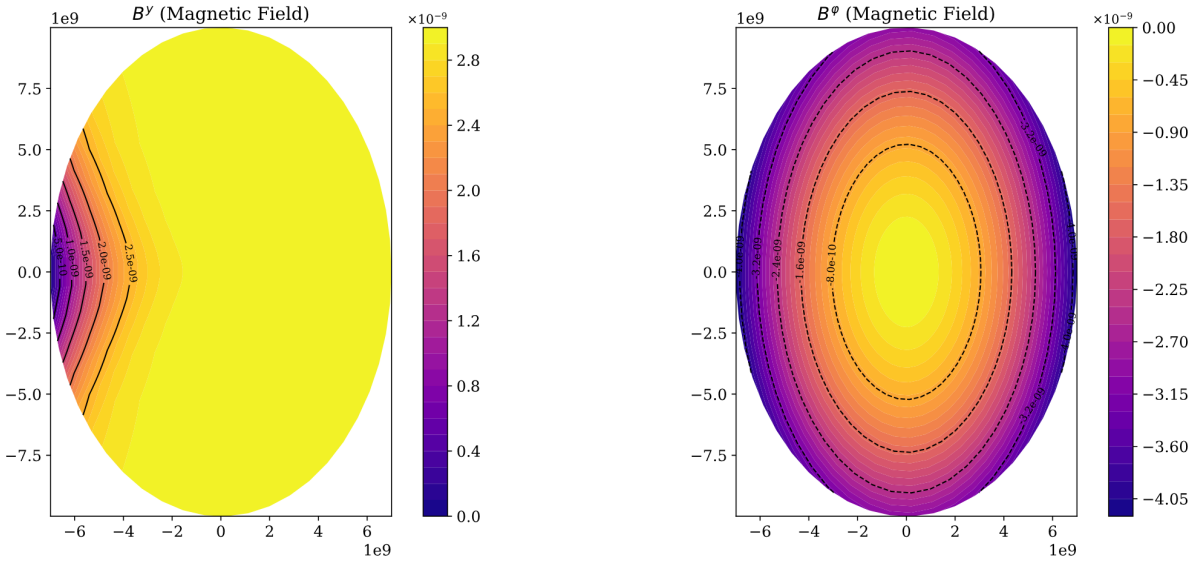


Figure 5.17: Magnetic field components in the transverse slice of the synthetic elliptical cylindrical Flux Rope with radial current.

The figures display the magnetic field components of the Flux Rope in the plane of $y = \text{const}$, with the applied perturbation included. As can be seen, both components exhibit dependence on both the radial and azimuthal coordinates. The following features are particularly highlighted:

- ◇ **Axial magnetic field:** Since the axial field component is directly influenced by the poloidal current through equation (4.31), it is the **only component affected** by the applied perturbation. A localized **magnetic anomaly emerges** around $\varphi = \pi$, caused by the Gaussian perturbation introduced earlier. This leads to

a **redistribution of the axial magnetic field** throughout the cross-section in order to support the injected radial current. As a result, the axial field exhibits a nearly constant maximum value across most of the domain, except in the vicinity of the perturbation, where a local minimum is observed.

◇ **Poloidal magnetic field:** Since the axial current remains unchanged from the previous synthetic simulation, the **poloidal magnetic field remains invariant**.

5.3.2 Current Density Fields in the Cross-Section

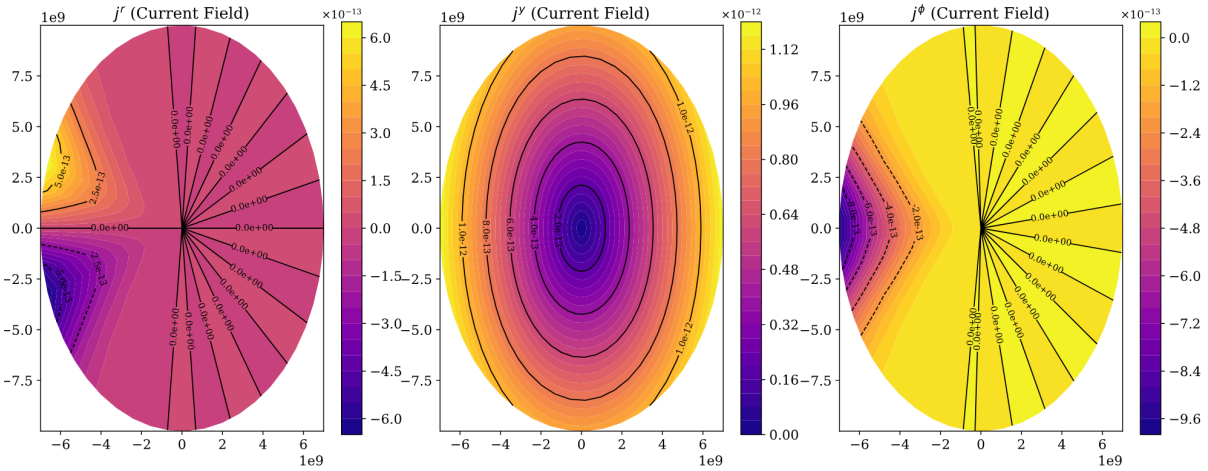


Figure 5.18: Current density components in the transverse slice of the synthetic elliptical cylindrical Flux Rope.

The formation of a radial current in the cross section is clearly observed. A closer analysis reveals the following behavior:

◇ **Radial current component:** This component results from the application of an operator to the poloidal current. The **presence of a non-constant poloidal component in the poloidal current leads to the emergence of this radial current**. A closer look reveals a region of positive current and a symmetrically placed region of negative current with respect to $z = 0$. This is a direct consequence of the current conservation law, as demonstrated in Section 4.8. In **a stationary configuration, the system must avoid monopole current sources**, implying that any current entering the Flux Rope must exit elsewhere. **This behavior is reflected in the antisymmetric sign change of the current distribution**.

◇ **Axial current component:** This component remains unchanged with respect to the previous model, as no modifications were applied to the axial current profile.

◇ **Poloidal current component:** This component is directly affected by the applied Gaussian perturbation. It is worth noting that **the profile remains locally flat outside the perturbation due to the chosen control parameter** σ. Since the maximum amplitude of the perturbation is of unit magnitude, it introduces only a small scale localized distortion into the system. The result is an almost negligible

current distribution outside the perturbed region, with a **concentrated feature centered at** $\varphi = \pi$, representing the effect of the Flux Rope interaction.

5.3.3 Lorentz Forces in the Cross-Section

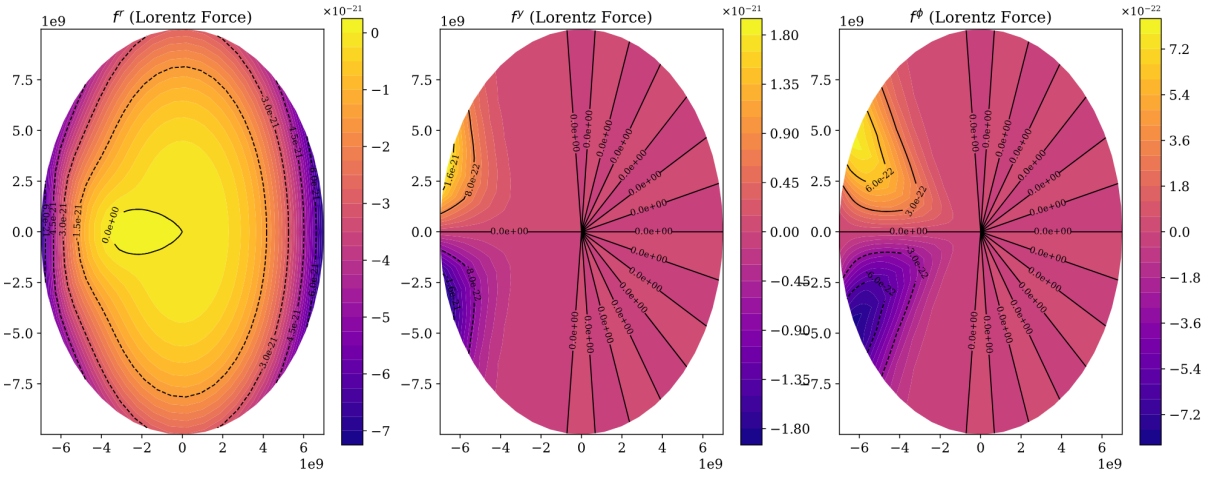


Figure 5.19: Radial component of the Lorentz force in the transverse section of the synthetic elliptical cylindrical Flux Rope.

Since the current density and magnetic field have both non-zero components in all directions, it is natural for Lorentz forces to emerge in this configuration. Examining each component in more detail reveals that:

- ◇ **Radial Lorentz force:** This component shows the **compressive behavior** already observed in the previous models, with a maxima located along the more elongated parts of the ellipse. However, the contour lines are now noticeably distorted toward the perturbation site, leading to a **reduced force contribution in the central region directed toward the perturbation**.
- ◇ **Axial Lorentz force:** This force introduces a **tangential stress along the Flux Rope** in the perturbation area that could potentially result in axial wrapping of the structure or in the development of instabilities that are beyond the scope of this work. Outside the perturbed region, the magnitude of the axial force remains negligible.
- ◇ **Poloidal Lorentz force:** This component produces a **compressive effect localized at the perturbation site**. The force acts poloidally toward $z = 0$, compressing it. This configuration tends to generate a **localized bulging or wrapping of the Flux Rope around the perturbation**. As in the axial case, the magnitude of the force outside the perturbation zone remains insignificant.

5.3.4 Perpendicular Current, Force-Freeness, and Magnetic Energy Density

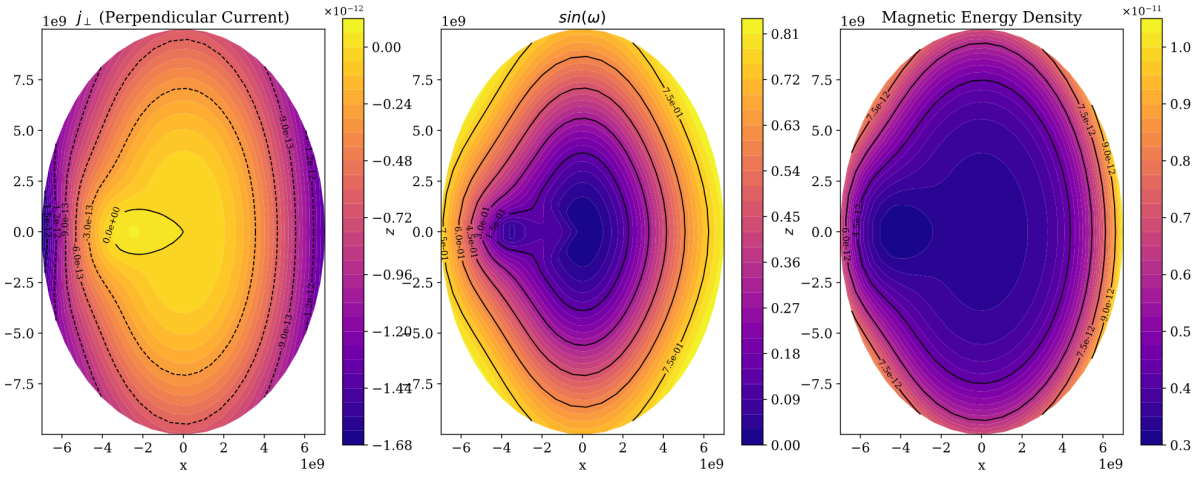


Figure 5.20: Distribution of perpendicular current $j_\perp$, misalignment factor $\sin(\omega)$, and magnetic energy density in the perturbed elliptical cylindrical Flux Rope with radial current. Although local variations arise due to the applied perturbation, the global structure remains similar to the unperturbed model. The core remains force-free, and energy density peaks remain consistent with the previous configuration, except for localized reductions near the perturbation site.

With the introduction of a radial current, one expects changes in physical quantities such as the perpendicular current, the misalignment factor, and the magnetic energy density, however, the **global structure is expected to remain qualitatively similar to the unperturbed case**. That is, if the core of the previous model exhibited a force-free configuration, it will still do so here, except in regions directly influenced by the perturbation.

- ◇ **Force-freeness:** The **central core preserves its force-free character**, as indicated by the low values of the misalignment factor $\sin(\omega)$. Also, the **force-free region now extends toward the direction of the perturbation**, consistent with the previously observed absence of radial Lorentz forces in that region.
- ◇ **Magnetic energy density:** The **magnetic energy density reaches its maximum along the elongated boundaries of the ellipse**, similar to earlier models. In contrast, a **clear minimum is observed both at the center and along the direction of the perturbation**, reflecting the local reduction in magnetic field strength caused by the injected radial current.

5.3.5 3D Simulation of Magnetic Field Lines Within the Flux Rope

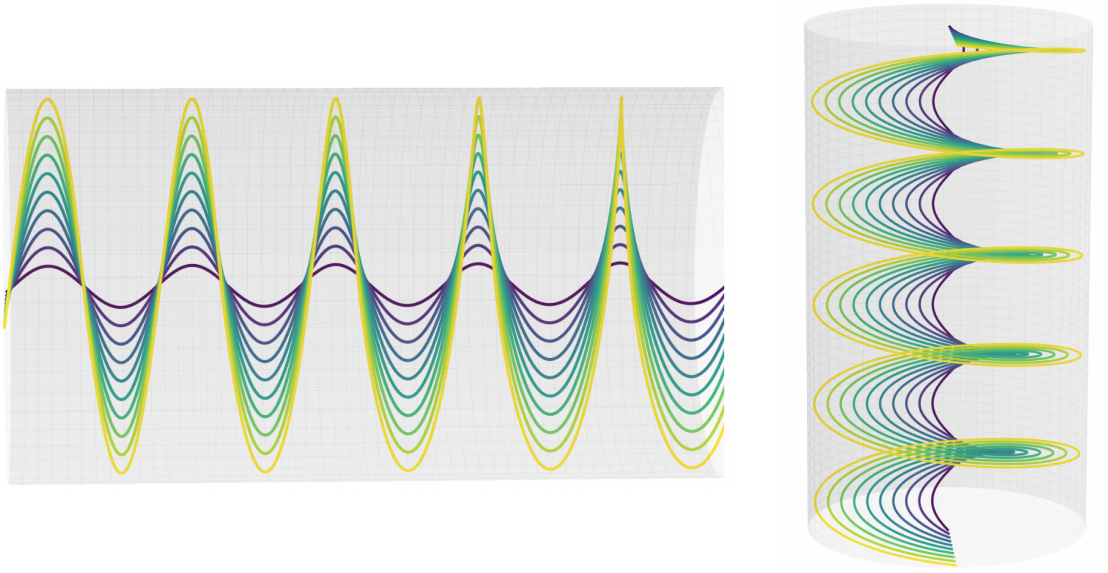


Figure 5.21: Top and side views of the confined synthetic elliptical cylindrical Flux Rope, illustrating the axial propagation of magnetic field lines.

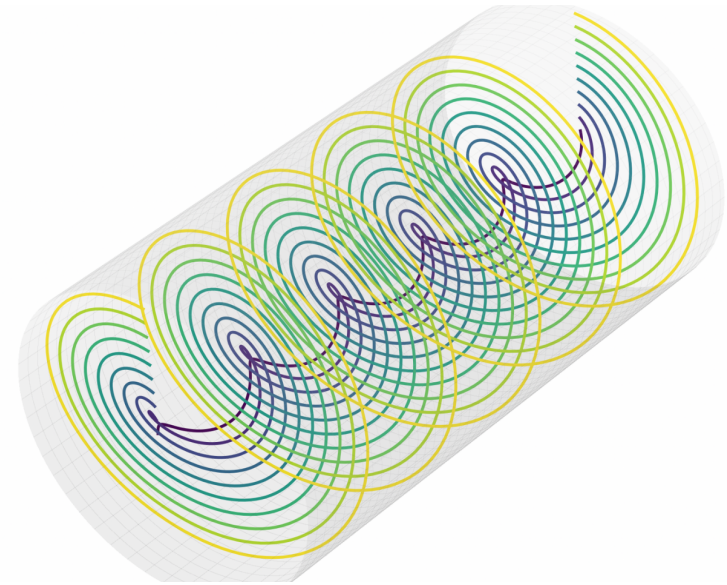


Figure 5.22: Oblique perspective of the Flux Rope showing the helical structure of the internal magnetic field lines.

The previous images display the 3D magnetic field lines of the perturbed elliptical Flux Rope. Unlike the unperturbed configuration, where the pitch angle remained constant due to the absence of a poloidal component, the **introduction of a localized perturbation breaks this symmetry**. As a result, **the pitch angle now varies along the field lines**, and clear distortions are observed as **field lines no longer form uniform helices but exhibit twisting/bending**.

5.3.6 Pitch Angle Variation of the Field Lines

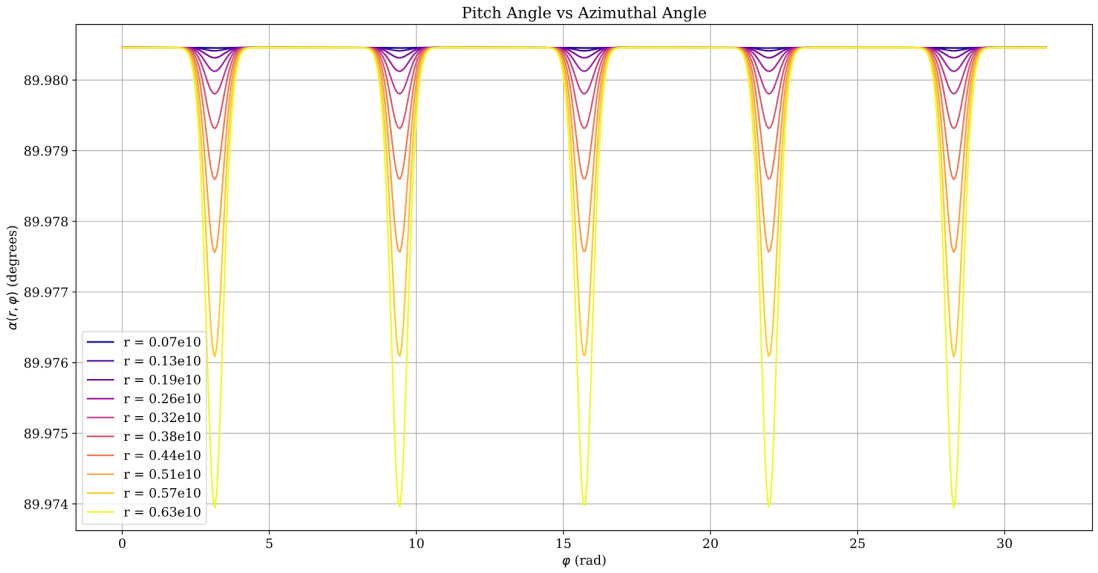


Figure 5.23: Pitch angle $\alpha(r, \varphi)$ as a function of poloidal angle φ for various radial positions within the Flux Rope.

This plot shows the pitch angle $\alpha(r, \varphi)$ as a function of the azimuthal angle φ for multiple radial positions int the perturbed. Unlike the circular/elliptical cylindrical case, where the pitch angle remained constant, **the presence of the localized perturbation (every 2π) introduces periodic distortions in the profile**. These sharp dips in pitch angle correspond to the position of the imposed Gaussian perturbation. This result confirms that **the field lines no longer follow uniform helices but instead exhibit localized variations in their inclination**.

5.3.7 Synthetic trajectories of the satelite and field magnitude along them

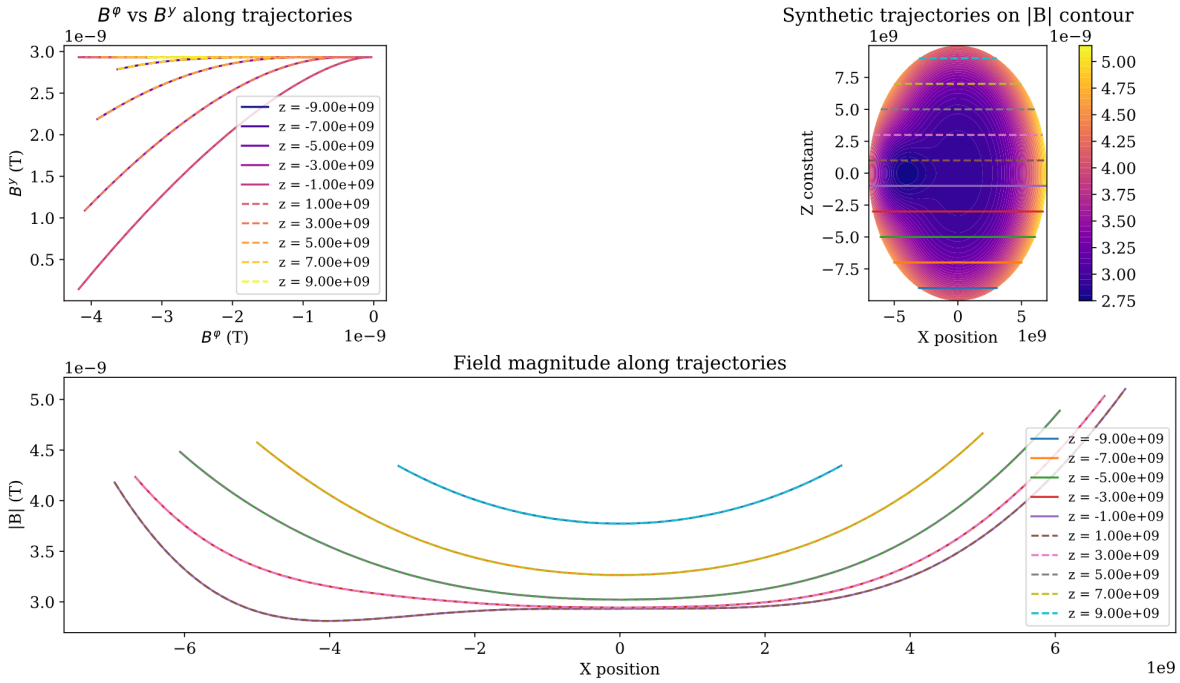


Figure 5.24: Synthetic spacecraft trajectories and corresponding magnetic field magnitude measurements in the elliptical cylindrical model. The lower plot shows the magnetic field magnitude along the x-axis for constant z-slices. The upper-left panel displays B^φ vs. B^y.

Figure 5.24 presents synthetic spacecraft trajectories and their corresponding magnetic field measurements in the perturbed elliptical Flux Rope configuration. Compared to the unperturbed case, a clear asymmetry emerges in the field behavior:

- ◇ In the lower plot, which shows the magnetic field magnitude $|B|$ along x for constant z-slices, the **profiles are no longer symmetric about** $x = 0$. The minima are shifted and differ in depth depending on the z-position, indicating a **loss of radial symmetry due to the perturbation**.
- ◇ In the upper-left panel, the B^φ vs. B^y curves, which **previously collapsed into a single aligned trajectory, now form different paths** for different z-slices. This highlights how the localized perturbation introduces different distortions in the poloidal and axial field components.
- ◇ The upper-right panel confirms the trajectory paths across $z = \text{const}$ planes. The **region around the perturbation shows deviation from the smooth elliptical contours seen previously**, confirming the localized **deformation of the magnetic field structure**.

These asymmetries illustrate how the introduction of a radial current and the resulting interaction change the internal structure of the Flux Rope, **breaking the ideal cylindrical symmetry and giving rise to more complex magnetic topologies**, precisely the type of behavior this new modeling approach aimed to capture.

6 . Thermodynamics of the Flux Rope

Current Flux Rope models, such as those proposed in Weiss et al. (2024) and Nieves-Chinchilla et al. (2023a) rely on two fundamental equations to determine the magnetic field components: Gauss's law for magnetism and the Ampère-Maxwell law. These allow the magnetic field to be derived from the corresponding current components, called in this thesis, generating functions. However, it is essential to emphasize that the physical system under study exhibits fluid-like behavior: composed of electrons and positive ions interacting each other. This naturally raises a deeper question:

- ◇ Could exploring the remaining governing equations provide not only greater internal consistency for the model but also a **more complete understanding of Flux Ropes?**
- ◇ Will this approach uncover hidden, **nontrivial relationships between the current generating functions** and **new constraints**?

Answering this question constitutes the core objective of the present section and will be fully addressed in the final sections of this work. Accordingly, the formulation of the thermodynamic model presented here will not introduce any additional free parameters beyond those already established in the circular cylindrical (CC) model when accounting for non-zero radial currents. The current density components are defined as:

$$\begin{aligned} j^r(r,\varphi) &= -\frac{\partial_\varphi\left(j_2^\varphi(\varphi)\right)}{r}\int_0^r r\, j_1^\varphi(r)\, dr \\ j^y(r) &= j_1^y(r) \\ j^\varphi(r,\varphi) &= j_1^\varphi(r)\cdot j_2^\varphi(\varphi) \end{aligned} \tag{6.1}$$

From these current components, all relevant thermodynamic quantities are derived, including the velocity field, pressure, temperature and mass density. Some previous efforts, such as Wang et al. (2015), have attempted to approach the internal thermodynamics of Flux Ropes by assuming a force-free configuration, as in the classical Lundquist (1950). However, it has already been established that such assumptions do not accurately capture the realistic behavior of these structures. Other models, like Mishra et al. (2020), introduce up to twelve degrees of freedom, making any attempt to reconstruct thermodynamic quantities from in-situ measurements very difficult due to the limited resolution and quality of the available data. So, **the objective here is not only to develop a physically consistent thermodynamic model, but to do so without introducing additional degrees of freedom beyond those already present in the non-force-free circular**

cylindrical configuration. The following assumptions underlie the formulation of the model:

- ◇ The system is in a **stationary regime**, so, all time derivatives are set to zero.
- ◇ physical quantities are **invariant under translations along the y-axis**.
- ◇ **No external forces** are considered beyond the currents and magnetic fields internally defined by the generating functions.

6.1 Formulation of the Governing Equations

6.1.1 Maxwell's Equations

As previously discussed, the magnetic field equations have been fully developed. The next step involves employing the electric field equations, Gauss's Law for the electric field and Faraday's law of induction, as well as the generalized Ohm's law, which provide additional constraints and insights into the system's behavior. In the stationary regime, these are expressed as:

$$\nabla \cdot \mathbf{E} = \frac{\rho_e}{\varepsilon_0} \tag{6.2}$$

$$\nabla \times \mathbf{E} = 0 \tag{6.3}$$

where ρ_e is the charge density and ε_0 is the vacuum permittivity. For a resistive case, the generalized Ohm's law reduces to:

$$\mathbf{j} = \sigma \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} \right) \tag{6.4}$$

where the term $\mathbf{E} + \mathbf{v} \times \mathbf{B}$ represents the total effective electric field in the moving plasma, and σ denotes the electrical conductivity of the medium, which is assumed to be constant throughout the analysis. In equation (6.4), the vector field $\mathbf{v}$ denotes the velocity of the fluid particles, representing the bulk motion of the plasma.

Plasma Quasineutrality Approximation

As seen in 2.5 quasineutrality assumption in plasma is highly reasonable. It implies that the ion number density is approximately equal to the electron number density in the medium, $n_e \approx n_i$. The significantly higher mobility of electrons compared to protons causes any charge imbalances to be rapidly neutralized, as the timescales associated with the phenomena under study are much longer than the microscopic timescales over which charge equilibrium is restored. Furthermore, the only regions where noticeable charge separation can occur are those with characteristic lengths on the order of the Debye length. In the case of a coronal mass ejection (CME), assuming an average electron temperature of 10^5K and an electron number density of $10^7 \mathrm{m}^{-3}$, the Debye length is given by:

$$\lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{2 n_e e^2}} \sim 1\text{m} \tag{6.5}$$

This value is significantly smaller than the characteristic scale of the system, $\lambda_D \ll R$ which tends to be in the order of $\sim 10^{11}$m, meaning that charge separation can only occur within extremely thin layers or highly localized regions. Therefore, the quasineutral approximation provides an excellent representation of the plasma behavior with a high degree of accuracy. This approximation further implies that the total internal charge of the CME is zero, since:

$$Q = \oint_V \rho_e dV \approx 0 \tag{6.6}$$

In this scenario, Gauss's law for the electric field yields

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{in}}}{\varepsilon_0} = 0 \tag{6.7}$$

The electrostatic field $\mathbf{E}$ must satisfy both

$$\nabla \times \mathbf{E} = 0 \tag{6.8}$$
$$\nabla \cdot \mathbf{E} = 0 \tag{6.9}$$

Equations (6.8) and (6.9) imply that $\mathbf{E}$ is both irrotational and divergence–free. Therefore, it admits a scalar potential ξ such that

$$\mathbf{E} = \nabla \xi \tag{6.10}$$

and substituting into $\nabla \cdot \mathbf{E} = 0$ gives the Laplace equation for the potential,

$$\nabla^2 \xi = 0 \tag{6.11}$$

with the initial value problem $\nabla \xi = \mathbf{E}$. One solves the resulting initial value problem by expanding the Laplacian operator in cylindrical coordinates, yielding:

$$\frac{1}{r}\, \partial_r (r\, \partial_r \xi) + \frac{1}{r^2}\, \partial_\varphi^2 \xi = 0 \tag{6.12}$$

To solve this equation, a separable ansatz function solution of the following form is proposed

$$\xi(r, \varphi) = R(r)\, \Phi(\varphi) \tag{6.13}$$

Substituting this ansatz into equation (6.12) and dividing by $R\Phi$ separates the variables:

$$\frac{1}{r\,R}\,\partial_r(r\,\partial_r R) + \frac{1}{\Phi\,r^2}\,\partial_\varphi^2\Phi = 0 \tag{6.14}$$

Once the variables are separated, two independent ordinary differential equations are obtained. The first depends only on the poloidal coordinate φ:

$$\frac{1}{\Phi}\partial_\varphi^2\Phi = -m^2 \tag{6.15}$$

where m^2 is the separation constant, chosen to be positive to ensure periodic solutions in φ. This leads to the standard form:

$$\partial_\varphi^2\Phi + m^2\Phi = 0 \tag{6.16}$$

The general solution to this linear homogeneous differential equation with constant coefficients is given by a linear combination of trigonometric functions:

$$\Phi(\varphi) = C_m\cos(m\varphi) + D_m\sin(m\varphi) \tag{6.17}$$

where C_m and D_m are constants determined by boundary conditions. The requirement $m \in \mathbb{Z}$ guarantees the 2π-periodicity that the function should preserve. For the radial component, the following equation is obtained:

$$\frac{1}{r}\partial_r\left(r\,\partial_r R\right) - \frac{m^2}{r^2}R = 0 \tag{6.18}$$

Multiplying by r^2, this can be rewritten as:

$$r^2\partial_r^2 R + r\partial_r R - m^2 R = 0 \tag{6.19}$$

This is a Cauchy-Euler differential equation, whose general solution depends on whether $m = 0$ or $m \neq 0$:

- ◇ For $m \neq 0$, the solution is a linear combination of power laws:

$$R(r) = A_m\,r^m + B_m\,r^{-m} \tag{6.20}$$

- ◇ For $m = 0$, the equation reduces to:

$$r^2\frac{\mathrm{d}^2R}{\mathrm{d}r^2} + r\frac{\mathrm{d}R}{\mathrm{d}r} = 0 \tag{6.21}$$

  whose general solution is:

$$R(r) = A_0 + B_0\ln r \tag{6.22}$$

  where A_0 and B_0 are constants.

Finally, due to the linearity of Laplace's equation, **the general solution is obtained by superposing all possible modes** m. Therefore, the global solution for the potential $\xi(r,\varphi)$ can be written as:

$$\boxed{\xi(r,\varphi) = A_0 + B_0 \ln r + \sum_{m=1}^{\infty} \left(A_m\, r^m + B_m\, r^{-m}\right) \left(C_m \cos(m\varphi) + D_m \sin(m\varphi)\right)} \tag{6.23}$$

The constants A_m, B_m, C_m, and D_m are determined by the specific boundary conditions of the physical problem and constitute (6.24). Since the solution must be well defined through the entire domain, including at $r = 0$, any terms that diverge, such as those proportional to $1/r$ or $\ln(r)$, must vanish. This ensures the existence and regularity of the function across the full domain.

$$\xi(r,\varphi) = A_0 + \sum_{m=1}^{\infty} \left(A_m r^m\right) \left(C_m \cos(m\varphi) + D_m \sin(m\varphi)\right) \tag{6.24}$$

This function is related to the electric field through:

$$\mathbf{E} = \nabla \xi \tag{6.25}$$

It should be noted that, on the boundary, the electric field must vanish due to boundary conditions and in order to maintain electric confinement. In contravariant components:

$$\begin{aligned} \nabla \xi = &\sum_{m=1}^{\infty} m A_m r^{m-1} \left(C_m \cos(m\varphi) + D_m \sin(m\varphi)\right) \mathbf{e}_r \\ &+ \sum_{m=1}^{\infty} m A_m r^{m-1} \left(D_m \cos(m\varphi) - C_m \sin(m\varphi)\right) \mathbf{e}_\varphi \end{aligned} \tag{6.26}$$

Evaluating at r = R, the following two conditions must be satisfied:

$$\sum_{m=1}^{\infty} m A_m \left(C_m \cos(m\varphi) + D_m \sin(m\varphi)\right) = 0 \tag{6.27}$$

$$\sum_{m=1}^{\infty} m A_m \left(D_m \cos(m\varphi) - C_m \sin(m\varphi)\right) = 0 \tag{6.28}$$

For both equalities to hold for all φ, it follows that $C_m = D_m = 0$ for all m. This implies that the solution reduces to:

$$\xi(r,\varphi) = A_0 \quad \Longrightarrow \quad \nabla \xi = \mathbf{E} = 0 \tag{6.29}$$

The conclusion is that the generalized Ohm's law takes the form:

$$\mathbf{j} = \sigma \mathbf{v} \times \mathbf{B} \tag{6.30}$$

6.1.2 Fluid Dynamics Equations

Since plasma also behaves as a fluid, it is necessary to impose a set of fundamental equations that govern its behavior. In this case, the continuity equation, representing mass conservation, and the momentum equation, describing the conservation of linear momentum of the fluid particles are considered. A stationary regime is assumed, and both viscous stresses and gravitational effects are neglected, as the dominant forces driving the phenomenon are the pressure gradient and the Lorentz force. Under these assumptions, in the steady-state limit, one can obtain:

$$\begin{aligned} \nabla \cdot (\rho \mathbf{v}) &= 0 \\ \rho(\mathbf{v} \cdot \nabla)\mathbf{v} &= -\nabla p + \mathbf{j} \times \mathbf{B} \end{aligned} \tag{6.31}$$

The concept of velocity in the description of the fluid

In fluid mechanics, the velocity in the governing equations is usually interpreted as the classical velocity of the species of particles composing the fluid. However, in plasma, multiple species can be found such as electrons, protons, and helium nuclei, which have different masses and charges and interact in distinct ways. It is therefore essential to clarify what is meant by this representative velocity of a "fluid element" in this description. Given the previously discussed quasi-neutrality and the fact that most of the plasma is composed of electrons and protons, a representative "fluid particle" can be thought as a volume element containing one electron and one proton traveling together. This interpretation accounts for the plasma mass, ensures quasi-neutrality, and provides a consistent and accurate framework for our description. Accordingly, the average velocity of this fluid element corresponds to the center of mass velocity of the electron-proton system.

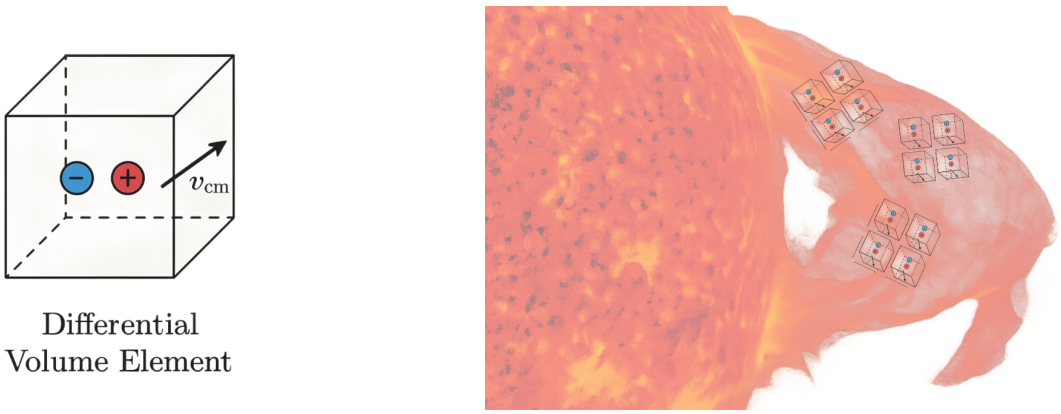


Figure 6.1: "Fluid particle" used in the description of the plasma and integration into a CME

We thus define the average velocity of a "fluid particle" as the center-of-mass velocity of the system. This is given by:

$$\mathbf{v} = \frac{m_e \mathbf{v}_e + m_p \mathbf{v}_p}{m_e + m_p} \tag{6.32}$$

Given that the mass of the proton is approximately 1000 times greater than that of the electron, this expression can be approximated by:

$$\mathbf{v} \approx \mathbf{v}_p + \frac{m_e}{m_p}\mathbf{v}_e \tag{6.33}$$

Since will be working in a region of the CME where the thermal velocity distribution is expected to be homogeneous as discussed in 2.4, away from the tails where suprathermal electrons may be present, the velocities of electrons and protons can be assumed to be of similar magnitude. Therefore, the expression further simplifies to:

$$\mathbf{v} \approx \mathbf{v}_p \tag{6.34}$$

6.1.3 Equation of State

To obtain a temperature distribution within the Flux Rope, it is essential to establish an equation of state or an energy balance. In this case, since it can be interpreted as a cloud of electrons and protons, a total plasma pressure can be defined. In this scenario, and given that pressure is a scalar quantity, the principle of superposition can be applied to compute the total pressure. The element of fluid is at a temperature T, which contributes a thermal pressure of nk_BT. Since dealing with both electrons and protons, the total thermal contribution becomes approximately:

$$P_{\text{thermal}} \approx 2nk_BT \tag{6.35}$$

Additionally, if there exists an internal velocity field $\mathbf{v}$, it contributes to the dynamic pressure, given by:

$$P_{\text{dynamic}} = \frac{1}{2}\rho|\mathbf{v}|^2 \tag{6.36}$$

Finally, the magnetic pressure resulting from the magnetic field $\mathbf{B}$ is:

$$P_{\text{magnetic}} = \frac{|\mathbf{B}|^2}{2\mu_0} \tag{6.37}$$

Therefore, the total pressure within the Flux Rope can be expressed as the superposition of all three contributions:

$$P_{\text{total}} = 2nk_BT + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{|\mathbf{B}|^2}{2\mu_0} \tag{6.38}$$

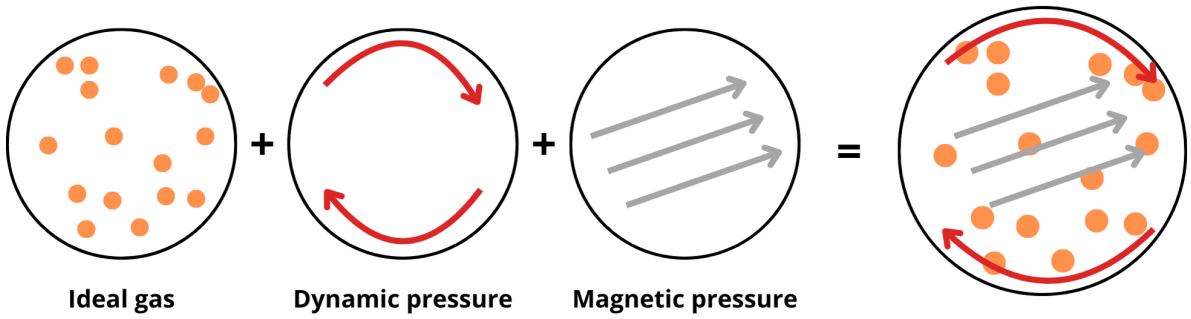


Figure 6.2: Schematic representation of the total pressure within a Flux Rope, resulting from the superposition of thermal pressure $2nk_BT$, dynamic pressure $\frac{1}{2}\rho|\mathbf{v}|^2$, and magnetic pressure $\frac{|\mathbf{B}|^2}{2\mu_0}$

However, in the momentum equation, the pressure term refers specifically to the thermal pressure, as the dynamic and magnetic pressure contributions appear on both sides of the equation.

$$\rho(\mathbf{v}\cdot\nabla)\mathbf{v} = -\nabla p - \nabla\left(\frac{|\mathbf{B}|^2}{2\mu_0}\right) + \frac{1}{\mu_0}(\mathbf{B}\cdot\nabla)\mathbf{B} \tag{6.39}$$

With the dynamic pressure:

$$\begin{aligned} &\rho\,(\mathbf{v}\cdot\nabla)\mathbf{v} = \nabla\left(\frac{1}{2}\rho|\mathbf{v}|^2\right) - \frac{1}{2}|\mathbf{v}|^2\nabla\rho - \rho\,\mathbf{v}\times(\nabla\times\mathbf{v}) \\ &\Rightarrow \underbrace{\nabla\left(\frac{1}{2}\rho|\mathbf{v}|^2 + p + \frac{B^2}{2\mu_0}\right)}_{P_{\text{total}}} = \frac{1}{\mu_0}(\mathbf{B}\cdot\nabla)\mathbf{B} + \frac{1}{2}|\mathbf{v}|^2\nabla\rho + \rho\,\mathbf{v}\times(\nabla\times\mathbf{v}) \end{aligned} \tag{6.40}$$

Therefore, once the pressure from the model has been obtained, it can be related to the temperature field as follows:

$$p = 2nK_BT \tag{6.41}$$

6.2 Exact analytical solution to the case $\sigma \to +\infty$

The methodology proposed for solving the internal thermodynamics of the Flux Rope has been optimized to determine all the relevant physical parameters. It is structured as follows:

- **Determination of the velocity field using the remaining electromagnetic equations**, in particular the generalized Ohm's Law (6.30). This equation establishes a relationship between the velocity field and the magnetic field although the price to pay is that it introduces an unknown scalar function that must be solved.
- **Application of fluid dynamic equations**, leading to the derivation of the den-

sity profile and the determination of the velocity and pressure fields through the conservation of mass and momentum equations.

- ◇ **Computation of the temperature field** using the equation of state.

This systematic methodology provides a standard and structured framework that proves especially useful when extending the solution to the resistive case, preventing the involved equations from becoming excessively complex.

6.2.1 Analytical Solution for the Velocity Field in the Superconducting Case

When resistivity is very low, in the limit of infinite conductivity ($\sigma \to \infty$), certain terms can be neglected. In the case of Ohm's Law (6.30), under the assumption of infinite conductivity:

$$\frac{\mathbf{j}}{\sigma} = \mathbf{v} \times \mathbf{B} \approx 0 \tag{6.42}$$

This implies that the velocity field is parallel to the magnetic field in all directions. The system behaves as a superconducting medium in which particles follow the magnetic field lines, and thus has no effect in slowing down the particles. The solution to this equation confirms that the fluid particles indeed move parallel to the magnetic field lines:

$$\mathbf{v} = \alpha(r, \varphi)\mathbf{B} \tag{6.43}$$

The contravariant components of the velocity field then become:

$$\begin{aligned} v^r &= 0 \\ v^\varphi(r, \varphi) &= \alpha(r, \varphi) B^\varphi(r) \\ v^y(r, \varphi) &= \alpha(r, \varphi) B^y(r, \varphi) \end{aligned} \tag{6.44}$$

Initially, there are no explicit constraints on the function α inside the Flux Rope cross-section; that is, α may vary over the disk. However, it must satisfy periodic boundary conditions in the angular direction (2π-periodicity). Additionally, note that this solution arises from the framework of ideal magnetohydrodynamics (MHD) and should be consistent with the description of magnetic field lines provided in 4.9.5. Specifically, the field lines were described by the relation:

$$\frac{dy}{r d\varphi} = \frac{B^y(r, \varphi)}{B^\varphi(r)} \tag{6.45}$$

If a segment of the spiral is taken in this form:

$$dy = \frac{B^y(r, \varphi)}{B^\varphi(r)}\, r d\varphi \tag{6.46}$$

and differentiate both sides with respect to time t, the instantaneous velocity is obtained:

$$\frac{dy}{dt} = \frac{B^y(r,\varphi)}{B^\varphi(r)} \frac{r\, d\varphi}{dt} \tag{6.47}$$

Expressed in terms of velocity components:

$$v^y = \frac{B^y(r,\varphi)}{B^\varphi(r)} v^\varphi \tag{6.48}$$

This is the same expression previously obtained in Equation (6.44), derived using the generalized Ohm's law. Therefore, the same relationship is achieved through two entirely different approaches: one based on a purely geometric interpretation, and the other using electromagnetic theory. This equivalence is a direct consequence of the ideal MHD assumptions and it is important to note that, in the resistive case, this relation no longer holds so directly. Additional parameters must be considered beyond the geometric interpretation.

6.2.2 Solution to the Fluid Mechanics Equations in the Superconducting Case

At this stage, the continuity equation and the momentum conservation equation are considered, respectively:

$$\begin{aligned} \nabla \cdot (\rho\, \mathbf{v}) &= 0 \\ \rho\, (\mathbf{v} \cdot \nabla)\mathbf{v} &= -\nabla P + \mathbf{f} \end{aligned} \tag{6.49}$$

Here, the vector $\mathbf{f}$ represents the Lorentz force in contravariant components. By substituting the velocity field obtained in Equation (6.44), the mass conservation equation can be rewritten as a constraint, which takes the form:

$$\partial_\varphi\, (\rho\, v^\varphi) = 0 \tag{6.50}$$

Momentum must be conserved in all directions, which leads to three equations, one per coordinate direction:

$$\begin{aligned} -\rho\, r\, (v^\varphi)^2 &= -\partial_r P + f^r \\ \rho\, v^\varphi\, \partial_\varphi v^\varphi &= -\frac{1}{r^2}\partial_\varphi P + f^\varphi \\ \rho\, v^\varphi\, \partial_\varphi v^y &= f^y \end{aligned} \tag{6.51}$$

To ensure the continuity condition is satisfied, Equation (6.50) is introduced into the momentum conservation equations, obtaining:

$$\begin{aligned} r\, \rho\, v^\varphi\, \partial_\varphi v^\varphi &= \partial_\varphi\, (\partial_r P - f^r) \\ (v^\varphi)^2\, \partial_\varphi \rho &= \frac{1}{r^2}\partial_\varphi P - f^\varphi \\ \partial_\varphi\, (\rho\, v^\varphi\, v^y) &= f^y \end{aligned} \tag{6.52}$$

Solving this system is a highly non-trivial task, as it represents a set of partial differential equations with unknown solution functions and no explicit constraints initially imposed. A useful starting point is to note that the solutions must be periodic in the poloidal component. Through a series of algebraic manipulations, the following expression is derived:

$$\frac{\partial_\varphi \rho}{\rho} = r\,\frac{f^\varphi - r^{-2}\partial_\varphi P}{f^r - \partial_r P} = \gamma(r,\varphi) \tag{6.53}$$

This equation relates the density field to the Lorentz force and the scalar pressure field. Both of these fields satisfy periodic boundary conditions, and let us define the right-hand side as $\gamma(r,\varphi)$. Thus, the general solution for the density field takes the form:

$$\rho(r,\varphi) = \rho_0(r)\,\exp\left(\int_0^\varphi \gamma(r,\varphi)\,d\varphi\right) \tag{6.54}$$

However, the density field must satisfy the periodicity condition $\rho(r,\varphi) = \rho(r,\varphi+2\pi)$ for all values of φ. For this to hold, the exponential term must yield a function that is constant or only r dependent. This conclusion arises from the fact that the integral of a periodic function is not necessarily periodic, therefore, the only way to ensure periodicity for any arbitrary function $\gamma(r,\varphi)$ is to require that $\gamma(r,\varphi) = 0$. This leads to the result that the **density field must be purely radial**: $\rho = \rho(r)$. Substituting this condition into the continuity equation (6.50), one can obtain:

$$\rho\,\partial_\varphi v^\varphi = \rho\,B^\varphi(r)\partial_\varphi \alpha(r,\varphi) = 0 \tag{6.55}$$

which implies that $\alpha(r,\varphi) = \alpha(r)$. Now, with the newly established constraints, it is possible to determine the pressure field using the following expression:

$$P(r,\varphi) = P_0 + \int_0^r \left(f^r + r\frac{B^\varphi f^y}{\partial_\varphi B^y}\right), dr \tag{6.56}$$

From this pressure distribution, the temperature field can be derived using the ideal gas law:

$$T(r,\varphi) = \frac{P(r,\varphi)}{2nk_B} \tag{6.57}$$

Moreover, for the density field and the scalar function α, the following relation holds:

$$\alpha^2\rho = \frac{f^y}{B^\varphi \partial_\varphi B^y} \tag{6.58}$$

Upon analyzing the right-hand side of the equation, one obtains:

$$\alpha^2\rho = \frac{f^y}{B^\varphi, \partial_\varphi B^y} = \frac{1}{\mu_0} \tag{6.59}$$

Since the product $\alpha^2\rho$ must depend solely on the radial coordinate r, as previously established, a trivial solution is adopted by assuming both scalar fields to be constant. Let M denote the total mass enclosed by the Flux Rope constituting the CME. Then, the total mass is given by:

$$M = \int_0^{2\pi} d\varphi \int_0^R r\rho(r,\varphi), dr \tag{6.60}$$

Assuming a uniform density, the density field reduces to:

$$\rho = \frac{M}{\pi R^2 L} \tag{6.61}$$

Finally, the value of the scalar field α becomes:

$$\alpha = \frac{1}{\sqrt{\rho\mu_0}} \tag{6.62}$$

Quantity	**Expression**
Radial velocity	$v^r = 0$
Azimuthal velocity	$v^\varphi(r) = \frac{1}{\sqrt{\rho\,\mu_0}}\, B^\varphi(r)$
Axial velocity	$v^y(r,\varphi) = \frac{1}{\sqrt{\rho\,\mu_0}}\, B^y(r,\varphi)$
Pressure field	$P(r,\varphi) = P_0 + \int_0^r \left(j^y B^\varphi - j^\varphi B^y + \frac{r(B^\varphi)^2}{\mu_0} \right) dr$
Temperature field	$T(r,\varphi) = \frac{P(r,\varphi)}{2nk_B}$
Density	$\rho = \frac{M}{\pi R^2 L}$
Alpha scalar	$\alpha = \frac{1}{\sqrt{\rho\,\mu_0}}$

Table 6.1: Physical properties in the ideal MHD limit ($\sigma \to +\infty$)

An unknown quantity can be identified, the static pressure P_0, which constitutes a global offset in the pressure function. This parameter can be estimated from observational data and is treated as a constant, its value will be discussed in detail later, but it typically lies on the order of nanopascals (pPa).

6.3 Analytical Solution for the Steady-State Case in the Resistive Regime

The resistive regime is particularly more complex because it inherently allows for magnetic reconnection processes. These phenomena occur when two magnetic field lines approach each other closely enough to merge, resulting in a final field configuration of lower energy than the initial one. By conservation of energy, the difference in magnetic energy is converted into internal energy, serving as a power source for CMEs and Flux Ropes. Although a resistive medium is a necessary condition for reconnection, it is not considered temporal variations in this steady state analysis; so no reconnection can happen. Nevertheless, the expressions derived here for the stationary resistive case may be extended in future work to include time dependent magnetic reconnection.

6.3.1 Analytical Solution for the Velocity Field in the Resistive Case

In the resistive regime, the assumption of infinite conductivity does not hold anymore. That is, particles can move due to magnetic field diffusion in the medium. This dependence shows how a resistive medium allows the deformation of magnetic field lines for transverse magnetic transport. While in the superconductive regime, particle movements were restricted to along the magnetic field lines, now one would expect particles could also transverse these lines, so that the Lorentz force can have a substantial effect on them. So, to take into account the hybrid phenomena:

$$\mathbf{j} = \sigma(\mathbf{v} \times \mathbf{B}) \tag{6.63}$$

In order to obtain the velocity field:

$$\mathbf{v} \times \mathbf{B} = \frac{\mathbf{j}}{\sigma} \tag{6.64}$$

and, by taking the vector product with $\mathbf{B}$, one can obtain:

$$\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) = \mathbf{B} \times \frac{\mathbf{j}}{\sigma} \tag{6.65}$$

Since

$$\mathbf{B} \times (\mathbf{v} \times \mathbf{B}) = |\mathbf{B}|^2 \, \mathbf{v} - (\mathbf{B} \cdot \mathbf{v}) \, \mathbf{B} \tag{6.66}$$

it is recognized that the velocity can be decomposed into components parallel and perpendicular to the magnetic field. In other words,

$$\mathbf{v} = \mathbf{v}_{\parallel} + \mathbf{v}_{\perp} \tag{6.67}$$

where $\mathbf{v}_{\parallel}$ is aligned with $\mathbf{B}$ and $\mathbf{v}_{\perp}$ lies in the vectorial subspace orthogonal to $\mathbf{B}$. In a purely orthogonal reference frame, the metric is diagonal, there are no cross-terms in the scalar product, and the velocity can be written as:

$$\mathbf{v} = \alpha\ \mathbf{B} - \frac{\mathbf{j}\times\mathbf{B}}{\sigma|\mathbf{B}|^2} = \underbrace{\alpha\,\mathbf{B}}_{\mathbf{v}_\parallel} - \underbrace{\frac{\mathbf{f}}{\sigma\,|\mathbf{B}|^2}}_{\mathbf{v}_\perp} \tag{6.68}$$

Here:

- $\mathbf{v}_\parallel = \alpha\,\mathbf{B}$, the component of the velocity parallel to the magnetic field; as in the superconducting case.
- $\mathbf{v}_\perp = \dfrac{\mathbf{f}}{\sigma\,|\mathbf{B}|^2}$, the component perpendicular to $\mathbf{B}$ arising from the balance of Lorentz and resistive forces in the new model.

Now, the contravariant components of the velocity take the form:

$$\begin{aligned} v^r &= -\frac{f^r}{\sigma|\mathbf{B}|^2} \\ v^y(r,\varphi) &= \alpha(r,\varphi)B^y(r,\varphi) - \frac{f^y}{\sigma|\mathbf{B}|^2} \\ v^\varphi(r,\varphi) &= \alpha(r,\varphi)B^\varphi(r) - \frac{f^\varphi}{\sigma|\mathbf{B}|^2} \end{aligned} \tag{6.69}$$

6.3.2 Solution to the Fluid Mechanics Equations in the Resistive Case

In the resistive case, new terms appear in the equations that make them more difficult to work with. However, the main difficulty is that now v^r is nonzero. Previously, any terms involving this component simply vanished because it was zero. For the continuity equation:

$$\partial_r(r\rho v^r) + \partial_\varphi(r\rho v^\varphi) = 0 \tag{6.70}$$

For the momentum equations, the three directions can be obtained as:

$$\begin{aligned} \rho\big(v^r\partial_r v^r + v^\varphi\partial_r v^r - r(v^\varphi)^2\big) &= -\partial_r P + f^r \\ \rho\big(v^r\partial_r v^\varphi + \tfrac{2}{r}v^r v^\varphi + v^\varphi\partial_\varphi v^\varphi\big) &= -\tfrac{1}{r^2}\partial_\varphi P + f^\varphi \\ \rho\big(v^r\partial_r v^y + v^\varphi\partial_\varphi v^y\big) &= f^y \end{aligned} \tag{6.71}$$

At first glance, this system may seem very complex to solve due to the number of unknowns involved. However, having carried out the superconducting solution actually allows to solve the resistive case. The complication arises from the term v^r, but it is already known, being a specific combination of the magnetic and current fields determined by the CC model. Consequently, v^r becomes a known inhomogeneity in the system rather than an additional unknown. The equation of state and the mass constraint are identical to those in the superconducting case. By enforcing poloidal periodicity, is is obtained again a constant density as the trivial solution:

$$\rho = \frac{M}{\pi R^2 L} \tag{6.72}$$

Introducing the resistive-case continuity equation (6.70) into the resistive-case momentum equations (6.71) and carrying out straightforward algebra yields a new expression for the parameter α, which now depends explicitly on position. Consequently, α becomes a scalar field rather than a constant.

$$\alpha(r,\varphi) = \alpha_0 + \frac{1}{\sigma B^{\varphi}(r)} \int_0^{\varphi} \left(\frac{1}{r}\partial_r \left(\frac{r f^r}{|\mathbf{B}|^2} \right) + \partial_{\varphi} \left(\frac{f^{\varphi}}{|\mathbf{B}|^2} \right) \right) d\varphi \tag{6.73}$$

The constant to be determined can be obtained by requiring that, in the superconductive limit, the same result as that obtained in 6.1 must be recovered. That is,

$$\lim_{\sigma \to +\infty} \alpha(r,\varphi) = \alpha_0 = \frac{1}{\sqrt{\rho\ \mu_0}} \tag{6.74}$$

Accordingly, the expression for the scalar field α is obtained as:

$$\alpha(r,\varphi) = \frac{1}{\sqrt{\rho\ \mu_0}} + \frac{1}{\sigma B^{\varphi}(r)} \int_0^{\varphi} \left(\frac{1}{r}\partial_r \left(\frac{r f^r}{|\mathbf{B}|^2} \right) + \partial_{\varphi} \left(\frac{f^{\varphi}}{|\mathbf{B}|^2} \right) \right) d\varphi \tag{6.75}$$

The pressure can be obtained by inverting the expression derived from the poloidal component of the momentum equation in 6.71, namely:

$$P(r,\varphi) = P_0 + r^2 \int_0^{\varphi} \left(f^{\varphi} - \rho \left(v^r \partial_r v^{\varphi} + \tfrac{2}{r} v^r v^{\varphi} + v^{\varphi} \partial_{\varphi} v^{\varphi} \right) \right) d\varphi. \tag{6.76}$$

The newly derived physical properties in the resistive regime can thus be expressed as follows:

Table 6.2: Physical properties in the resistive MHD regime with position-dependent $\alpha(r,\varphi)$

Quantity	**Expression**
Radial velocity	$v^r = -\dfrac{f^r}{\sigma\,\lvert\mathbf{B}\rvert^2}$
Azimuthal velocity	$v^{\varphi}(r,\varphi) = \alpha(r,\varphi)\, B^{\varphi}(r,\varphi) - \dfrac{f^{\varphi}}{\sigma\,\lvert\mathbf{B}\rvert^2}$
Axial velocity	$v^{y}(r,\varphi) = \alpha(r,\varphi)\, B^{y}(r,\varphi) - \dfrac{f^{y}}{\sigma\,\lvert\mathbf{B}\rvert^2}$
Pressure field	$P(r,\varphi) = P_0 + r^2 \int_0^{\varphi} \left(f^{\varphi} - \rho\left(v^r \partial_r v^{\varphi} + \tfrac{2}{r} v^r v^{\varphi} + v^{\varphi} \partial_{\varphi} v^{\varphi} \right) \right) d\varphi$
Temperature field	$T(r,\varphi) = \dfrac{P(r,\varphi)}{2nk_B}$

Continued on next page

Quantity	Expression
Density	$\rho = \dfrac{M}{\pi R^2 L}$
Alpha scalar	$\alpha(r,\varphi) = \dfrac{1}{\sqrt{\rho\,\mu_0}} + \dfrac{1}{\sigma B^{\varphi}(r)} \int_0^{\varphi} \left(\dfrac{1}{r}\partial_r \left(\dfrac{r f^r}{\lvert\mathbf{B}\rvert^2} \right) + \partial_{\varphi} \left(\dfrac{f^{\varphi}}{\lvert\mathbf{B}\rvert^2} \right) \right) d\varphi$

6.4 Application to a Synthetic Flux Rope

The physical parameters used in this model correspond to typical coronal mass ejection (CME) conditions observed at 1 astronomical unit (1 AU). These include:

- ◇ Electron number density: $n_e \sim 1 \times 10^7\,\mathrm{m}^{-3}$, averaged in situ measurements of ICMEs Richardson and Cane (2010) and typical values reported in Coronal Mass Ejection Group, University of New Hampshire (n.d.).
- ◇ Base thermal pressure: $P_0 \sim 10^{-12}\,\mathrm{Pa}$, as estimated by Klein and Burlaga (1982).

From these inputs, a thermodynamically synthetic Flux Rope is obtained (with $\sigma \to +\infty$), from which the main parameters that a satellite would measure, directly related to the provided observations are derived. In this case study, the averaged observations of Coronal Mass Ejection Group, University of New Hampshire (n.d.) and Nieves-Chinchilla et al. (2012) are compared with the synthetic trajectories. It is worth noting that, when the satellite crosses the Flux Rope, stagnation properties emerge. For the velocity distribution in the slice, the obtained fields are:

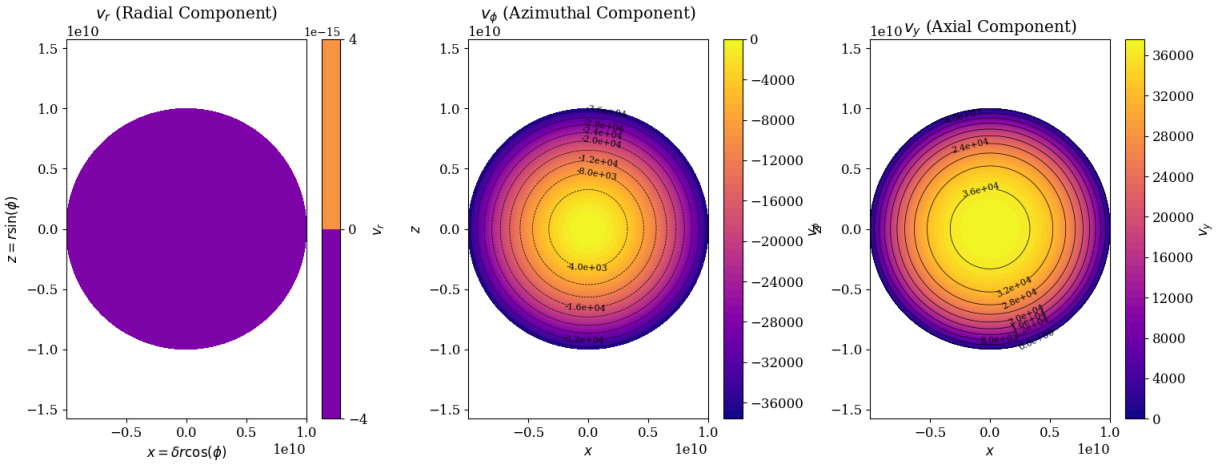


Figure 6.3: Radial (left), azimuthal (center), and axial (right) velocity components within the Flux Rope cross-section. The azimuthal and axial components exhibit a clear radial structure consistent with the expected internal magnetic field configuration.

This implies that the poloidal and axial velocity components are each on the order of $\sim$ 10% of the solar wind speed($\sim$ 400km/s $\to\sim$ 440km/s), fully consistent with the

velocity increase observed upon entering the magnetic obstacle, as shown in Coronal Mass Ejection Group, University of New Hampshire (n.d.).

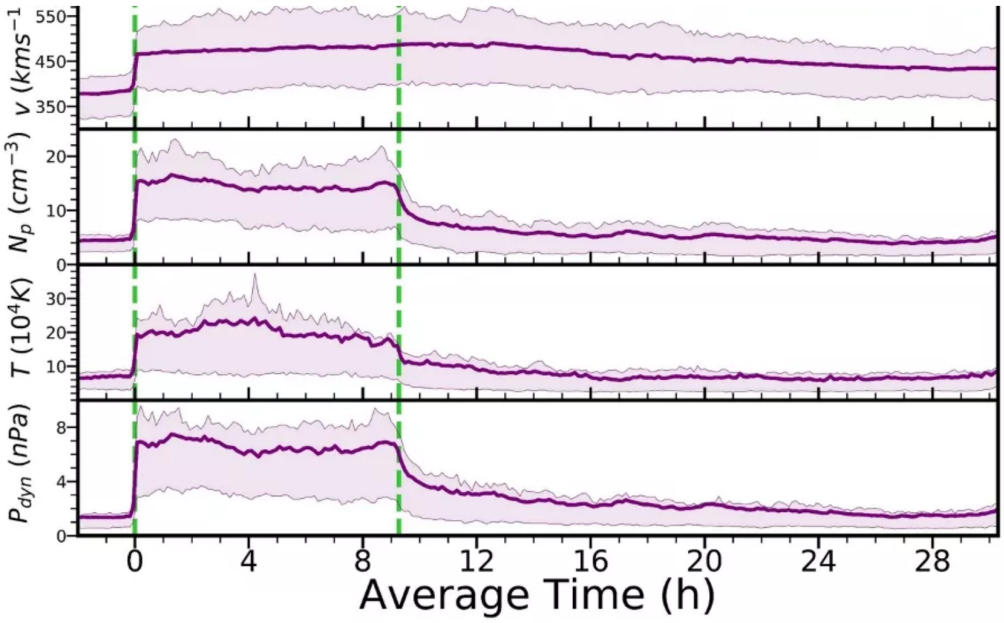


Figure 6.4: Observed average in situ profiles of CME passage at 1 AU. From top to bottom: plasma velocity, number density, temperature, and thermal pressure. The central dashed green line marks the magnetic cloud center and **the speed discontinuity of about** $\sim 10\%$ **can be observed**. Adapted from Coronal Mass Ejection Group, University of New Hampshire (n.d.).

Below is the thermal pressure and temperature (non-stagnation properties) obtained inside the Flux Rope:

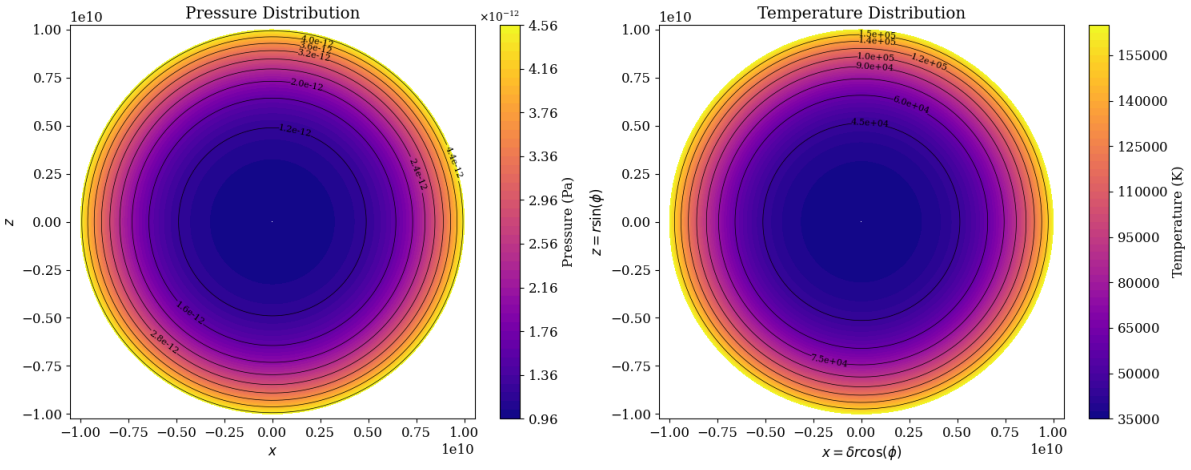


Figure 6.5: Cross-sectional maps of thermal pressure (left) and temperature (right) within the synthetic Flux Rope without considering stagnation effects.

However, if the velocity component responsible for stagnation, the one measured by the sensor is included, both the correct order of magnitude and similar behavior to those exhibited in Figure 6.4 and Nieves-Chinchilla et al. (2012) are recovered:

$$p_{\text{stag}} \approx p + \frac{\rho}{2}|\mathbf{v}|^2 \sim \text{nPa} \tag{6.77}$$

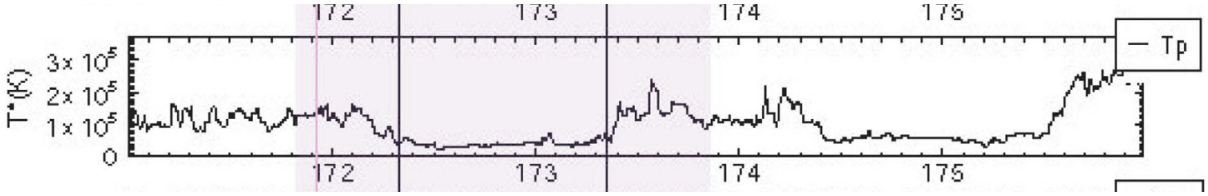


Figure 6.6: Non averaged stagnation temperature obtained from in situ observations Nieves-Chinchilla et al. (2012)

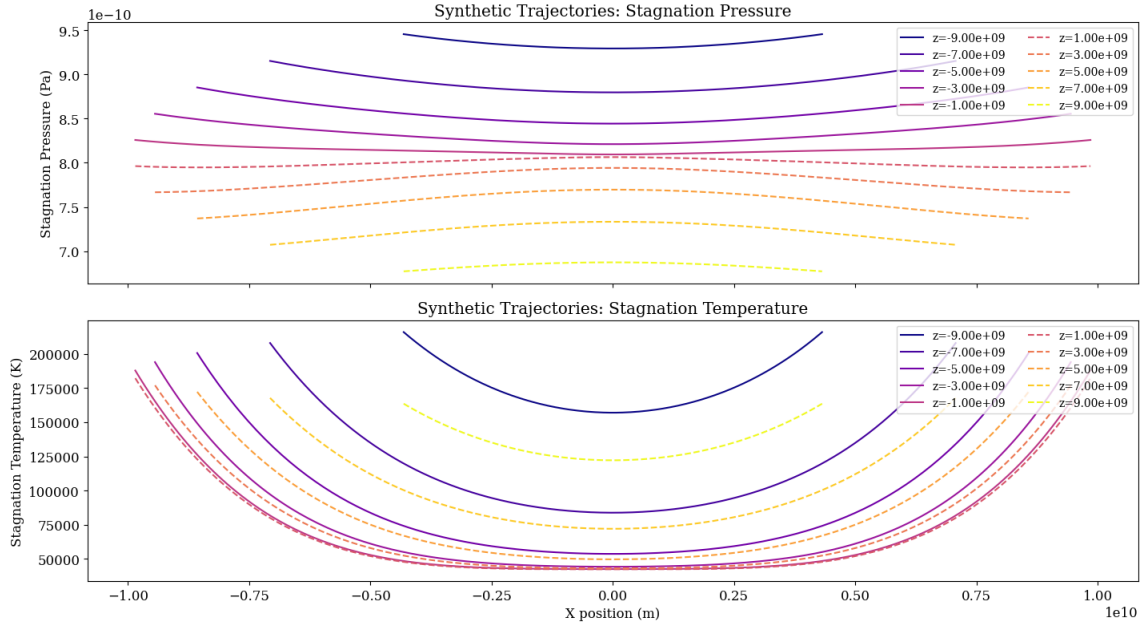


Figure 6.7: Stagnation pressure (top) and stagnation temperature (bottom) profiles along synthetic trajectories crossing the Flux Rope at different fixed heights z. The solar wind is assumed to be aligned with the z-axis, and its interaction with the internal velocity field produces localized enhancements in stagnation properties. Each curve corresponds to a distinct trajectory at a different z-position, revealing the axial symmetry of the stagnation profiles.

As a partial conclusion, this approach performs extremely well: the **agreement between the modeled fields and the in situ measurements is remarkably good, both in terms of profile shape and physical magnitude**. The synthetic model captures the essential features of Flux Rope thermodynamics, particularly the characteristic enhancement of stagnation temperature and pressure as a spacecraft traverses the magnetic structure. This modeling framework thus offers not only a powerful tool for understanding Flux Rope dynamics but also a robust basis for constraining model parameters during fitting procedures. In particular, it enables a consistent coupling between thermodynamic observables, such as temperature, pressure, or density and the magnetic field structure. This becomes especially useful in situations where certain magnetic or velocity profiles appear locally flat or ambiguous: by using the thermodynamic data obtained in-situ, one can extract meaningful physical constraints that may not be evident from the magnetic field alone. **Incorporating this type of synthetic diagnostic modeling into future**

fitting algorithms could significantly enhance the accuracy and robustness of CME reconstructions.

7 . Conclusions and Future Work

7.0.1 Main conclusions of the theory developed

The motivation for this work began with a fundamental question: **could the inclusion of a radial current not only improve our understanding of magnetic structures, but also enable us to model their interactions more comprehensively?** Or would it merely add another degree of freedom that complicates the system without offering physical insight? At the beginning, there were concerns around the additional complexity that such a current component might introduce. However, the slow development of the underlying theoretical framework revealed that the inclusion of a radial current is not only physically consistent, but in fact essential for a more complete description of Flux Rope structures. Far from being a redundant complication, **it emerged as a key component for modeling magnetic field phenomena that could not be explained previously through traditional force-free models**. The resulting formulation introduces only one additional degree of freedom into the system: the poloidal current becomes $j^{\varphi}(r) \rightarrow j^{\varphi}(r, \varphi)$. This single extension is enough to unlock Lorentz force components, specifically, the poloidal and axial components that were previously null under the assumption of force-freeness, while also enabling the emergence of a non-zero radial current. This new model represents a significant step forward in the analytical modeling of Flux Ropes. It provides a framework capable of describing:

- **Magnetic interactions** between magnetically confined structures via current coupling
- **Discontinuities in magnetic field** components that were previously inaccessible under classical assumptions
- And a **deeper understanding of the internal magnetic topology of Flux Ropes**

The framework developed offers a compelling extension to existing analytical models, preserving their usability while expanding their descriptive power. Looking forward, this model opens up promising ideas for future exploration, particularly in **applying this additional degree of freedom to interpret spacecraft data** and **constrain fits in observational studies**. Its ability to link theory with observed features of Flux Ropes structures makes it a powerful tool in advancing heliophysical modeling and our understanding of this magnetic objects.

On the other hand, once this theory was developed, a natural question arose: given that the Flux Rope described by this model does not fulfill a minimum energy condition, since

it is not in a terminal phase of its life cycle, **could there exist additional constraints on the currents arising from unused magnetohydrodynamic (MHD) equations?** The challenge, then, was to determine whether a thermodynamic model could reveal deeper relationships among the currents in the circular case Nieves-Chinchilla et al. (2016), without introducing any new degrees of freedom and whether such a framework could assist those performing magnetic field fittings by providing physical constraints. By developing the theory in a steady-state, resistive and also conductive regime under quasi-neutral conditions, it was possible to reach a remarkable result: **the model not only accurately reproduced the correct orders of magnitude of observed quantities via synthetic fields, but also captured the morphological structure of the stagnation properties as seen by spacecraft**. The **agreement between synthetic and observational profiles** was very good, both in shape and amplitude, suggesting that this approach is not merely theoretically but also practically applicable. Therefore, this framework not only extends the physics embedded in analytical models of Flux Ropes, but also **provides a robust tool for the scientific community engaged in magnetic field reconstruction and interpretation**. Its capacity for constraining current distributions and predicting observable quantities of the thermodynamics makes it a most promising prospect for further development in heliophysical modeling as well as furthering our comprehension of heliospheric Flux Ropes.

7.0.2 What to Expect from Future Work and Next Steps

From my perspective, one of the most critical directions for future research is to analyze the effects of introducing a radial current on the structural dynamics of Flux Ropes. The emergence of non-radial Lorentz forces leads to tangential stresses that may be responsible for triggering instabilities such as warping, sausage modes, or kink deformations in the magnetic structure. These force components arise as a direct consequence of the radial current and suggest an **stability analysis** to understand their physical implications on the evolution and topology of the Flux Rope. In parallel, it would be **beneficial to develop a catalog of new radial current profiles capable of modeling a broader range of physical scenarios**. These may include the response of the Flux Rope to external magnetic perturbations, finite jump discontinuities in the field and asymmetries in current distributions. Such a framework would significantly enhance the analytical flexibility of the model and open the door to more realistic reconstructions of observed events. From a thermodynamic standpoint, future work could involve **testing ansatz type solutions with temporal separability to study the relaxation behavior of Flux Ropes as they evolve toward a force-free state**. These time-dependent extensions would allow the scientific community to characterize how internal stresses and thermodynamic quantities decay under non-ideal conditions. Additionally, it would be valuable to **build a comprehensive database of synthetic spacecraft trajectories based on the thermodynamic model**. This **could provide the magnetic field fitting community with a powerful tool of constrained reconstructions using observational data**. The current thermodynamic framework can also be **extended to extract energetically relevant quantities**, such as internal energy, entropy variations, or localized heat accumulation.

7.0.3 Other Contributions to the Team and Personal Milestones

I'd like to dedicate this section to taking a moment of reflection on my own growth over the last few months of being at NASA Goddard. I was tasked with improving the radial-current analytical model of Nieves-Chinchilla et al. (2018) when I joined the team. After following the formulation of the theory, a scientific paper is in the process of being written up jointly with Dr. Teresa Nieves Chinchilla [ORCID] and Dr. Laura Balmaceda [ORCID]. This work further extends the effort towards creating more physically realistic analytical models of Flux Ropes. After that, I undertook the initiative of investigating the model's thermodynamic consequences, as explained in the previous chapter. This task was highly challenging and involved a lot of extensive mathematical formulation. A second paper is in the process of being written up and addresses the full formulation of the thermodynamic theory and the synthetic spacecraft profiles obtained therefrom. This work is expected to make a significant contribution toward the topic area as analytical MHD models with realistic thermodynamics lack abundance. Although the analytical solution relies on the geometry presented in Nieves-Chinchilla et al. (2016), its structure can differ for more realistic configurations, however, the information extracted from the solution should provide a good basis for investigating more complex magnetic topologies. Outside of the Flux Ropes, I have actively participated in a project led by Dr. Cynthia López-Portela [ORCID] and with the collaboration Stefano Boccelli [ORCID] on the propagation of blobs of plasma in the interplanetary medium. In this activity, we are comparing the evolution of blob volume, mass, and density using a CME-based propagation model. In the process of doing this work, I provided a new analytical formulation describing the extension and variation of blobs in their passage throughout interplanetary media. These findings have turned out to be of particular insight and are being formulated into a scientific paper that is in preparation. Lastly, in an experiment oriented capacity, I assisted in the SONTRAC project (Instrument for the Detection of Fast Neutrons) by resolving a critical diffraction problem that was compromising the measurement capacity of the instrument. Specifically, I helped block the Fraunhofer diffraction that existed in the device by preparing and implementing a 3D-printed mask from a conductive PLA material. The mask acted as a wave guideline structure and served the purpose of suppressing unwanted diffraction. Consequently, the team managed to achieve accurate measurement across both low and high gain channels of the sensor array; these future months will stay in touch for collaborations not only in SONTRAC but LOONS mission too.

Figure 7.1: Left: High-gain signal response of the SONTRAC sensor array after implementation of a custom 3D-printed conductive mask to mitigate Fraunhofer diffraction. The mask successfully eliminates diffraction across the majority of the array, resulting in saturation of only a single channel (pixel 27), even under high-gain conditions. Right: Cleanroom activities at NASA Goddard during the preparation and inspection of the diffraction-suppression components.

8. Methodology and Planning

This thesis followed a structured timeline combining model development, scientific writing, and collaborations. After a literature review in September–October 2024, I developed the CC and EC models in early November, followed by the radial current extension and physical analysis through December. After a short winter break, I wrote the EC paper (Jan–Feb), developed the thermodynamic model (Feb–Mar), and drafted its paper (Mar–Apr), while also writing this thesis from mid-March to mid-May. In parallel, I collaborated on a study of interplanetary blobs (Feb–May) and supported the SONTRAC team at NASA (Dec–Apr).

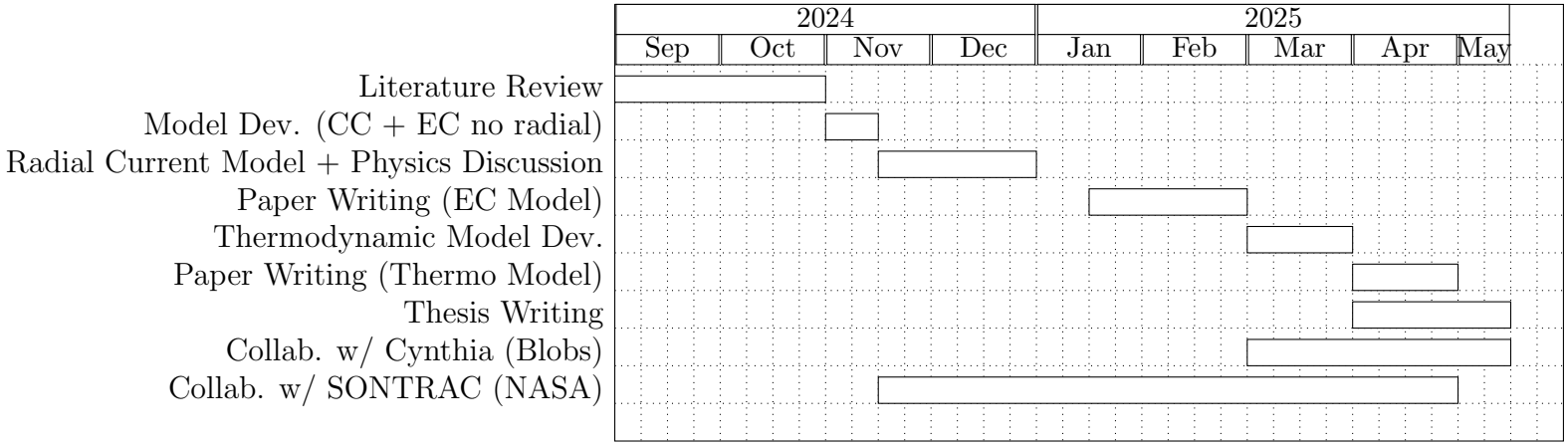


Figure 8.1: Gantt chart showing the planning of the development of this thesis.

9 . Sustainability Report

9.1 Sustainability matrix

Table 9.1 summarises the environmental, economic and social dimensions of the three project phases required (development, execution, and risks and limitations). The paragraphs after the table discuss every part in detail.

	Development of the thesis	**Execution / life-cycle of the project**	**Risks and limitations**
Environmental	◇ **Travel by air:** 4 transatlantic flight legs generate 3.5 t CO_2.Extracted from Carbon Footprint Ltd (n.d.) ◇ **Travel by car:** 4 992 mi in a 2013 Nissan Altima add 1 t CO_2. Extracted from Carbon Footprint Ltd (n.d.)	◇ Theoretical work with negligible operational footprint; future numerical services for space weather forecasting may add emissions.	◇ If future simulations scale up in resolution and parameters, energy consumption could grow sharply.

	Development of the thesis	**Execution / life-cycle of the project**	**Risks and limitations**
Environmental	◇ **Laptop use:** 7h/day^{-1}, 6 days/week^{-1} consume 80.6 kWh $\rightarrow$ 32 kg CO_2 ◇ **Total footprint during development:** about 4.5 t CO_2		
Economic	◇ Flights: \$1200 for the two round trips ◇ Housing: \$900 per month from October to mid-May totals \$6750 ◇ Car share: \$2500; fuel: \$582 (166 gal $\times$ \$3.50) ◇ Car insurance: \$2000 ◇ Food: \$300 per month, giving \$2250; gym membership: \$150 ◇ Electricity, water, gas and trash management: \$750 ◇ **Total development cost: about \$16182**	◇ Due to the theoretical nature of the work, the project's operational phase is hosted under NASA's Heliophysics Science and Space Exploration programme.	◇ Funding gaps may arise when political priorities shift; across-the-board cuts to NASA's budget could ultimately cancel these initiatives.

	Development of the thesis	Execution / life-cycle of the project	Risks and limitations
Social	◇ Inclusive and gender-neutral language is used throughout the thesis	◇ Early warnings enhance resilience for power-grid operators, airlines, insurance providers, and critical radio-frequency systems	◇ An inaccurate space-weather forecast could lead to regional blackouts, communication-equipment failures, and other societal disruptions

Table 9.1: Sustainability matrix of the present thesis

9.1.1 Environmental impact

Development. The largest individual contribution to the project's emissions is air travel. Four Washington–Spain flights (two there and back, economy class) emit around $3.5\,\mathrm{tCO_2}$, already representing around 78 % of the emissions expended in the development phase. Daily travel between residence and NASA Goddard contributes a further $1.0\,\mathrm{tCO_2}$ (approximately 22 %) if driving in a 2013 Nissan Altima with an average emission rate of $125\,\mathrm{g/km^{-1}}$. Laptop usage continually throughout the day for seven days a week from October 2024–May 2025 contributes 80.6 kWh, which is a relatively modest $32\,\mathrm{kg/CO_2}$ (sub-1 % of the total). Adding these gives a total development emissions of around $4.5\,\mathrm{tCO_2}$. Measures taken to reduce these emissions include keeping research trips concentrated to the minimum involving air travel. Future initiatives might decrease the emissions even more by conducting one transatlantic trip and using a low-emission car or public transport when commuting is necessary.

Execution. Since the thesis presents a purely conceptual model, the operational phase is currently being run on current NASA facilities and hence does not put any significant additional environmental burden. If the model eventually become part of an operational space weather forecasting service, its contribution will then be dominated by high performance computation cycles.

Risks and limitations. The most pressing environmental challenge is a possible spike in compute demand if more resolution physics or degrees of freedom are included in the model. Such an extension potentially would add a great amount of electricity consump-

tion. Active monitoring of the resultant energy intake and the purchasing of renewable-energy are suggested as a way of keeping the operating profile reasonable.

9.1.2 Economic impact

Development. Direct expenses for the thesis comes to roughly $16182. The largest share is housing, $6750 for eight months, which represents about 42 % of the total. Car costs amount to $5082, or 31 %.Living expenses (food, gym, utilities) add another $3150, roughly 20 %. Finally, flights account for $1200, or just 7 % of the overall budget. Keeping travel to the minimum necessary was the main cost-containment strategies adopted during development.

Execution. At present the model runs on existing NASA heliophysics infrastructure, so operating costs are negligible. If the code is transitioned to a real-time forecasting service, annual expenses will increase.

Risks and limitations. Future funding is sensitive to shifts in NASA's budget priorities. Regular cost reviews and the pursuit of mixed funding (agency + private industry) are recommended to mitigate these risks.

9.1.3 Social impact

Development. Inclusive, gender-neutral language and accessible figures were used throughout the thesis.

Execution. If deployed, the model would provide earlier space weather warnings, boosting resilience for power operators, airlines, insurers, and radio-frequency systems.

Risks and limitations. Incorrect forecasts could erode public trust or trigger unnecessary shutdowns; timely validation and clear communication are therefore essential.

9.2 Ethical implications

The project addresses a clear public-safety need and follows the UPC Ethical Code and since the project addresses merely theoretical analysis of Flux Ropes, there is no direct human-subject and environmental intervention involved.

9.3 Relationship with the Sustainable Development Goals

- **SDG3:Good health and well-being**: better space weather forecasts help protect flight crews, passengers and satellite operators from radiation related health risks.

- ◇ **SDG7:Affordable and clean energy**: greater resilience of power grids.
- ◇ **SDG9:Industry, innovation and infrastructure**: innovative protection of critical infrastructure through advanced modelling and early warning systems.
- ◇ **SDG11:Sustainable cities and communities**: reliable electricity and communication networks support the sustainability and safety of urban areas.
- ◇ **SDG15:Life on land**: preventing large scale grid failures minimises cascading environmental impacts, such as uncontrolled emissions from backup generators and potential harm to terrestrial ecosystems.

9.4 Conclusions

The project's main environmental consumer is the 4.5tCO_2 emitted during development, driven almost entirely by air travel; operational impacts are currently negligible. Direct costs amount to about \$16k, while the potential societal benefit, more reliable space, weather warnings makes it totally worth it. If future work limits flights, computation and secures stable funding, the model can deliver high scientific and public value with a minimal, well-managed footprint.

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